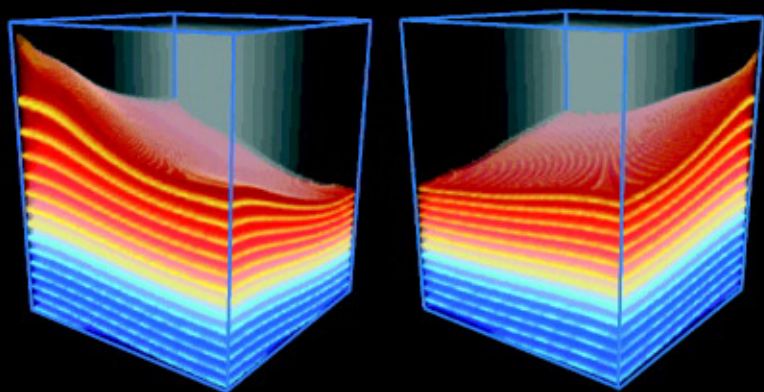




Liquid Sloshing Dynamics

Theory and Applications

Raouf A. Ibrahim

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Liquid Sloshing Dynamics

Theory and Applications

The problem of liquid sloshing in moving or stationary containers remains of great concern to aerospace, civil, and nuclear engineers; physicists; designers of road tankers and ship tankers; and mathematicians. Beginning with the fundamentals of liquid sloshing theory, this book takes the reader systematically from basic theory to advanced analytical and experimental results in a self-contained and coherent format. It presents liquid sloshing effects on space vehicles, storage tanks, road vehicle tanks and ships, and elevated water towers under ground motion.

The book is divided into four sections. Part I deals with the theory of linear liquid sloshing dynamics; Part II addresses the nonlinear theory of liquid sloshing dynamics, Faraday waves, and sloshing impacts; Part III presents the problem of linear and nonlinear interaction of liquid sloshing dynamics with elastic containers and supported structures; and Part IV considers the fluid dynamics in spinning containers and microgravity sloshing.

This book will be invaluable to researchers and graduate students in mechanical and aeronautical engineering, designers of liquid containers, and applied mathematicians.

Raouf A. Ibrahim is a professor of mechanical engineering at Wayne State University. From 1963 until 1971, he worked as a research engineer at the Aerospace Research Center of the rockets industry in Egypt, then gained his Ph.D. in Mechanical Engineering from the University of Edinburgh in 1974. From 1976 to 1979, he worked as a senior research specialist at Sakr Factory in Cairo and as an adjunct assistant professor at Cairo University. In 1979, he moved to the United States and worked at Shaker Research Corporation, before joining Texas Tech University in 1980 as assistant, associate, and then full professor. In 1987, he joined Wayne State University and continued his research activities in nonlinear random vibration, liquid sloshing dynamics, friction-induced vibration, and flutter of aeroelastic structures. In 1994 he was named the Arthur Carr Professor of Engineering and in 1995 he was awarded the Board of Governors Outstanding Professor Award. He has published more than 90 papers in refereed journals, and a research monograph. He is a Fellow of the ASME and Associate Fellow of the AIAA.

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CAMBRIDGE UNIVERSITY PRESS

Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore, São Paulo

Cambridge University Press

The Edinburgh Building, Cambridge CB2 2RU, UK

Published in the United States of America by Cambridge University Press, New York

www.cambridge.org

Information on this title: www.cambridge.org/9780521838856

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First published in print format 2005

ISBN-13 978-0-511-12364-1 eBook (Adobe Reader)

ISBN-10 0-511-12364-7 eBook (Adobe Reader)

ISBN-13 978-0-521-83885-6 hardback

ISBN-10 0-521-83885-1 hardback

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Foreword

When I was a very young boy I was enchanted by airplanes. The very idea that such a machine, with no apparent motions of its own – except, of course, for that tiny rotating thing at the front – could fly through the air was amazing. I could see that birds and insects could all fly with great dexterity, but that was because they could flap their wings, and thus support their weight as well as maneuver. And fish could even “fly” through water by motions of their body. How exciting it was then, to begin to learn something about how objects interact with the fluids surrounding them, and the useful consequences of those flows. That ultimately led, of course, to the broad study of fluid dynamics, with all of its wonderful manifestations.

There is hardly a single aspect of our daily lives, and indeed even of the entire universe in which we live, that is not in some way governed or described by fluid dynamics – from the locomotion of marine animals to the birth and death of distant galaxies. As a major field of technical and scientific knowledge, there are vast bodies of literature devoted to almost every facet of fluid behavior: laminar and turbulent flows, discontinuous (separated) flows, vortex flows, internal waves, free surface waves, compressible fluids and shock waves, multi-phase flows, and many, many others. With such a countless array of fluid phenomenon before us, what then leads to the focus of the present work?

Almost since the earliest philosophers began the development of what we now call “rational mechanics,” curiosity about how liquids behave when contained in some form of vessel has been strong. Beginning with the oscillations of water in lakes and harbors occurring as the result of earthquakes, similar phenomena surround us at almost every turn, arising from all aspects of our modern technology and civilization as well as from nature herself. Thus, geophysicists and seismologists, engineers, mathematicians, and many other scientific workers have devoted themselves to the study of these fascinating subjects for many years. In terms of modern technology, the behavior of liquids in contained systems is of importance in the tankage of ocean-going ships and offshore platforms, in airplanes, automotive vehicles, railway cars, nuclear power plants, and, of course, rockets and spacecraft, and countless others as well. One would think, however, that the general question of the behavior of liquids in such systems is a rather simple one to answer – not so. Our everyday experience in carrying a cup of coffee or a bowl of soup may be frustrating unless we are very careful as to how we move, but may still deceive us into believing that the “sloshing” of the liquid is simple. Indeed, depending on the amplitudes and frequencies of the motions of the container, the responses of the liquid free surface can be almost bewildering in their complexity!

Interest in this subject came almost to a crescendo, of course, with the advent of large liquid propelled rockets nearly half a century ago. At that time almost no one imagined that so many

forms of liquid behavior could manifest themselves or that such a tremendous body of technical literature would evolve as a consequence of trying to describe, understand, and predict these fascinating phenomena. Over the years, a fairly large number of monographs and literature summaries appeared in an effort to catalog what is known, and while these have each been useful in their own way, none has been exhaustive. In this work, the author has tried to do just that.

This is a monumental work in that it attempts to cover almost every aspect of liquid sloshing dynamics and cites approximately three thousand references. I would hope that its readers will find in it whatever they are looking for – if not, I do not know where else they should look and so they might have to discover it for themselves, perhaps using the methods and analytical tools described herein.

*H. Norman Abramson
San Antonio, March 2004*

Acknowledgment

The story of this book goes back to my first job as a Research Engineer at the Rocket Research Center, Cairo, Egypt, in 1963. I was involved in estimating the trajectories of liquid propellant rockets. This task required the knowledge of the actual values of mass moment of inertia and center of mass versus the time of rocket trajectory. The difficulty was mainly in the liquid part which constitutes almost 85% of the total weight of the rocket. I used to ignore the contribution of the liquid propellant in roll oscillations. This inability on my part motivated me to enroll for the Masters degree (by research) at Cairo University. Professor M. I. Rashid, my M.S. thesis advisor, encouraged me to conduct an extensive literature review on the entire subject of liquid propellant sloshing dynamics in addition to the main task of solving the Navier–Stokes equations to estimate the effective boundary layer thickness in roll due to tank walls and immersed pipes. Upon completion of my Masters degree, I was awarded a scholarship to study for my Ph.D. at the University of Edinburgh under the supervision of Professor A. D. S. Barr to work on the autoparametric interaction of liquid sloshing dynamics with supported structure dynamics. At that time I compiled over 900 references on liquid sloshing dynamics and related topics.

When I joined academia at Texas Tech University in 1981, I continued my research on liquid sloshing under random excitation with the help of my graduate students Drs Soundararajan, Hun Heo, Wenlung Li, J. Gau, and Messrs G. Lattore and R. Henrieck. The work was supported by the National Research Foundation and dealt with the random response of liquid free surface under different types of random excitation. When I moved to Wayne State University, the work was extended jointly with Dr Valery Pilipchuk to consider the liquid sloshing impact supported by the National Science Foundation (NSF). Dr Mohamed El-Sayad joined me to conduct his Ph.D. on the same problem. During the same time, Professor Takashi Ikeda spent several weeks at Wayne State University to conduct Monte Carlo simulation on parametric random excitation of liquid containers supported on elastic structures in the presence of internal resonance. Both Professor Ikeda and Dr Pilipchuk joined me to write an extensive literature review on liquid sloshing dynamics citing over 1200 references. Upon completing the review article I was strongly motivated to write the present book to cover every aspect of the subject. However, as soon as I began, I found the task was formidable and beyond my own capacity. It was my wife, Sohair, who encouraged me to accept the challenge and to complete the task at my own pace. I thank her so deeply for her patience and long suffering during the last four years. Indeed without her support I could not have completed the book.

Upon completion of the first draft of the manuscript, I asked Dr Norman Abramson, who I consider to be the father of liquid sloshing, to write a Foreword for the book. May God grant

him health and many years. I extend my sincere thanks to my colleagues and friends who have helped me. In particular I thank Professors Victor Berdichevsky at Wayne State University, and Professor W. Eidel of the Universitat der Bundeswehr, Germany, who helped me prepare the proof of some results.

Last but not least, I would like to thank the editorial staff of Cambridge University Press. In particular, I thank Dr Phil Meyler who handled the editorial process, and Ms Emily Yossarian, Ms Carol Miller, and Mr Keith Westmoreland who have been working with me during the production process.

Introduction

Sloshing means any motion of the free liquid surface inside its container. It is caused by any disturbance to partially filled liquid containers. Depending on the type of disturbance and container shape, the free liquid surface can experience different types of motion including simple planar, nonplanar, rotational, irregular beating, symmetric, asymmetric, quasi-periodic and chaotic. When interacting with its elastic container, or its support structure, the free liquid surface can exhibit fascinating types of motion in the form of energy exchange between interacting modes. Modulated free surface occurs when the free-liquid-surface motion interacts with the elastic support structural dynamics in the neighborhood of internal resonance conditions. Under low gravity field, the surface tension is dominant and the liquid may be oriented randomly within the tank depending essentially upon the wetting characteristics of the tank wall.

The basic problem of liquid sloshing involves the estimation of hydrodynamic pressure distribution, forces, moments and natural frequencies of the free-liquid surface. These parameters have a direct effect on the dynamic stability and performance of moving containers.

Generally, the hydrodynamic pressure of liquids in moving rigid containers has two distinct components. One component is directly proportional to the acceleration of the tank. This component is caused by the part of the fluid moving with the same tank velocity. The second is known as “convective” pressure and represents the free-surface-liquid motion. Mechanical models such as mass-spring-dashpot or pendulum systems are usually used to model the sloshing part.

A liquid’s motion inside its container has an infinite number of natural frequencies, but it is the lowest few modes that are most likely to be excited by the motion of a vehicle. Most studies have therefore concentrated on investigating forced harmonic oscillations near the lowest natural frequencies, predicted by the fluid field linear equations. However, nonlinear effects result in the frequency of maximum response being slightly different from the linear natural frequency and dependent on amplitude. Nonlinear effects include amplitude jump, parametric resonance, chaotic liquid surface motion, and nonlinear sloshing mode interaction due to the occurrence of internal resonance among the liquid sloshing modes. The nonlinearities associated with free-surface motion inside moving containers are different from those nonlinear water waves in ocean and canals. The theory of nonlinear dispersive waves originated by Stokes (1947) and solitons¹ observed by Russell (1844) is well documented in Debnath (1994) and will not be addressed in this book.

¹ The solitary wave represents not a periodic wave, but the propagation of a single isolated symmetrical hump of unchanged form.

Analytical solutions are limited to regular geometric tank shapes such as cylindrical, and rectangular. The nature of sloshing dynamics in cylindrical tanks is better understood than for prismatic tanks. However, analytical techniques for predicting large-amplitude sloshing are still not fully developed. Such loads are extremely important in the design stage of supporting structures and internal components of vehicle tanks. In addition, much of the sloshing technology developed for space applications is not applicable to road tankers. The reason is that more emphasis has been placed on frequencies and total forces as they relate to control system requirements. Accordingly, the effects of local peak impact pressure on structural requirements have not been studied to any extent. Further, the excitation amplitudes considered in space applications are too small for road vehicle simulation. To avoid catastrophic sloshing in space vehicles, the control system frequencies, the vehicle elastic structure frequencies, and the fluid-slosh frequencies must be fairly widely separated.

Sloshing phenomena in moving rectangular tanks can usually be described by considering only two-dimensional fluid flow if the tank width is much smaller than its breadth. Sloshing in spherical or cylindrical tanks, however, is usually described by three-dimensional flow. Tanks with two-dimensional flow are divided into two classes: low and high liquid fill depths. The low fill depth case is characterized by the formation of hydraulic jumps and traveling waves for excitation periods around resonance. At higher fill depths, large standing waves are usually formed in the resonance frequency range. When hydraulic jumps or traveling waves are present, extremely high impact pressures can occur on the tank walls. Impact pressures are only measured experimentally and cannot be estimated theoretically or numerically.

Early attempts to analyze liquid waves in oscillating containers include those of Hough (1895), Honda and Matsushita (1913), Jeffries (1924), Sen (1927), Goldsborough (1930), Westergaard (1933), Binnie (1941, 1955), K. W. Smith (1947, 1956), C. Smith (1948), Taylor (1950, 1954), Luskin and Lapin (1952), Schy (1952), Moiseev (1952a,b,c,d, 1953, 1954, 1956), Senda and Nakagawa (1954), Nakagawa (1955, 1956), Birkhoff (1956), Narimanov (1956, 1957c), Okhotsimski (1956), Heinrich and Kaufman (1956), Sretanskii (1956, 1957), Housner (1957), Krein and Moiseev (1957), and Shved (1959). The fundamental theory of liquid surface waves is documented in several references (see, e.g., Lamb, 1945, Stoker, 1957, Kochin, *et al.*, 1964, Thomson, 1965, Brodkey, 1967 and Barber and Ghey, 1969). Different aspects of the subject were reviewed by Sloane (1960), Cooper (1960), Abramson (1961a, 1963b, 1966b, 1968), Ring (1964), Abramson and Kana (1967), Brown (1982b), Rammerstofer, *et al.* (1990), and Ibrahim, *et al.* (2001). Some research monographs entirely devoted to different space problems of liquid sloshing include Abramson (1966a), Moiseev (1968), Moiseev and Rumyantsev (1968), Babskii, *et al.* (1976a), Narimanov, *et al.* (1977), Myshkis, *et al.* (1987, 1992), Walter (1987), and Monti (2001).

The problem of liquid sloshing in moving or stationary containers remains of great concern to aerospace, civil, and nuclear engineers, physicists, designers of road tankers and ship tankers, and mathematicians. Civil engineers and seismologists have been studying liquid sloshing effects on large dams, oil tanks and elevated water towers under ground motion. They also mounted liquid tanks on the roofs of multistory buildings as a means to control building oscillations due to earthquakes. Since the early 1960s, the problem of liquid sloshing dynamics has been of major concern to aerospace engineers studying the influence of liquid-propellant sloshing on the flight performance of jet vehicles, and new areas of research activities have emerged. The modern theory of nonlinear dynamics has indeed promoted

further studies and uncovered complex nonlinear phenomena. These include rotary sloshing, Faraday waves, nonlinear liquid sloshing interaction with elastic structures, internal resonance effects, stochastic sloshing dynamics, hydrodynamic sloshing impact dynamics, g-jitter under microgravity field, dynamics of liquid bridges, cross-waves, and spatial resonance. The dynamic stability of liquefied natural gas tankers and ship cargo tankers, and liquid hydrodynamic impact loading are problems of current interest to the designers of such systems. In populated cities, gasoline and other flammable liquid tankers are prone to rollover accidents while entering and exiting highways. This is a difficult mathematical problem to solve analytically, since the dynamic boundary condition at the free surface is nonlinear and the position of the free surface varies with time in a manner not known a priori. A liquid free surface in partially filled containers can experience a wide spectrum of motions such as planar, non-planar, rotational, quasi-periodic, chaotic, and disintegration. Other important contributions include the development of digital computer codes to solve complex problems that were difficult to handle in the past.

The purpose of this book is to present the basic theories and results developed in different applications in a coherent format. It highlights the major achievements and results reported in the literature. It begins with the fundamentals of liquid sloshing theory to take the reader systematically from the basic theory to advanced analytical and experimental results. Each chapter includes an extensive literature review for its major sections, followed by analytical description and main results. The book is supported by an extensive bibliography of technical journal papers, NASA reports, research monographs, M.S. and Ph.D. theses, and conference proceedings.

The book can be divided into four major parts. Part I deals with the theory of linear liquid sloshing dynamics (Chapters 1 to 3). Part II addresses the nonlinear theory of liquid sloshing dynamics (Chapters 4, 6, and 7). Note that Chapter 5 is in Part II but links linear and nonlinear equivalent mechanical models, and thus it also belongs to Part I. Part III presents the problem of linear and nonlinear interaction of liquid sloshing dynamics with elastic containers and supported structures (Chapters 8 to 10). Part IV considers the fluid dynamics in spinning containers and microgravity sloshing (Chapters 11 and 12). The book is closed by an extensive list of references that exceeds 2600.

The fluid field equations are developed in the first chapter with the purpose of determining the natural frequencies and the corresponding mode shapes in different container geometries. The analytical results are compared with those measured experimentally. Within the framework of the linear theory, the hydrodynamic loads and moments generated under harmonic excitations are evaluated in Chapter 2. A unique feature in this chapter is the study of forced excitation of magnetic fluids. Analytical treatments of some nontraditional tank geometries that were regarded as difficult in the past are presented.

Chapter 3 presents the influence of viscous damping and sloshing suppression devices in closed containers. The damping and modal analysis in a circular cylindrical container is analytically treated. The influence of sloshing suppression devices is addressed based on reported experimental results. The basic concept of Stokes boundary layers over an oscillating flat plate is introduced as a foundation to the estimation of inertia and damping parameters. It introduces the contribution of periodic boundary layers of fluids, in upright circular containers experiencing roll oscillations, to the system inertia and damping parameters. The analysis is extended to determine the contribution of immersed rods and pipes.

Chapter 4 deals with weakly nonlinear sloshing dynamics under lateral excitation. The associated phenomena include nonplanar unstable motion of the free surface accompanied by rotation of the nodal diameter (rotary sloshing) and chaotic sloshing. These phenomena can be uncovered using the theory of weakly nonlinear oscillations for quantitative analysis and the modern theory of nonlinear dynamics for stability analysis. In circular containers, the problem is strongly analogous to the dynamics of a spherical pendulum whose support point experiences lateral sinusoidal excitation. Some nonlinear theorems have been developed to predict not only rotary sloshing, but also other complex motions such as period doubling and chaos. The main sources of nonlinearity in the fluid field equations are the free-surface boundary conditions. The study is devoted to circular cylinders and rectangular containers. There is a critical fluid depth, which separates two regimes of liquid nonlinearity, hard and soft. A unique feature in this chapter is the random excitation of rigid containers, which has a direct application to liquid containers subjected to earthquakes. The chapter includes the problem of self-induced sloshing usually encountered in nuclear reactors. Self-induced sloshing implies the interaction between the liquid free-surface dynamics and submerged fluid jet dynamics.

Chapter 5 introduces the concept of equivalent mechanical models. A realistic representation of the liquid dynamics inside closed containers can be approximated by an equivalent mechanical system. The equivalence is taken in the sense of equal resulting forces and moments acting on the tank wall. By properly accounting for the equivalent mechanical system, the problem of overall dynamic system behavior can be formulated more simply. For small oscillations in which the fluid free surface remains planar without rotation of its nodal diameter the equivalent mechanical model can be represented by a set of pendulums or mass-spring-dashpot systems. For relatively large-amplitude oscillations, in which the free liquid surface experiences nonplanar motion, the equivalent mechanical model is a simple pendulum describing relatively large motion. To model nonplanar and rotational sloshing one has to replace the simple pendulum by a compound or spherical pendulum. The theory of the spherical pendulum will be presented for promoting our understanding of rotary sloshing. For strongly nonlinear motion associated with hydrodynamic pressure impacts, the modeling can be achieved by an equivalent pendulum of high restoring force. This modeling will be considered in details in Chapter 7.

Chapter 6 deals with the problem of parametric sloshing or Faraday's waves. Parametric sloshing refers to the motion of the liquid free surface due to an excitation applied perpendicular to the plane of the undisturbed free surface. Generally, parametric oscillation occurs in dynamical systems as a result of time-dependent variation of such parameters as inertia, damping, or stiffness. The early research efforts focused on the estimation of instability boundaries of liquid in the presence of viscous and surface tension forces or in their absence. While the damping forces raise the threshold excitation amplitude, they do not bring the free-surface amplitude into a bounded value. It is the inherent nonlinearity of the surface-wave amplitude that can limit the growth of the amplitude. Both nonlinear theory and experimental tests revealed "soft" or "hard" spring characteristics depending on the liquid depth ratio. With the advent of the modern theory of nonlinear dynamics, many complex surface dynamic phenomena have been uncovered. These include modal competition, quasi-periodic motion, chaotic motion, and mixed mode motion in the presence of internal resonance. These phenomena are presented for circular and rectangular tanks and the corresponding motion was identified in bifurcation diagrams under slow variations of certain control parameters such

as excitation amplitude and frequency. Most of these phenomena may take place under random parametric excitation in terms of excitation bandwidth and spectral density function.

Chapter 7 presents the basic concepts of the theory of hydrodynamic slamming and impacting. The material of this chapter has not been addressed in other research monographs on liquid sloshing dynamics. It begins with the analogy of shock waves that take place in a gas column. Analytical expressions of hydrodynamic impact forces and moments acting on the tank walls, due to sinusoidal pitching excitation, are given in closed approximate forms and compared with experimental measurements. Under lateral excitation, the modal interaction is presented. Hydrodynamic pressure impacts due to sudden acceleration of the tank are displayed according to experimental measurements. A phenomenological modeling of impact forces and damping forces is proposed and used in the analytical modeling of elastic structures carrying liquid containers. The impact forces are replaced by a high-power force, which covers both elastic and rigid impact cases. Three nontraditional analytical techniques, namely sawtooth time transformation, nonsmooth coordinate transformation, and Lie group transformations are briefly outlined. An application of a mechanical system simulating an elastic structure carrying a liquid container that allows high-power impact forces will be adopted. The response will be obtained using the method of multiple scales and the results will be compared with those obtained by nontraditional techniques. The analysis involves the simultaneous occurrence of parametric resonance and internal resonance conditions. Sloshing impact in ship tankers and road tankers are addressed together with a brief account of numerical algorithms used for large surface motion.

Chapter 8 presents the linear problem of liquid interaction with its elastic container. The first step in studying this interaction is to estimate the linear eigenvalue problem and the linear response to external excitations. The coupling may take place between the liquid free-surface dynamics and either the tank bending oscillations or the breathing modes (or shell modes). Breathing vibrations of the tank are essentially radial, such that both flexure and stretching deformations of the wall occur while the longitudinal axis of the tank remains straight. The mathematical formulation of interacting forces is formulated for a general motion of the liquid containers. The linear analysis deals mainly with the estimation of the coupled natural frequencies of liquid–structure systems. Specific cases are addressed. These include: (i) rigid tank walls with an elastic bottom modeled as either a stretched membrane or elastic plate, (ii) a rigid bottom with elastic walls experiencing either bending deformation or breathing vibrations, and (iii) interaction with a completely elastic container. The nonlinear interaction under external and parametric excitations is treated in Chapter 9. The nonlinear interaction covers three different groups: (1) nonlinear free-surface interaction with linear shell deformation, (2) linear free-surface interaction with nonlinear shell deformation, and (3) nonlinear surface interaction with nonlinear shell deformation. Storage tanks interacting with their liquid and the numerical algorithms developed to solve for the liquid–tank interaction are reviewed. Chapter 10 deals with the interaction of elastic support structure and tuned liquid sloshing absorbers. This problem was not addressed in other books. This type of interaction takes place between the free-liquid-surface motion and the supported elastic structure dynamics based on the assumption that the liquid container is rigid. Under the base motion of the supporting structure, the fluid container experiences motion in a certain trajectory governed by the excitation and the liquid response. The free-liquid-surface motion will result in hydrodynamic forces that are fed back to the supporting structure. Civil engineers took advantage of this by

using liquid tanks placed on the top of large buildings to act as vibration absorbers. A detailed treatment of sloshing absorbers such as tuned liquid-sloshing absorbers, and liquid-column vibration absorbers is presented. Nonlinear interaction of liquid free surface with support structure is then treated in the presence of parametric excitation, internal resonance, and combination resonance, under deterministic and random excitations.

Chapter 11 presents the problem of liquid dynamics in spinning tanks. This problem is usually encountered in the study of stability and control of rockets, space vehicles, liquid-cooled gas turbines, centrifuges, earth with its oceans and fluid core. The spinning and unavoidable wobbling and precession of spacecraft vehicles cause the contained liquid to oscillate, and thus generate dynamic forces, which can destabilize the spacecraft. Note that the kinetic energy of the spinning spacecraft is dissipated by the liquid motion and thus destabilizes the vehicle and causes it to tumble. Spacecraft vehicles are designed to spin in order to gain gyroscopic stiffness during the transfer from low earth orbit. This spin helps to control the location of liquid propellant in its container. Another class of problems deals with the dynamics and stability of rotating rigid bodies as they are applied to the evolution of celestial bodies and astronavigation control. Stabilization is achieved when the spacecraft spins about its axis of minimum moment of inertia. A satellite that spins about its axis of minimum moment of inertia may experience instability if energy is dissipated. This is similar to a spinning top on a rough surface and as a result of friction the top's nutation angle increases as it seeks to conserve angular momentum. Other topics addressed in this chapter include the dynamics of fluid-filled spinning containers, inviscid fluid in a partially filled upright cylinder, inertia waves in rotating fluid, and rotating liquids in microgravity.

The last chapter deals with the physics and dynamics of liquid sloshing in microgravity. In view of the diversity of microgravity fluid physics, this chapter will address selected topics supported by literature reviews of the main problems. It begins with the mechanics of free liquid surface under microgravity, stability of static and dynamic free liquid surface, and contact line and contact angle. The modal analysis includes the estimation of free-surface natural frequencies and associated surface shapes under micro- and zero-gravity environments. The problem of forcing sloshing for slipping and anchored contact lines will be addressed for simple cases. The influence of g-jitter and liquid handling are discussed briefly. Capillary systems including Marangoni flow and liquid bridges will be treated for static and dynamic cases. The chapter also includes the problem of the thermocapillary effects of fluid flow and liquid bridges, and the sloshing of cryogenics, and briefly discusses hydroelastic oscillations under microgravity.

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Part I

Linear sloshing dynamics

Fluid field equations and modal analysis in rigid containers

1.1 Introduction

The theory of liquid sloshing dynamics in partially filled containers is based on developing the fluid field equations, estimating the fluid free-surface motion, and the resulting hydrodynamic forces and moments. Explicit solutions are possible only for a few special cases such as upright cylindrical and rectangular containers. The boundary value problem is usually solved for modal analysis and for the dynamic response characteristics to external excitations. The modal analysis of a liquid free-surface motion in a partially filled container estimates the natural frequencies and the corresponding mode shapes. The knowledge of the natural frequencies is essential in the design process of liquid tanks and in implementing active control systems in space vehicles. The natural frequencies of the free liquid surface appear in the combined boundary condition (kinematic and dynamic) rather than in the fluid continuity (Laplace's) equation.

For an open surface, which does not completely enclose the field, the boundary conditions usually specify the value of the field at every point on the boundary surface or the normal gradient to the container surface, or both. The boundary conditions may be classified into three classes (Morse and Feshbach, 1953):

- (1) the *Dirichlet boundary conditions*, which fix the value of the field on the surface;
- (2) the *Neumann boundary conditions*, which fix the value of the normal gradient on the surface; and
- (3) the *Cauchy conditions*, which fix both value of the field and normal gradient on the surface.

Each class is appropriate for different types of equations and different boundary surfaces. For example, Dirichlet conditions on a closed surface uniquely specify a solution of Laplace's equation inside the closed surface.

The variational formulation based on Hamilton's principle is regarded as the most powerful tool for developing the fluid field equations. This approach has been proposed and used by Lawrence, *et al.* (1958), Troesch (1960), Bogoryad (1962), Borisova (1962), Petrov (1962a,b,c), Moiseev (1964), Moiseev and Petrov (1966), Luke (1967), Whitham (1967), Lukovskii (1967, 1976), Moiseev and Rumyantsev (1968), Limarchenko (1978a, 1980, 1983b), Lukovskii and Timokha (1992, 1995), and Rocca, *et al.* (1997). The method of integral equations was adopted for containers whose wetted walls are not straight vertical but curved such as spherical containers and horizontal cylindrical containers (see, e.g., Budiansky, 1960 and McIver, 1989). Some analytical and approximate approaches to estimating the sloshing frequencies were developed by Housner (1963a), Evans (1990), and Evans and Linton (1993).

Henrici, *et al.* (1970) presented an extensive treatment of liquid sloshing in a half-space bounded above by a rigid plane that contains either a circular or infinite-strip aperture. They

obtained both upper and lower bounds of the natural frequencies. Troesch and Troesch (1972) and Miles (1972) discussed some features of the spectrum of the eigenvalues of liquid sloshing in a half-space with an emphasis on their upper bounds. Banning, *et al.* (1966) built an apparatus for demonstrating the dynamics of liquid sloshing.

The dynamic behavior of liquid propellant free surface was addressed by Ehrlich (1959), Abramson (1961b, 1965), Eulitz and Glaser (1961), Eulitz (1963), Bonneau (1964), Buchanan and Bugg (1966), Fontenot (1968), Martin (1971), and Dodge and Garza (1971). The modal analysis in a circular cylindrical container was originally treated by Poisson (1828) but the results were not interpreted because the theory of Bessel's function was not sufficiently developed at that time. The equations of motion of a liquid in rigid rectangular and rigid circular tanks of uniform depth and with linearized boundary conditions were also given by Rayleigh (1887), Steklov (1902), and Lamb (1945). The solution of the Laplace equation using the method of separation of variables is somewhat less powerful for cases where the liquid depth is variable and other methods, such as the Ritz method, should be used. Bratu (1971) studied the oscillations of liquid masses in reservoirs.

The free-surface mode shapes for containers with axial symmetry were determined by Borisova (1962), Bonneau (1964), Moiseev and Petrov (1965, 1966), Pfeiffer (1967a,b), Einfeldt, *et al.* (1969), McNeil and Lamb (1970), Henrici, *et al.* (1970), Pshenichnov (1972), and Boyarshina and Koval'chuk (1986) determined the normal modes and natural frequencies of the free surface in an inclined cylinder. Trotsenko (1967) studied the liquid oscillations in a cylindrical tank with annular baffle. For a spherical tank, the problem is analytically more complex and approximate solutions for the natural frequencies were obtained by Bauer (1958a), Budiansky (1960), Leonard and Walton (1961), Riley and Trembath (1961), Lukovskii (1961a,b), Chu (1964a), Boudet (1968), McIver (1989), El-Rahib and Wagner (1981), and Bauer and Eidel (1989b). The natural frequency of horizontal circular canals and spherical containers is determined from an integral equation, which is usually discretized into a matrix form for numerical calculations (Barnyak, 1997).

Bauer (1964b) and Mooney, *et al.* (1964a,b) analyzed the free-surface oscillations in a quarter tank and in a tank with annular sector cross-section. In both cases, the natural frequency of the free surface was found to have the same expression as the circular cylindrical tank but with different roots of the Bessel function. The liquid sloshing frequencies for different container geometries were evaluated by Miles (1964, 1972), Kuttler and Sigillito (1969), Fox and Kuttler (1981, 1983), Meserole and Fortini (1987), and McIver and McIver (1993). The influence of movable devices and internal pipes on the natural frequencies of the free surface was determined by Siekmann and Chang (1971b) and Drake (1999). Bauer and Eidel (1999c) considered different configurations of cylindrical containers.

Based on the two-dimensional analysis of liquid motion in rectangular tanks, the natural frequency depends essentially on the liquid depth to width ratio. The effect of liquid depth is diminished as the mode order increases. Graham and Rodriguez (1952) solved the three-dimensional velocity potential for which the natural frequency depends on the three major dimensions of the fluid. Ghali (1965) determined the nonlinear dependence of the natural frequencies on the wave motion amplitude. The influence of damping on the natural frequency was studied experimentally by Ghali (1965), Scarsi and Brizzolara (1970), Scarsi (1971), and Schilling and Siekmann (1980). It was found that for higher viscosities (of kinematic viscosity $\nu = 2.5$ poise) the resonance frequency is slightly higher than the predicted value for an ideal liquid.

The purpose of this chapter is to develop the fluid field equations with reference to an inertial frame and with respect to a moving coordinate system. The variational approach is demonstrated as a tool to derive the boundary-value problem in one treatment. The modal analysis of liquid free-surface motion is formulated for different tank shapes. The analysis includes the estimation of the velocity potential function, fluid free-surface natural frequencies and mode shapes. The influence of surface tension is included in some cases. However, its effect is dominant in a microgravity field as will be demonstrated in Chapter 12. Note that analytical solutions in a closed form are only obtained for regular tank shapes whose walls are upright straight. For other tank geometries with variable depth, one can determine the natural frequencies and mode shapes either experimentally or numerically.

1.2 Fluid field equations

The analytical description of the fluid field equations is documented for different cases of tank geometries by Ewart (1956), Bauer (1962a, 1966c, 1969a), Lomen (1965a), Abramson (1966a), Ibrahim (1969), Khandelwal (1980), Kornecki (1983), and Bauer (1999). The general equations of motion for a fluid in closed containers can be simplified by assuming the container rigid and impermeable. Furthermore, the fluid is assumed inviscid, incompressible, and initially irrotational. Capillary or surface tension effects will be ignored in a gravitational field. However, the effect of surface tension will be introduced for some simple cases. The free-surface oscillations can be generated by giving an initial impulse, or an initial disturbance to the free surface.

This section considers the general case of a tank moving along some trajectory in space. The formulation is applicable to free and forcing liquid free-surface oscillations. It is convenient to refer the fluid motion to a moving coordinate system as the variables are measured relative to the moving frame. It is also useful to write the fluid equations of motion with reference to stationary and moving coordinates as shown in Figure 1.1. In the present analysis, the tank is allowed to move in planar curvilinear motion without rotation. Let $O'X'Y'Z'$ be the stationary Cartesian coordinate frame. The Euler equations of motion of the fluid are written in the vector form

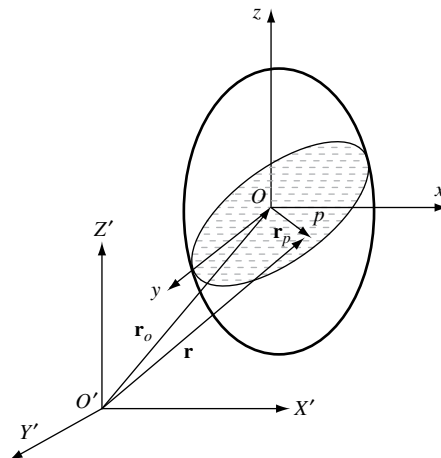


Figure 1.1 Moving liquid container showing inertia and moving coordinates.

$$\frac{\partial}{\partial t} \mathbf{q} + (\mathbf{q} \cdot \nabla) \mathbf{q} = -\frac{1}{\rho} \nabla P - \nabla(gZ') \quad (1.1)$$

where $\mathbf{q}$ is the fluid velocity, $\partial \mathbf{q} / \partial t$ is the local acceleration of the flow at the point whose coordinates are not allowed to vary (this acceleration is measured by a fixed observer), $(\mathbf{q} \cdot \nabla) \mathbf{q}$ is the convective acceleration for a fluid particle drifting with the stream at a velocity $\mathbf{q}$ in the flow direction (this acceleration is measured by an observer moving with the particle p), P is the fluid pressure, ρ is the fluid density, gZ' is the gravitational potential, and ∇ is an operator given for different coordinate frames in the appendix to this chapter. Note that the convective acceleration $(\mathbf{q} \cdot \nabla) \mathbf{q}$ may also be written in the form (Thomson, 1965, p. 44)

$$(\mathbf{q} \cdot \nabla) \mathbf{q} = \frac{1}{2} \nabla q^2 - \mathbf{q} \times (\nabla \times \mathbf{q}) = \frac{1}{2} \nabla q^2 \quad (1.2)$$

For irrotational flow the curl of the velocity vanishes, that is, $\nabla \times \mathbf{q} = 0$.

For irrotational fluid motion, there exists a velocity potential function, Φ , whose (negative) gradient gives the fluid velocity,

$$\mathbf{q} = -\nabla \Phi \quad (1.3)$$

The negative sign is optional and in some cases it may be removed provided the analysis preserves the sign convention. Introducing relations (1.2) and (1.3) into equation (1.1) gives

$$\nabla \left(\frac{P}{\rho} + \frac{1}{2} q^2 + gZ' - \frac{\partial \Phi}{\partial t} \right) = 0 \quad (1.4)$$

Upon integrating equation (1.4) one obtains

$$\frac{P}{\rho} + \frac{1}{2} q^2 + gZ' - \frac{\partial \Phi}{\partial t} = C(t) \quad (1.5)$$

where $C(t)$ is an arbitrary function of time.

Equation (1.5) is the general form of Kelvin's equation for an unsteady fluid flow. In this equation the potential function Φ is a function of space and time, and its derivative with respect to time measures the unsteadiness of the flow. However, $\partial \Phi / \partial t$ is interpreted as the work done on a unit mass of the fluid whose coordinates are (X, Y, Z) . Furthermore, equation (1.5) is only valid for incompressible flow for which the continuity condition $\nabla \cdot \mathbf{q} = 0$ yields Laplace's equation, which after introducing equation (1.2) takes the form

$$\nabla^2 \Phi = 0 \quad (1.6)$$

Let $Oxyz$ be another coordinate frame fixed to the tank such that the Oxy plane coincides with the undisturbed free surface. Let $\mathbf{V}_0$ be the velocity of the origin O relative to the fixed origin O' . In this case, the time rate of change of the velocity potential Φ at a point fixed in the stationary frame $O'X'Y'Z'$ as measured by an observer in the moving frame $Oxyz$ is $(\partial / \partial t - \mathbf{V}_0 \cdot \nabla) \Phi$, since this point will appear to have a velocity $-\mathbf{V}_0$ with respect to the observer. Accordingly, the pressure equation (1.5) takes the form

$$\frac{P}{\rho} + \frac{1}{2} q^2 + gZ' - \frac{\partial \Phi}{\partial t} + \mathbf{V}_0 \cdot \nabla \Phi = C(t) \quad (1.7a)$$

The fluid particle velocity $\mathbf{q}_{\text{rel}}$ relative to the moving coordinate is

$$\mathbf{q}_{\text{rel}} = \mathbf{q} - \mathbf{V}_0 = -\nabla\Phi - \mathbf{V}_0 \quad (1.8)$$

Expressing $\mathbf{q}$ in terms of $\mathbf{q}_{\text{rel}}$ and $\mathbf{V}_0$, using relation (1.8), gives

$$\frac{P}{\rho} + \frac{1}{2}q_{\text{rel}}^2 + gZ' - \frac{\partial\Phi}{\partial t} - \frac{1}{2}V_0^2 = C(t) \quad (1.7b)$$

Equation (1.7a) is written in terms of the total fluid velocity as measured by the fixed coordinate and equation (1.7b) is given in terms of the fluid relative velocity to the tank moving coordinates. At the free surface, the pressure is equivalent to the ambient pressure or can be set to zero in equation (1.7a). This gives the dynamic boundary condition

$$\frac{1}{2}(\nabla\Phi \cdot \nabla\Phi) + g\eta - \frac{\partial\Phi}{\partial t} + \mathbf{V}_0 \cdot \nabla\Phi = 0 \quad (1.9)$$

where the function $C(t)$ has been absorbed in the potential function, Φ .

The vertical velocity of a fluid particle located on the free surface $z = \eta(r, \theta, t) = \eta(x, y, t)$ should be equated to the vertical velocity of the free surface itself. This condition is known as the kinematic free-surface condition and is given by the following expression

$$-\frac{\partial\Phi}{\partial z} = \frac{\partial\eta}{\partial t} + \mathbf{q}_{\text{rel}} \cdot \nabla\eta \quad (1.10)$$

At the wetted rigid wall and bottom, the velocity component normal to the boundary must have the same value of the corresponding velocity component of the solid boundary at the point in question. For example, if the tank is allowed to move in the vertical plane then the velocity vector in terms of Cartesian and cylindrical coordinates may be written in the form, respectively

$$\mathbf{V}_0 = \dot{X}_0 \mathbf{i} + \dot{Z}_0 \mathbf{k} \quad (1.11a)$$

$$\mathbf{V}_0 = (\dot{X}_0 \cos \theta) \mathbf{i}_r - (\dot{X}_0 \sin \theta) \mathbf{i}_\theta + \dot{Z}_0 \mathbf{i}_z \quad (1.11b)$$

The boundary conditions at the wall and bottom for Cartesian and cylindrical coordinates are, respectively,

$$-\frac{\partial\Phi}{\partial z} \Big|_{z=-h} = \dot{Z}_0, \quad -\frac{\partial\Phi}{\partial x} \Big|_{x=a} = \dot{X}_0 \quad (1.12a)$$

$$-\frac{\partial\Phi}{\partial z} \Big|_{z=-h} = \dot{Z}_0, \quad -\frac{\partial\Phi}{\partial r} \Big|_{r=R} = \dot{X}_0 \cos \theta \quad (1.12b)$$

It is possible to split the total velocity potential function, Φ , into a disturbance potential function, $\tilde{\Phi}$, and a potential function, Φ_o , which defines the motion of the tank, that is,

$$\Phi = \tilde{\Phi} + \Phi_o \quad (1.13)$$

The function Φ_o can be determined by integrating equation (1.11) as

$$\Phi_o = -\dot{X}_0 r \cos \theta - \dot{Z}_0 z - \frac{1}{2} \int (\dot{X}_0^2 + \dot{Z}_0^2) dt \quad (1.14)$$

Introducing (1.13) and (1.14) into the free-surface boundary conditions gives

$$\frac{1}{2} (\nabla \tilde{\Phi} \cdot \nabla \tilde{\Phi}) + (g + \ddot{Z}_0) \eta - \frac{\partial \tilde{\Phi}}{\partial t} + \ddot{X}_0 r \cos \theta = 0 \quad (1.15)$$

$$-\frac{\partial \tilde{\Phi}}{\partial z} = \frac{\partial \eta}{\partial t} - \frac{\partial \eta}{\partial r} \frac{\partial \tilde{\Phi}}{\partial r} - \frac{1}{r^2} \frac{\partial \eta}{\partial \theta} \frac{\partial \tilde{\Phi}}{\partial \theta} \quad (1.16)$$

The corresponding conditions for a rectangular tank are

$$\frac{1}{2} (\nabla \tilde{\Phi} \cdot \nabla \tilde{\Phi}) + (g + \ddot{Z}_0) \eta - \frac{\partial \tilde{\Phi}}{\partial t} + \ddot{X}_0 x = 0 \quad (1.17)$$

$$-\frac{\partial \tilde{\Phi}}{\partial z} = \frac{\partial \eta}{\partial t} - \frac{\partial \eta}{\partial x} \frac{\partial \tilde{\Phi}}{\partial x} - \frac{\partial \eta}{\partial y} \frac{\partial \tilde{\Phi}}{\partial y} \quad (1.18)$$

One may introduce the effect of surface tension, σ , by including the pressure change across the displaced free liquid surface as described by the Laplace–Young equation

$$p_s = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (1.19)$$

where R_1 and R_2 are the principal radii of curvature. The complete formulation of the boundary value problem in terms of the disturbance potential function is summarized as follows:

(1) *for a cylindrical container:*

$$\nabla^2 \tilde{\Phi} = 0 \quad (1.20a)$$

$$\left. \frac{\partial \tilde{\Phi}}{\partial r} \right|_{r=R} = 0, \quad \left. \frac{\partial \tilde{\Phi}}{\partial z} \right|_{z=-h} = 0 \quad (1.20b, c)$$

$$\frac{1}{2} (\nabla \tilde{\Phi} \cdot \nabla \tilde{\Phi}) + (g + \ddot{Z}_0) \eta - \frac{\partial \tilde{\Phi}}{\partial t} + \frac{\sigma}{\rho} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \ddot{X}_0 r \cos \theta = 0, \quad \text{at } z = \eta(r, \theta, t) \quad (1.20d)$$

$$-\frac{\partial \tilde{\Phi}}{\partial z} = \frac{\partial \eta}{\partial t} - \frac{\partial \eta}{\partial r} \frac{\partial \tilde{\Phi}}{\partial r} - \frac{1}{r^2} \frac{\partial \eta}{\partial \theta} \frac{\partial \tilde{\Phi}}{\partial \theta}, \quad \text{at } z = \eta(r, \theta, t) \quad (1.20e)$$

The curvature κ for cylindrical coordinates is given by the expression

$$\begin{aligned} \kappa &= - \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \\ &= - \frac{\eta_{rr} (1 + (\eta_\theta^2/r^2)) + (1 + \eta_r^2) ((\eta_r/r) + (\eta_{\theta\theta}/r^2)) - 2 \eta_r (\eta_\theta/r^2) (\eta_{rr} + (\eta_\theta/r))}{[1 + \eta_r^2 + (\eta_\theta^2/r^2)]^{3/2}} \end{aligned} \quad (1.21)$$

This expression can be linearized in the form

$$\kappa = - \left[\eta_{rr} + \frac{\eta_r}{r} + \frac{\eta_{\theta\theta}}{r^2} \right] \quad (1.22)$$

(2) for a rectangular container:

$$\nabla^2 \tilde{\Phi} = 0 \quad (1.23a)$$

$$\left. \frac{\partial \tilde{\Phi}}{\partial x} \right|_{x=\pm a/2} = 0, \quad \left. \frac{\partial \tilde{\Phi}}{\partial y} \right|_{y=\pm b/2} = 0, \quad \left. \frac{\partial \tilde{\Phi}}{\partial z} \right|_{z=-h} = 0 \quad (1.23b, c, d)$$

$$\frac{1}{2} (\nabla \tilde{\Phi} \cdot \nabla \tilde{\Phi}) + (g + \ddot{Z}_0) \eta - \frac{\partial \tilde{\Phi}}{\partial t} + \frac{\sigma}{\rho} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \ddot{X}_0 x = 0, \quad \text{at } z = \eta(x, y, t) \quad (1.23e)$$

$$-\frac{\partial \tilde{\Phi}}{\partial z} = \frac{\partial \eta}{\partial t} - \frac{\partial \eta}{\partial x} \frac{\partial \tilde{\Phi}}{\partial x} - \frac{\partial \eta}{\partial y} \frac{\partial \tilde{\Phi}}{\partial y}, \quad \text{at } z = \eta(x, y, t) \quad (1.23f)$$

The curvature, κ , for cylindrical coordinates is given by the expression

$$\kappa = - \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = - \frac{\eta_{xx} (1 + \eta_y^2) + \eta_{yy} (1 + \eta_x^2) - 2\eta_x \eta_y \eta_{xy}}{\left[1 + \eta_x^2 + \eta_y^2 \right]^{3/2}}. \quad (1.24)$$

This expression can be linearized in the form

$$\kappa = - [\eta_{xx} + \eta_{yy}] \quad (1.25)$$

For other container geometries, such as spherical, prolate and oblate spherical, and elliptic containers, the continuity equation and other related operators are listed in the appendix to this chapters. Note that the velocity potential function, $\tilde{\Phi}$, must satisfy Laplace's equation, $\nabla^2 \tilde{\Phi} = 0$, which is a linear partial differential equation. The nonlinearity in the boundary value problem only exists in the free-surface boundary conditions on $z = \eta$. If one is interested in the modal analysis then one should drop the nonlinear and nonconservative terms from the free-surface boundary conditions. If the potential function is obtained analytically in a closed form, then the natural frequencies of the fluid free surface are obtained by using the dynamic free-surface condition based on the fact that $\tilde{\Phi}$ is harmonic in time. Another powerful approach is to use the variational formulation together with the Rayleigh–Ritz method. The next section describes an alternative approach based on the variational principle.

1.3 Variational formulation

The variational approach is based on establishing the superlative of a certain function that describes the system behavior. The Lagrangian, $L = T - V$, has to be minimized (or maximized), where T and V are the kinetic and potential energies of the system, respectively. The variational principle, or Hamilton's principle, is

$$\delta I = \delta \int_{t_1}^{t_2} (T - V) dt = 0 \quad (1.26)$$

Basically Hamilton's principle states that the actual path in the configuration space yields the value of the definite integral stationary with respect to all arbitrary variations of the path between two instants of time t_1 and t_2 provided the path variations vanish at these two end points. For any actual motion of the system, the system will move so that the time average of the difference between the kinetic and potential energies will be a minimum. This formulation is very powerful since it brings in one statement the fluid field equations and the associated boundary conditions. Substituting for the kinetic energy, $T = \int_v (\rho/2) |\nabla\Phi|^2 dv$, and potential energy, $V = \int_S (\eta/2)(\rho g \eta dS)$, where v is the fluid volume, and S is the fluid free surface, in equation (1.26), and taking the variation gives

$$\begin{aligned} \delta I &= \delta \int_{t_1}^{t_2} dt \left\{ \int_v \frac{\rho}{2} |\nabla\Phi|^2 dv - \int_S \frac{\eta}{2} (\rho g \eta dS) \right\} \\ &= \rho \int_{t_1}^{t_2} dt \left\{ \int_v \nabla\Phi \nabla\delta\Phi dv - g \int_S \eta \delta\eta dS \right\} = 0 \end{aligned} \quad (1.27)$$

The volume integral can be transformed into a surface integral using Green's formula, $\int_v \nabla\Phi \nabla\delta\Phi dv = \int_S \Phi \frac{\partial\delta\Phi}{\partial n} dS$. Furthermore, one may use the relationship

$$\nabla\Phi = \mathbf{n} \frac{\partial\Phi}{\partial n} = -\mathbf{n} \frac{\partial\eta}{\partial t}$$

(see, e.g., Thomson, 1965), where $\mathbf{n}$ is the unit vector along the normal at the point in question to the equi-potential surface of Φ . In this case, the variational takes the form

$$\rho \int_{t_1}^{t_2} dt \left\{ \int_S \left\{ \Phi \frac{\partial\delta\Phi}{\partial n} - g\eta\delta\eta \right\} dS \right\} = -\rho \int_{t_1}^{t_2} dt \left\{ \int_S \left\{ \Phi \delta \frac{\partial\eta}{\partial t} + g\eta\delta\eta \right\} dS \right\} = 0$$

Integrating by parts gives

$$\rho \int_{t_1}^{t_2} \int_S \left\{ -\frac{\partial\Phi}{\partial t} + g\eta \right\} \delta\eta dS dt = 0$$

This statement yields the linearized dynamic free surface condition

$$-\frac{\partial\Phi}{\partial t} + g\eta = 0 \quad (1.28)$$

Moiseev and Rumyantsev (1968) introduced the Neumann operator H , which makes the velocity potential, Φ , harmonic inside the fluid volume domain, v . The harmonic property is based on the fact that the integral of free-surface velocity, $\dot{\eta}(s)$, vanishes over the free surface, that is, $\int_S \dot{\eta}(s) ds = 0$. The function Φ satisfies the following conditions on the fluid boundary

$\frac{\partial\Phi}{\partial n} = 0$ at the container walls, and $\frac{\partial\Phi}{\partial n} = -\dot{\eta}$ at the free surface.

In this case, one can write the correspondence as $\Phi = H\dot{\eta}$, where H is the integral operator

$$H\dot{\eta} = \int_S H(s, v) \dot{\eta}(s) ds = \Phi(v) \quad (1.29)$$

Note that the kernel is the Green function of Neumann's problem for the region v . This kernel is known to be symmetric (see, e.g., Mikhlin, 1964, and Gunter, 1965). The kernel is logarithmic in the case of the plane problem, that is, when the volume is reduced to a plane surface. The kernel is a polar function in the case of the three-dimensional problem. Thus, H is a fully continuous selfadjoint operator, and one can write the following representation

$$\Phi(v) = H \frac{\partial \Phi}{\partial z} = -H \frac{\partial \eta}{\partial t} \quad (1.30)$$

Taking the time derivative for both sides and using equation (1.28), gives

$$g\eta + H \frac{\partial^2 \eta}{\partial t^2} = 0 \quad (1.31)$$

The average energy function can be written in terms of the Neumann operator in the form

$$I_1 = \frac{\rho}{2} \int_{t_1}^{t_2} \int_S \left\{ H \left(\frac{\partial \eta}{\partial t} \right)^2 - g\eta^2 \right\} ds dt \quad (1.32)$$

Alternatively, one can write the average energy function in terms of the scalar potential function in the form, after using equation (1.28)

$$I = \int_{t_1}^{t_2} dt \left\{ \int_v \frac{\rho}{2} |\nabla \Phi|^2 dv - \frac{1}{g} \int_S \Phi^2 ds \right\} \quad (1.33)$$

Equations (1.32) and (1.33) can be used for estimating the natural frequencies for the liquid free surface. At the free surface, S , both the velocity potential function and the free-surface wave height can be expressed in terms of time and space as follows

$$\Phi(s, t) = F(s) \cos \omega t, \quad \text{and} \quad \eta(s, t) = G(s) \sin \omega t \quad (1.34)$$

where ω is the free-surface natural frequency. Substituting equations (1.34) in the average energy functions given by equations (1.32) and (1.33) and integrating over time t from $t_1 = 0$ to $t_2 = 2\pi/\omega$, gives

$$I_1 = \lambda \int_S H G \cdot G ds - \int_S G^2 ds \quad (1.35)$$

$$I_2 = \int_v \frac{\rho}{2} |\nabla F|^2 dv - \lambda \int_S F^2 ds \quad (1.36)$$

where $\lambda = \omega^2/g$. The natural frequencies of the fluid free surface are determined by using the Rayleigh–Ritz method. The method is based on introducing a linear combination of a

complete set of functions whose coefficients form a set of linear variational parameters. For example, the function F may be expressed by the following summation of the trial functions f_n

$$F = \sum_{n=1}^N a_n f_n \quad (1.37)$$

Substituting equation (1.37) into equation (1.36), it follows from equation (1.27) that the coefficients a_n should satisfy a system of algebraic homogeneous equations for the coefficients a_n by setting $\partial I / \partial a_n = 0$, for $n = 1, 2, \dots, N$. This yields the following set of equations

$$\sum_{m=1}^N a_{mn} (A_{mn} - \lambda B_{mn}) = 0 \quad (1.38)$$

where $A_{mn} = \int_v \nabla f_n \cdot \nabla f_m dv$, $B_{mn} = \int_S f_n f_m ds$, $A_{nm} = A_{mn}$, and $B_{nm} = B_{mn}$.

A nontrivial solution of equations (1.38) exists only if the determinant of the coefficients of a_n vanishes

$$|A_{mn} - \lambda B_{mn}| = 0 \quad (1.39)$$

Equation (1.39) is the frequency equation. The first term can be determined by the variational method. The procedure involves the inclusion of the first few terms of the series (1.37) and solving the secular equation exactly, increasing the number of terms and solving again. The process continues until the values of the natural frequencies do not change. In view of the absence of any rigorous method of convergence and the difficulty of selecting the coordinate functions, the process may yield some errors. Morse and Feshbach (1953) introduced a number of perturbational techniques to determine the natural frequencies of dynamical systems using the Rayleigh–Ritz method. Moiseev (1970) and Moiseev and Rumyantsev (1968) indicated that the value of λ_1 is “only slightly sensitive” to the selection of the functions f_n . This means that if the function F_1 , which brings about a minimum of the functional λ_1 , is replaced by another function F_1^* (such that $\int \nabla F_1 \nabla F_1^* dv \neq 0$) then the value of λ_1 will not change significantly.

The system of coordinate functions $\{f_n\}$ can be selected quite roughly. It is only necessary to make sure that this system is complete. Hence, it is expedient to take these functions as the eigenfunctions of the problem of liquid sloshing in some volume that contains the specified volume but has a simple shape. This will be demonstrated for the case of liquid sloshing in conical containers later in this chapter.

1.4 Normal modes in an upright cylindrical container

Circular cylindrical tanks appear as bare containers or with partitioning walls. One can raise the natural frequencies of the fluid by partitioning the tank with longitudinal walls, or by introducing annular rings or a concentric tank. This section summarizes the main results of different tank geometries shown in Figures 1.2(a)–(d).

1.4.1 Bare wall cylindrical tank

With reference to Figure 1.2(a), a possible solution of the Laplace equation, which satisfies the wall and bottom boundary conditions listed in relations (1.20b, c), is

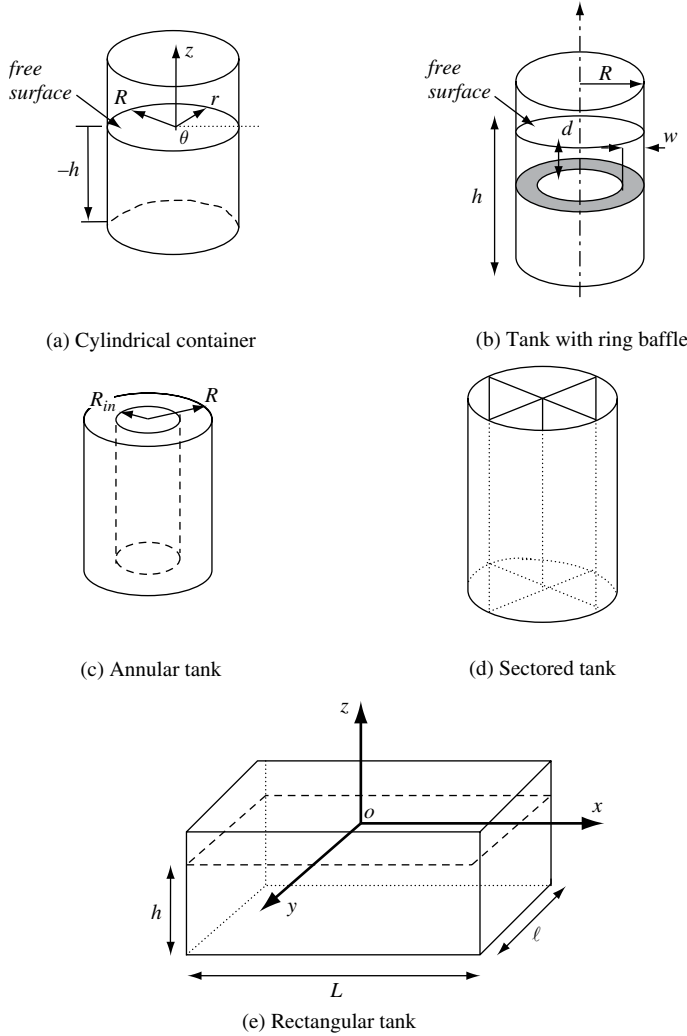


Figure 1.2 Different tank geometries.

$$\tilde{\Phi}(r, \theta, z, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [\alpha_{mn}(t) \cos m\theta + \beta_{mn}(t) \sin m\theta] J_m(\lambda_{mn}r) \frac{\cosh[\lambda_{mn}(z+h)]}{\cosh \lambda_{mn}h} \quad (1.40)$$

where α_{mn} and β_{mn} are time dependent to be determined from the free-surface initial conditions, $J_m(\cdot)$ is the Bessel function of the first kind of order m , $\lambda_{mn} = \xi_{mn}/R$ are the roots of $\partial J_m(\lambda_{mn}r)/\partial r|_{r=R} = 0$. Bauer (1963a) developed a set of tables and graphs for zeros of cross product Bessel functions. These roots are also well documented in Abramowitz and Segun (1968).

The analysis can be significantly simplified if the fluid field equations are linearized for small displacements. The normal mode frequencies are determined from the linearized free-surface boundary condition

$$\frac{\partial \tilde{\Phi}}{\partial t} - g\eta = 0 \quad (1.41)$$

Differentiating once with respect to time and using equation (1.3) gives

$$\frac{\partial^2 \tilde{\Phi}}{\partial t^2} + g \frac{\partial \tilde{\Phi}}{\partial z} = 0 \quad (1.42)$$

If the functions α_{mn} and β_{mn} are expressed as harmonics, $\sin \omega_{mn}t$, one can obtain the natural frequencies of the liquid free-surface by substituting equation (1.40) into (1.42)

$$\omega_{mn}^2 = \frac{g\xi_{mn}}{R} \tanh(\xi_{mn}h/R) \quad (1.43a)$$

The above expression approaches a constant value for $h/R > 2$, given by the following relation

$$\omega_{mn}^2 = \frac{g\xi_{mn}}{R}, \quad \text{for } \xi_{mn}h/R \geq 2.65 \quad (1.43b)$$

If one considers surface tension, the linearized dynamic free-surface condition combined with the kinematic condition (1.42) gives, after using Laplace's equation,

$$\frac{\partial^2 \hat{\Phi}}{\partial t^2} + g \frac{\partial \hat{\Phi}}{\partial z} + \frac{\sigma}{\rho} \frac{\partial^3 \hat{\Phi}}{\partial z^3} = 0 \quad (1.44)$$

In this case the natural frequency is given by the expression

$$\omega_{mn}^2 = \left[\frac{g\xi_{mn}}{R} + \frac{\sigma\xi_{mn}^3}{\rho R^3} \right] \tanh(\xi_{mn}h/R) \quad (1.45)$$

This result is valid for the slip contact line and it reveals that the surface tension causes an increase in the normal mode frequencies.

The fluid surface elevation η measured from the undisturbed free surface is obtained from equation (1.41) using (1.40)

$$\eta = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [\bar{\alpha}_{mn} \cos m\theta + \bar{\beta}_{mn} \sin m\theta] J_m(\lambda_{mn}r) \cosh(\lambda_{mn}h) (\omega_{mn} \cos \omega_{mn}t) \quad (1.46)$$

where $\bar{\alpha}_{mn}$ and $\bar{\beta}_{mn}$ are constant coefficients to be determined from initial conditions. The symmetric mode shape takes the form

$$\eta(r, \theta, t) = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \bar{\alpha}_{mn} \cos m\theta J_m(\lambda_{mn}r) \cosh(\lambda_{mn}h) (\omega_{mn} \cos \omega_{mn}t) \quad (1.47)$$

For the first mode, $m=0$, the motion is symmetric about the origin in the form of angular ridges and furrows. In this case, the roots of $dJ_0(\xi_{0n}r/R)/dr|_{r=R}=0$ are $\xi_{0n}=3.832, 7.0156, 10.173, \dots, \pi(n+0.25)$. The corresponding nodal circles are determined by setting $\eta=0$, or $J_0(\xi_{0n}r/R)=0$, which gives $\xi_{0n}r/R=2.404, 5.517, 8.648, \dots$. The corresponding radii of nodal circles are $r=0.628R, 0.786R, 0.85R$.

For asymmetric sloshing modes the wave height takes the form

$$\eta(r, \theta, t) = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [\bar{\beta}_{mn} \sin m\theta] J_m(\lambda_{mn}r) \cosh(\lambda_{mn}h) (\omega_{mn} \cos \omega_{mn}t) \quad (1.48)$$

For the first mode, $m = 1$, the roots of $dJ_1(\xi_{1n}r/R)/dr|_{r=R} = 0$ are $\xi_{1n} = 1.841, 5.335, 8.535, 11.205, 14.850, \dots$, and $\xi_{1n} = \xi_{1(n-1)} + \pi$, for $n > 5$. In this case, we have a nodal diameter perpendicular to the direction of rocking the container.

McCarty and Stephens (1960) found excellent agreement between the analytical and measured values of the natural frequency for the first three modes. They also included other results obtained by Eulitz (1958). Werner and Coldwell (1961) experimentally validated the analytical models for the inertias and natural frequencies of fuel sloshing in circular cylindrical tanks. Further analysis of liquid sloshing in cylindrical tanks was given by Kuttler and Sigillito (1984).

Bauer (1992d) considered the case of strong surface tension for which the contact line remains anchored at its location, that is, $\eta(R, \theta, t) = 0$. This condition results in the following free-surface displacement

$$\eta(r, \theta, t) = e^{i\omega t} \bar{\eta}(r, \theta) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{\alpha_{mn}}{i\omega} \lambda_{mn} \cos m\theta \sinh(\lambda_{mn}h) J_m(\lambda_{mn}r) e^{i\omega t} \quad (1.49)$$

The dynamic condition takes the form

$$\frac{\partial^2 \bar{\eta}}{\partial r^2} + \frac{1}{r} \frac{\partial \bar{\eta}}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \bar{\eta}}{\partial \theta^2} - \frac{\rho g}{\sigma} \bar{\eta} = i \frac{\omega \rho}{\sigma} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \alpha_{mn} \cos m\theta \cosh(\lambda_{mn}h) J_m(\lambda_{mn}r) \quad (1.50)$$

The solution of this partial differential equation must preserve the volume of the fluid

$$\int_0^{2\pi} \int_0^R \bar{\eta}(r, \theta) r \, dr \, d\theta = 0 \quad (1.51)$$

Bauer (1992d) obtained the solution of equation (1.50) in the form

$$\begin{aligned} \bar{\eta}(r, \theta) = & \sum_{m=0}^{\infty} C_m I_m(\beta r/R) \cos m\theta - i \frac{\omega}{g} \alpha_{00} \\ & - i \frac{\omega}{\sigma} R^2 \rho \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \alpha_{mn} \frac{\cosh(\lambda_{mn}h)}{(\xi_{0n}^2 + \beta^2)} J_m(\lambda_{mn}r) \cos m\theta \end{aligned} \quad (1.52)$$

where I_m is the modified Bessel function, $\beta^2 = \rho g R^2 / \sigma$, and C_m are constants of integration.

For axisymmetric oscillations, $\partial/\partial\theta = 0$, $m = 0$, equation (1.52) takes the form

$$\bar{\eta}(r, \theta) = C_0 I_0(\beta r/R) - i \frac{\omega}{g} \alpha_{00} - i \frac{\omega}{\sigma} R^2 \rho \sum_{n=1}^{\infty} \alpha_{0n} \frac{\cosh(\lambda_{0n}h)}{(\xi_{0n}^2 + \beta^2)} J_0(\lambda_{0n}r) \quad (1.53)$$

The stick condition $\eta(R, \theta, t) = 0$ gives

$$C_0 I_0(\beta) - \frac{\varpi}{\beta^2} \bar{\alpha}_{00} - \varpi \sum_{n=1}^{\infty} \bar{\alpha}_{0n} \frac{\cosh(\lambda_{0n}h)}{(\xi_{0n}^2 + \beta^2)} J_0(\xi_{0n}) = 0 \quad (1.54)$$

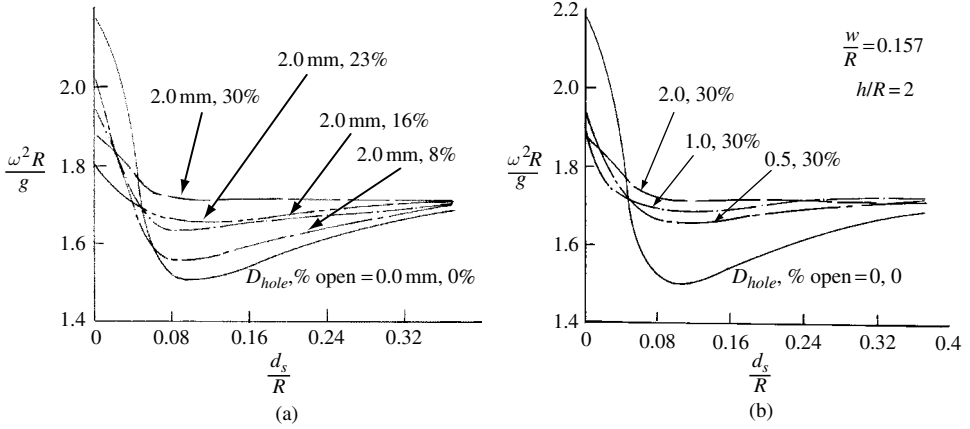


Figure 1.3 (a) Effect of % perforation on the liquid natural frequencies as function of baffle depth, (b) effect of perforation hole size on the liquid natural frequencies as function of baffle depth. (Garza and Abramson, 1963)

where $\varpi = \omega / \sqrt{\sigma / (\rho R^3)}$, and $\bar{\alpha}_{0n} = i\alpha_{0n} / (R\sqrt{\sigma / (\rho R^3)})$, $n = 0, 1, 2, \dots$

Expanding the modified Bessel function I_0 into a Dini-series

$$I_0(\beta r/R) = \frac{2}{\beta} I_1(\beta) + \sum_{n=1}^{\infty} \frac{2\beta I_1(\beta) J_0(\lambda_{0n}r)}{(\xi_{0n}^2 + \beta^2) J_0(\lambda_{0n})} \quad (1.55)$$

Introducing expansion (1.54) into (1.52) and comparing coefficients of the Bessel–Fourier series with those of equation (1.49) gives

$$2I_1(\beta)\beta C_0 - \varpi \bar{\alpha}_{00} = 0 \quad (1.56a)$$

$$2\beta C_0 \frac{I_1(\beta)}{J_0(\xi_{0n})} \varpi - \bar{\alpha}_{0n} [\varpi^2 \cosh(\lambda_{0n}h) - \xi_{0n}^2 (\xi_{0n}^2 + \beta^2) \sinh(\lambda_{0n}h)] = 0 \quad (1.56b)$$

Equations (1.54) and (1.56) constitute $(n+2)$ homogeneous algebraic equations for the constants C_0 , $\bar{\alpha}_{00}$ and $\bar{\alpha}_{0n}$, $n = 1, 2, \dots$. Setting the truncated determinant of the coefficients to zero gives the frequency equation of the axisymmetric natural oscillations. For asymmetric oscillations one should follow the same procedure.

1.4.2 Cylindrical tank with ring baffles

Usually ring baffles, see Figure 1.2(b), are introduced to control the free surface motion and to raise its natural frequency. Garza and Abramson (1963) conducted a series of experimental tests on a circular cylindrical tank with ring baffles and found the free-surface natural frequencies depend on the ring baffle area and its location, d_s , below the free-surface level. Figures 1.3(a) and (b) shows the dependence of the liquid first mode natural frequency on the location of the baffle d_s/R for baffle width, w , to tank radius ratio, $w/R = 0.157$. Figure 1.3(a) shows the influence of percentage-perforated area for a hole diameter of 2.0 mm, while Figure 1.3(b) shows the effect of the hole diameter on the natural frequency of the free surface for a fixed opening ratio. It is seen that the liquid frequency exhibits a maximum value when the

baffle is located at the fluid free surface $d_s/R=0$. It then decreases to a minimum value near a baffle depth close to $d_s/R=0.1$. This observed drop might be attributed to the fact that the baffle restricts the motion of the liquid to that of an equivalent shallow tank of the same diameter. The effect diminishes as the baffle moves to a larger depth. Above depth ratio 0.1 the frequency increases gradually until it reaches the first resonant frequency for a bare-wall cylindrical tank. Another feature may be observed for baffle depths greater than 0.06, and with holes of 2 mm diameter, is that the free-surface frequency increases as the percentage of perforated area is increased.

In some cases, it may be desirable to maintain the highest possible resonant frequency with a given baffle system. This can be achieved by using ring baffles with an axial spacing that must be less than 0.08.

1.4.3 Annular tank

Introducing a concentric tank of radius R_{in} , as shown in Figure 1.2(c), divides the fluid flow into two regions, the concentric tank and the ring tank. Bauer (1960b, 1961c) studied the effect of a concentric tank and derived the following velocity potential function

$$\begin{aligned} \tilde{\Phi}(r, \theta, z, t) = & \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos m\theta + B_{mn} \sin m\theta] \sin \omega_{mn} t \frac{C_m(\xi_{mn} r/R)}{Y'_m(\xi_{mn})} \\ & \times \frac{\cosh[\xi_{mn}(z+h)/R]}{\cosh \xi_{mn} h/R} \end{aligned} \quad (1.57)$$

where a prime denotes differentiation with respect to r , $Y_m(\cdot)$ is the Bessel function of the second kind of order m , ξ_{mn} are zeros of the determinant

$$\Delta_m(\xi_{mn}) = \begin{vmatrix} J'_m(\xi_{mn}) & Y'_m(\xi_{mn}) \\ J'_m(k\xi_{mn}) & Y'_m(k\xi_{mn}) \end{vmatrix} = 0 \quad (1.58)$$

$k = R_{in}/R$, and $C_m(\xi_{mn} r/R)$ is given by the expression

$$C_m(\xi_{mn} r/R) = J_m(\xi_{mn} r/R) Y'_m(\xi_{mn}) - J'_m(\xi_{mn} r/R) Y_m(\xi_{mn} r/R) \quad (1.59)$$

The roots of $\Delta_m(\xi_{mn})=0$ are required to determine the liquid natural frequencies. McMahon (1894) gave asymptotic expansions for most of the roots. Curves showing the roots for $m=0, 1, 2, 3$, and 4 are given by Kirkham (1958). The dominant mode in most applications is the first mode, $m=1$, because it exerts most fluid forces onto the tank wall. The corresponding natural frequencies are given by the expression

$$\omega_{1n}^2 = \frac{g}{R} \xi_{1n} \tanh(\xi_{1n} h/R) \quad (1.60)$$

The fluid free-surface displacement measured from the undisturbed surface is:

$$\eta = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos m\theta + B_{mn} \sin m\theta] \omega_{mn} \cos \omega_{mn} t \frac{C_m(\xi_{mn} r/R)}{Y'_m(\xi_{mn})} \quad (1.61)$$

Table 1.1 Roots of the determinant $\Delta_m(\xi_{mn}) = 0$ for different values of radii ratio k

$k \backslash n$	0.01	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	1.8408	1.8035	1.7051	1.5821	1.4618	1.3537	1.2621	1.1824	1.1134	1.0531
1	5.3291	5.1371	4.9608	5.1374	5.6592	6.5649	8.0411	10.592	15.778	11.4469
2	8.5305	8.1992	8.4331	9.8083	10.6833	12.7066	15.8013	21.0041	31.4513	62.8481
3	11.6951	11.3588	12.1650	13.6837	15.8481	18.9427	23.6239	31.4557	47.1504	94.2645
4	14.8461	14.6344	15.9932	18.1159	21.0488	25.2025	31.4633	41.9189	62.8510	125.675
5	17.9898	17.9864	19.8616	22.5707	26.2641	31.4721	39.3076	52.3845	78.5549	157.088
6	21.1289	21.3837	23.7502	27.0369	31.4859	37.7459	47.1552	62.8523	94.2601	188.502
7	24.2647	24.8081	27.6498	31.5092	36.7118	44.0223	55.0047	73.3212	109.966	219.917
8	27.3979	28.2497	31.5563	35.9855	41.9403	50.3005	62.8553	83.791	125.673	251.332
9	30.5288	31.7027	35.4675	40.4643	47.1704	56.5797	70.7066	94.2612	141.379	282.747

Source: Bauer (1963a).

The first ten zeros of ξ_{1n} are listed for different values of radii ratio k in Table 1.1. Figure 1.4(a) and (b) shows the natural frequencies in the inner and outer tanks for radii ratios $k = 0.5$ and 0.62 , respectively. For these ratios, the liquid masses are approximately equal and the natural frequencies are spread favorably in such a way that the relative phase shift between the inner and outer tanks makes it possible to reduce the total hydrodynamic sloshing forces and moments.

1.4.4 Cylindrical quarter tank

When an upright cylindrical tank is divided by two orthogonal walls in the form of a crucifix as shown in Figure 1.2(d) each quarter can be treated independently. In this case the boundary conditions at the tank solid boundaries are

$$\left. \frac{\partial \tilde{\Phi}}{\partial r} \right|_{r=R} = 0, \quad \left. \frac{\partial \tilde{\Phi}}{\partial z} \right|_{z=-h} = 0, \quad \left. \frac{1}{r} \frac{\partial \tilde{\Phi}}{\partial \theta} \right|_{\theta=0, \pi/2} = 0 \quad (1.62)$$

in addition to the free-surface condition (1.42). Bauer (1963e, 1964d) solved the boundary value problem and obtained the following velocity potential function

$$\tilde{\Phi}(r, \theta, z, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos 2m\theta] \sin(\omega_{mn}t) \frac{\cosh[\xi_{mn}(z+h)/R]}{\cosh \xi_{mn}h/R} J_{2m}(\xi_{mn}r/R) \quad (1.63)$$

The natural frequency of the fluid free surface is

$$\omega_{mn}^2 = \frac{g}{R} \xi_{mn} \tanh(\xi_{mn}h/R), \quad m, n = 0, 1, 2, \dots \quad (1.64)$$

The corresponding wave height is

$$\eta(r, \theta, z, t) = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos 2m\theta] \omega_{mn} \cos(\omega_{mn}t) J_{2m}(\xi_{mn}r/R) \quad (1.65)$$

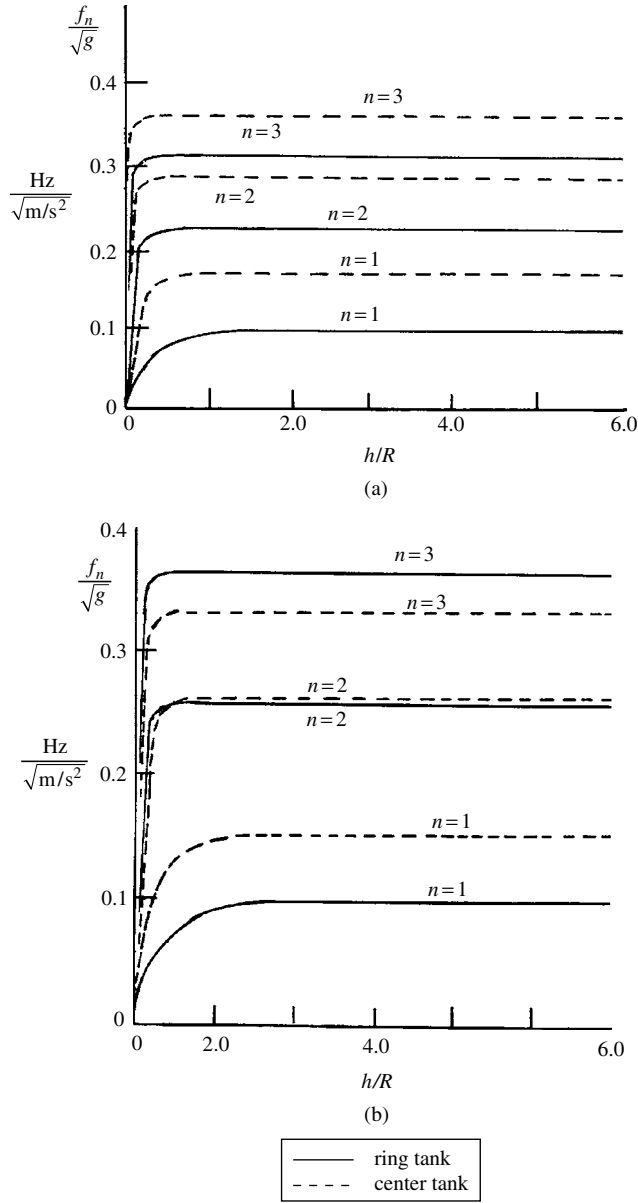


Figure 1.4 (a) Dependence of natural frequencies on liquid depth ratio in annular tank $k = R/R_m = 0.5$, $R = 128$, (b) dependence of natural frequencies on liquid depth ratio in annular tank $k = R/R_{in} = 0.62$, $R = 128$. (Bauer, 1960b)

where ξ_{mn} are the roots of $dJ_{2m}(\xi_{mn} r/R)/dr|_{r=R} = 0$, and are given in Table 1.2

Sector walls perforation effect

Note that if the tank with perforated sector walls is under lateral excitation, the resonance frequency is significantly affected by the excitation amplitude due to liquid intermixing from one sector to another. Because of the complexity of liquid sloshing in sector tanks with

Table 1.2 *Roots of $dJ_{2m}(\xi_{mm} r/R)/dr|_{r=R}=0$*

$n \backslash m$	0	1	2	3	4	5	6	7	8	9
0	3.832	3.054	5.318	8.105	9.648	11.716	13.821	15.917	18.104	20.189
1	7.016	6.706	9.282	11.735	14.115	16.448	18.745	21.015	23.264	25.495
2	10.173	9.969	12.682	15.268	17.774	20.223	22.629	25.002	27.347	29.670
3	13.324	13.370	15.964	18.637	21.587	23.761	26.246	28.694	31.112	33.504
4	16.471	16.348	19.196	21.932	24.587	27.182	29.729	32.237	34.712	37.160
5	19.616	19.513	22.401	25.184	27.889	30.535	33.131	35.689	38.212	40.707
6	22.760	22.672	25.590	28.410	31.155	33.842	36.481	39.079	41.643	44.178
7	25.904	25.826	28.768	31.618	34.397	37.118	39.792	42.426	45.052	47.595
8	29.047	28.978	31.939	34.813	37.620	40.371	43.075	45.740	48.371	50.971
9	32.189	32.127	35.104	38.000	40.830	43.607	46.338	49.030	51.687	54.315

Source: Bauer (1963e).

perforated walls, the natural frequencies were measured experimentally by Abramson, *et al.* (1962a), Garza (1964), and Abramson and Garza (1965). The measured results revealed a decrease of resonance frequencies as the size of perforated holes increased. Figure 1.5(a), (b) and (c) shows the dependence of the first mode natural frequency parameter $\omega_1^2 R/g$ on an equivalent Reynolds number $d2R\sqrt{gd}/(\nu X_0)$, where ν is the fluid kinematic viscosity, d is the diameter of perforations, and X_0 is the excitation amplitude, for 23% and 30% open. Figure 1.5(a), (b) and (c) are obtained for 45° , 60° , and 90° sector tanks, respectively. It is seen that the values of the resonance frequency could be maintained equal to or greater than that of solid wall sector up to a certain Reynolds number, above which the resonance frequencies drop to experimental values for an uncompartmented cylindrical tank.

1.5 Normal modes in a rectangular container

The fluid behavior in rectangular tanks may be treated as two- or three-dimensional flow depending on the tank length. If the length is very long, $\ell \rightarrow \infty$, a two-dimensional flow treatment is adequate. Consider first the case where the motion is two-dimensional along the x -axis, see Figure 1.2(e). In this case, the potential function $\tilde{\Phi}$ must satisfy the Laplace equation

$$\frac{\partial^2 \tilde{\Phi}}{\partial x^2} + \frac{\partial^2 \tilde{\Phi}}{\partial z^2} = 0 \quad (1.66)$$

The solution of equation (1.66) subject to the boundary conditions $\frac{\partial \tilde{\Phi}}{\partial x}|_{x=\pm L/2} = 0$ and $\frac{\partial \tilde{\Phi}}{\partial z}|_{z=-h} = 0$, is

$$\tilde{\Phi}(x, y, z, t) = \sum_{m=1}^{\infty} [\alpha_m(t) \cos k_m x + \beta_m(t) \sin k_m x] \cosh[k_m(z+h)] \quad (1.67)$$

where $k_m = (2m-1)\pi/L$ for asymmetric modes, and $k_m = 2m\pi/L$ for symmetric modes. Following the steps of the previous case, the natural frequencies of the free surface are given by the expression

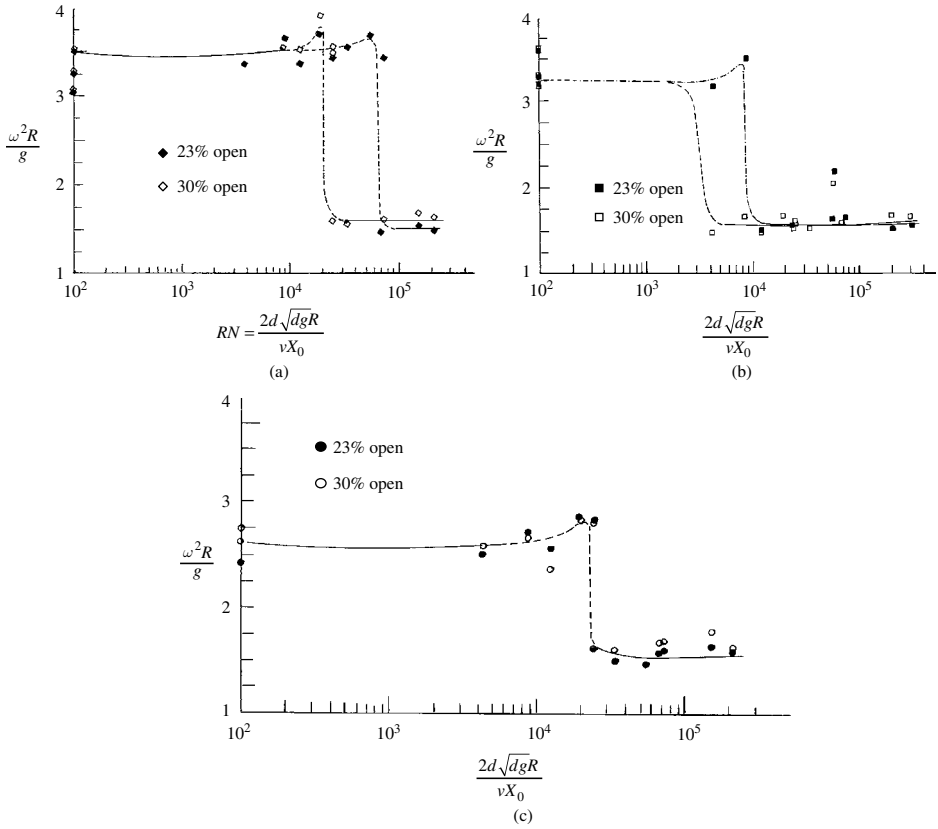


Figure 1.5 (a) Dependence of the lowest liquid resonant frequency on Reynolds number for 45°-sector perforated tank, (b) dependence of the lowest liquid resonant frequency on Reynolds number for 60°-sector perforated tank, (c) dependence of the lowest liquid resonant frequency on Reynolds number for 90°-sector perforated tank. (Abramson and Garza, 1965)

$$\omega_m^2 = gk_m \tanh(k_m h) \quad (1.68)$$

For deep liquids, $h/L > 2$, $\tanh(k_m h) \approx 1$.

If the surface tension is considered, then, for a slipping contact line, the natural frequency takes the form

$$\omega_m^2 = \left[gk_m + \frac{\sigma}{\rho} k_m^3 \right] \tanh(k_m h) \quad (1.69)$$

For the case of an anchored contact line, one may follow the same procedure as that of the circular cylindrical tank.

The surface elevation is

$$\eta(x, t) = \frac{1}{g} \sum_{m=1}^{\infty} [\bar{\alpha}_m \cos k_m x + \bar{\beta}_m \sin k_m x] \omega_m \cos \omega_n t \cosh(k_m h) \quad (1.70)$$

This equation combines symmetric and asymmetric standing waves. The symmetric mode shape is

$$\eta(x, t) = \frac{1}{g} \sum_{m=1}^{\infty} [\bar{\alpha}_m \cos k_m x] \omega_m \cos \omega_n t \cosh(k_m h) \quad (1.71)$$

At a given value of x the surface of a liquid moves up and down, and for a given value of time t , the form of the surface is a cosine curve. A wave of this type is not propagated.

The asymmetric mode shape is given by the expression

$$\eta(x, t) = \frac{1}{g} \sum_{m=1}^{\infty} [\bar{\beta}_m \sin k_m x] \omega_m \cos \omega_n t \cosh(k_m h) \quad (1.72)$$

With reference to Figure 1.2(e), Graham and Rodriguez (1952) solved the three-dimensional velocity potential

$$\tilde{\Phi}(x, y, z, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \bar{\alpha}_{mn}(t) \cos\left(\frac{2m\pi x}{L}\right) \cos\left(\frac{2n\pi y}{\ell}\right) \cosh[k_{mn}(z + h)] \quad (1.73)$$

where L and ℓ are the tank width and breadth, respectively, $k_{mn} = \pi \sqrt{((2m)^2/L^2) + ((2n)^2/\ell^2)}$, m and n are positive integers. The corresponding free-surface natural frequency is

$$\omega_{mn}^2 = g k_{mn} \tanh(k_{mn} h) \quad (1.74)$$

If the surface tension is considered then expression (1.74) takes the following form for the case of a slipping contact line

$$\omega_{mn}^2 = \left[g k_{mn} + \frac{\sigma}{\rho} k_{mn}^3 \right] \tanh(k_{mn} h) \quad (1.75)$$

The wave height η can be written in the form

$$\eta(x, y, t) = \frac{1}{g} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \bar{\alpha}_{mn} \omega_{mn} \cos\left(\frac{2m\pi x}{L}\right) \cos\left(\frac{2n\pi y}{\ell}\right) \cosh(k_m h) \quad (1.76)$$

Note that the time dependence of the wave height results in a variation of the position of the container's center of mass (see, e.g., Lorell, 1952). Bauer and Eidel (2000) considered the normal modes of liquid-free-surface oscillation in a long rectangular tank with structural obstructions.

Several techniques have been developed to measure the wave height of free liquid surface during sloshing (Whittenbury, *et al.*, 1959, Wilner, *et al.*, 1960, Crews and Earth, 1963, Kana, 1964b, Muenz and Marcello, 1964, Sturtevant, 1966 and Lindstrom, *et al.*, 1969). The fundamental natural frequency in a rectangular tank was measured experimentally by Addington (1960) and the measured results were within 5% accuracy compared to the analytical values. Addington also examined the effect of baffles on the natural frequencies and suggested that the baffles should be positioned in a region of the highest velocity, such that the surface waves are trapped but allow dissipation of energy at the shallowest depth possible.

1.6 Normal modes in containers with variable depth

The analysis of liquid sloshing dynamics in containers with curved walls or walls that are not parallel to the container longitudinal axis is not a simple task. Examples of such containers include conical, spherical, horizontal cylinder (circular canal), toroidal (ring with circular cross-section) tanks. Generally, these cases do not have a closed form solution and the natural frequencies are estimated numerically or by using approximate techniques. For example, the actual geometry of the container can be approximated by a geometry that permits an exact solution provided the free-surface area and liquid volume are preserved. Alternatively, one may select a proper coordinate system in which most of the boundaries of the liquid form coordinate surfaces such that the boundary conditions are satisfied exactly. Budiansky (1960) used a sophisticated integral equation technique and obtained the liquid natural frequency for cases of nearly empty, half full, and nearly full tanks. Petrov (1961, 1962a,b,c, 1963) considered different problems of liquid oscillations in horizontal cylindrical tanks and arbitrary tanks. Further analytical and experimental investigations were carried out by McCarty and Stephens (1960), McCarty, *et al.* (1960), Mikishev and Dorozhkin (1961), Stofan and Armstead (1962), Sumner (1963, 1966), and Barron and Chang (1989).

This section presents liquid modal analyses in different tank geometries. These include canals with 45°-straight walls, horizontal cylinders, spherical tanks, prolate and oblate spheroidal tanks, conical containers, toroidal containers, upright elliptic tanks, and paraboloid containers.

1.6.1 Canal with 45°-straight walls

This classical case was considered by Lamb (1945, Section 258) who analyzed two-dimensional oscillations of a liquid bounded by two walls each inclined at 45° to the vertical z -axis. With reference to Figure 1.6(a), the walls are defined by the two straight-line equations, $z = \pm y$. Lamb assumed the following velocity potential function for symmetric modes

$$\Phi(y, z, t) = A(\cosh ky \cos kz + \cos ky \cosh kz) \sin \omega t \quad (1.77)$$

Introducing this function in the free-surface boundary condition (1.42) yields the frequency equation

$$\omega^2(\cosh ky \cos kh + \cos ky \cosh kh) = gk(-\cosh ky \sin kh + \cos ky \sinh kh) \quad (1.78)$$

This condition requires

$$\omega^2 \cos kh = -gk \sin kh \quad \text{and} \quad \omega^2 \cosh kh = gk \sinh kh \quad (1.79)$$

This means that the wave number k is obtained by solving the transcendental equation

$$\tanh kh = -\tan kh \quad (1.80)$$

Having obtained the values of k , one can determine the corresponding natural frequencies from one of the expressions of relations (1.79), that is,

$$\omega^2 = gk \tanh kh \quad (1.81)$$

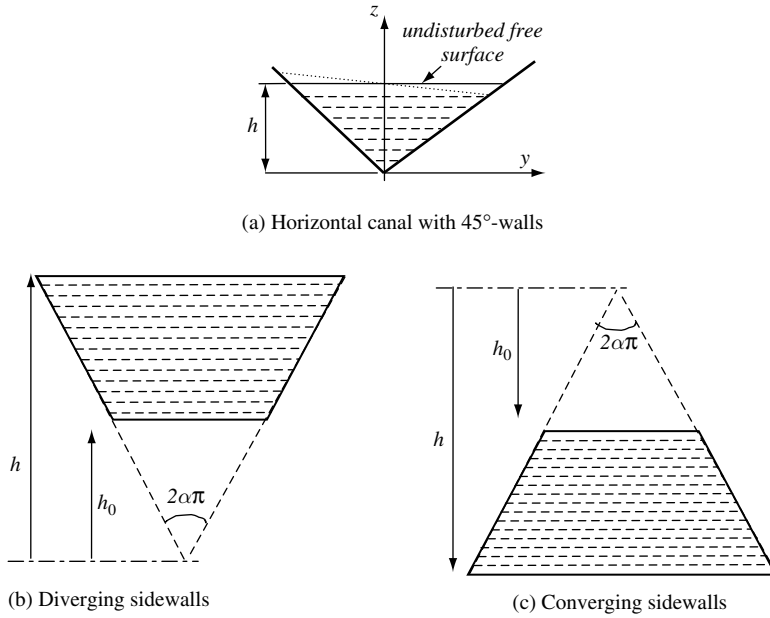


Figure 1.6 Horizontal canals with diverging and converging sidewalls.

The first root of equation (1.80) is $kh = 2.365$ and the corresponding natural frequency is

$$\omega = 1.5244\sqrt{g/h} \quad (1.82)$$

For asymmetric sloshing modes the following velocity potential function may be considered

$$\Phi(y, z, t) = -A(\sinh ky \sin kz + \sin ky \sinh kz) \sin \omega t \quad (1.83)$$

The dynamic free-surface condition requires the following relations

$$\omega^2 \sin kh = gk \cos kh \quad \text{and} \quad \omega^2 \sinh kh = gk \cosh kh \quad (1.84)$$

The corresponding wave number, k , is given by solving the transcendental equation

$$\tanh kh = \tan kh \quad (1.85)$$

For the first mode, the free surface is always plane and one can set $\tan kh \approx kh$, and from relations (1.84) the first natural frequency of the asymmetric sloshing mode is

$$\omega^2 = g/h \quad (1.86)$$

For wedge-shaped containers, Kelland (1839) considered the symmetric natural frequencies in a 90°-wedge-shaped container, while Greenhill (1887) estimated the asymmetric frequencies. Macdonald (1894) analyzed the free-surface oscillations in a container of apex angle 120°. Bauer (1981c, 1999) treated different geometries of wedge-shaped containers of finite length, ℓ , and arbitrary apex angle $2\pi\alpha < 2\pi$. Without going into detail, the following are the basic formulas for the natural frequencies of diverging and converging sidewalls shown in Figure 1.6(a).

For a wedged-shaped container with diverging sidewalls, Figure 1.6(b), the natural frequencies of the free surface are

$$\omega_{nm}^2 = \frac{n\pi}{\ell} \left[g + \frac{\sigma}{\rho R^2} \left(\frac{n^2 \pi^2 h^2}{\ell^2} + \frac{m^2}{4\alpha^2} \right) \right] \times \frac{\left[I'_{m/2\alpha}(n\pi h/\ell) K'_{m/2\alpha}(n\pi h_0/\ell) - I'_{m/2\alpha}(n\pi h_0/\ell) K'_{m/2\alpha}(n\pi h/\ell) \right]}{\left[I_{m/2\alpha}(n\pi h/\ell) K'_{m/2\alpha}(n\pi h_0/\ell) - I'_{m/2\alpha}(n\pi h_0/\ell) K_{m/2\alpha}(n\pi h/\ell) \right]} \quad (1.87)$$

where h and h_0 are the heights of the upper and lower levels of the liquid volume, I and K are the modified Bessel functions of the first and second kinds, respectively, and R is the radius of curvature of the free surface.

For the case of a container with converging sidewalls, Figure 1.6(c), the gravitational term g is replaced by $-g$, and h by h_0 . If the free-surface oscillations do not depend on the angular coordinate of the wedge, the natural frequencies are obtained by setting $m = 0$, and one may use the well-known equalities $I'_0 = I_1$ and $K'_0 = -K_1$. For free-surface oscillations independent of the axial coordinate, $n = 0$, and in terms of the ratio $k = h_0/h$ the natural frequencies for the diverging walls are

$$\omega_{m0}^2 = \frac{m}{2\alpha h} \left[g + \frac{\sigma}{\rho R^2} \left(\frac{m^2}{4\alpha^2} - 1 \right) \right] \left[\frac{1 - k^{m/\alpha}}{1 + k^{m/\alpha}} \right] \quad (1.88)$$

For a converging wall, the natural frequencies are

$$\omega_{m0}^2 = \frac{m}{2\alpha h k} \left[g + \frac{\sigma}{\rho R^2 k^2} \left(\frac{m^2}{4\alpha^2} - 1 \right) \right] \left[\frac{1 - k^{m/\alpha}}{1 + k^{m/\alpha}} \right] \quad (1.89)$$

In all cases, the surface tension results in an increase of the fluid natural frequencies.

1.6.2 Horizontal cylindrical and spherical containers

Horizontal cylinders and spherical tanks partially filled with liquid are difficult to analyze to determine the free-liquid natural frequencies and mode shapes. The difficulty arises due to the fact the walls are not straight and parallel to the axis of symmetry. Budiansky (1960) and Chu (1964b) used conformal mapping to formulate the problem as an integral equation, which was discretized for numerical solution. The method was restricted to particular liquid filling conditions. The restriction in the analysis is due to the lack of Green's function of the second kind (Neumann function) for the spherical bowl. Green's function of the first kind for spherical containers is documented in Bateman (1944). It is not an easy task to obtain a simple expression for Green's function of the second kind because the normal derivative on the spherical surface is a combination of two derivatives in toroidal coordinates (see the appendix). Chu (1964a) devised a numerical scheme to determine the kernel function, which forms one component of the Neumann function. The method allows the estimation of the natural frequencies of a free-liquid surface in a spherical tank to any arbitrary liquid filling.

Moiseev and Petrov (1965) developed a variational technique and estimated the liquid modal analysis for the lowest frequency as a function of liquid depth. Kuttler and Sigillito

(1984) estimated the liquid natural frequencies in a horizontal cylindrical tank following an approach similar to the Rayleigh–Ritz method and the bounds of the eigenvalues of elliptic operators (Kutler and Sigillito, 1969, 1978). McIver (1989) selected coordinate systems in which the container walls and the free surface coincide with coordinate lines or a coordinate surface. The eigenvalue problem was formulated in terms of integral equations, which were solved numerically. For a horizontal cylindrical container, McIver used two-dimensional bipolar coordinates and obtained a homogeneous Fredholm integral equation of the second kind. For a spherical container, he used toroidal coordinates and formulated the eigenvalue problem in terms of a pair of coupled integral equations. This complicated formulation is mainly due to the fact that Laplace’s equation is not fully separable when expressed in toroidal coordinates.

Barnyak (1997) and Barnyak and Barnyak (1996) determined the normal mode frequencies of viscous liquid partially filling a circular horizontal cylinder.

Horizontal cylinder

Lamb (1945, Sections 72 and 259) presented an energy approach to determine the natural frequency of the first transverse mode of liquid free surface in a half-filled horizontal cylinder, that is, $h = -R$. He obtained the following expression for the velocity potential function

$$\Phi(r, \theta) = \frac{1}{2} \omega r^2 \sin 2\theta - \sum A_{2n+1} \sin(2n+1)\theta \left(\frac{r}{R}\right)^{(2n+1)} \quad (1.90)$$

where $A_{2n+1} = (-1)^{n+1} (\omega R^2 / \pi) \{ (1/(2n-1)) - (2/(2n+1)) + (1/(2n+3)) \}$.

The kinetic energy of the fluid free surface experiencing small oscillations is

$$\begin{aligned} T &= -\frac{\rho}{2} \int_s \frac{\partial \Phi}{\partial n} ds = -\rho \omega \int_0^R \Phi(r, \pi/2) r dr \\ &= \frac{1}{2} \pi \rho R^4 \left(\frac{4}{\pi^2} - \frac{1}{4} \right) \omega^2 \end{aligned}$$

From this expression one can write the effective mass moment of inertia of the liquid about the center of the cylinder

$$I_{o(\text{eff})} = \pi \rho R^4 \left(\frac{4}{\pi^2} - \frac{1}{4} \right)$$

which is much less than the moment of inertia of the frozen liquid.

The potential energy is

$$U = \frac{1}{3} \rho g R^3 \theta^2$$

Equating the maximum kinetic energy to the maximum potential energy, and assuming sinusoidal oscillation, gives

$$\omega_1^2 = \frac{8\pi}{48 - 3\pi^2} \frac{g}{R} = 1.366 \, 56 \frac{g}{R} \quad (1.91)$$

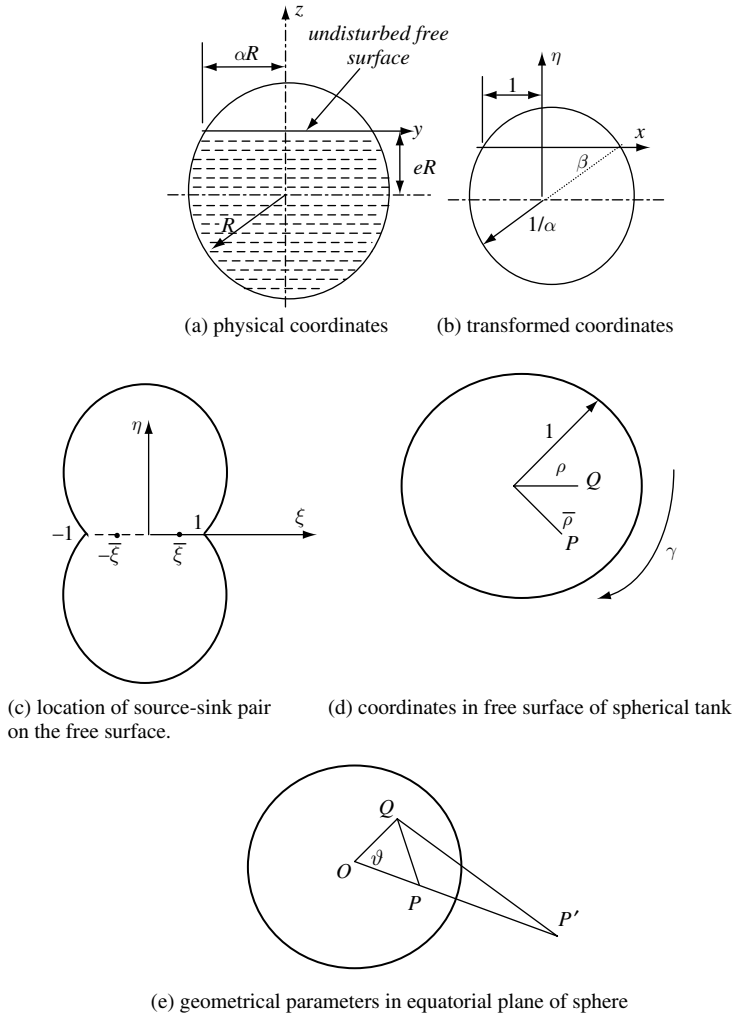


Figure 1.7 Geometries and coordinate system of horizontal canal and spherical tank.

Conformal mapping With reference to Figure 1.7(a), Budiansky (1960) formulated the boundary value problem of free-liquid-surface asymmetric motion in the two-dimensional cross-section of a horizontal canal. In terms of the x, y coordinates and the velocity potential function $\phi(x, y) \sin \omega t$, the flow field equations are

$$\nabla^2 \phi(x, y) = 0 \quad (1.92)$$

subject to the boundary conditions:

At the walls (given by (93) and the free surface condition (94))

$$\partial \phi / \partial n = 0 \quad (1.93)$$

where n is the normal to the wetted surface. The kinematic and dynamic free-surface conditions are combined to give the free-surface condition

$$\frac{\partial \phi}{\partial y} = \frac{\omega^2}{g} \phi. \quad (1.94)$$

Introducing the following coordinate transformation

$$\xi = x/\alpha R, \quad \eta = y/\alpha R \quad (1.95)$$

where the factor $\alpha = a/R$, $0 \leq \alpha \leq 1$, and a is half the width of the free surface on the x -axis. Figure 1.7(b) shows the new coordinates and the angle β , defined by the relationship $\sin \beta = eR/R = e$, $-1 \leq e \leq 1$. The angle β also defines the fluid depth and it varies from $\pi/2$ for the completely filled condition to $-\pi/2$ for the empty case. In view of the geometrical symmetry, the natural modes of the free surface involve free-surface displacements that are either symmetric or anti-symmetric. The analysis is restricted for the anti-symmetric modes since the symmetric modes are not generated under lateral excitation. The free-surface condition (1.94) becomes

$$\frac{\partial \phi}{\partial \eta} = \alpha \lambda \phi, \quad \text{at } \eta = 0 \quad (1.96)$$

where $\lambda = \omega^2 R/g$ are the associated natural frequencies.

Budiansky considered the fluid-filled container bounded by the original wetted surface plus its reflection about the ξ as shown in Figure 1.7(c). A two-dimensional sink of strength $2w(\bar{\xi}) d\bar{\xi}$ at $\bar{\xi}$ and a source of equal strength at $-\bar{\xi}$ over a strip of length $d\bar{\xi}$ are introduced to describe the motion of the free surface, where $w(\bar{\xi})$ represents the mode shape of the free surface over the strip $d\bar{\xi}$ at $\bar{\xi}$. The resulting velocity potential function for the internal flow is

$$\phi(\xi, \eta) = \int_0^1 w(\bar{\xi}) \Phi(\xi, \eta; \bar{\xi}) d\bar{\xi} \quad (1.97)$$

where $\Phi(\xi, \eta; \bar{\xi})$ is the normalized velocity potential function whose value at $\xi = 0$ is zero, that is, $\Phi(0, \eta; \bar{\xi}) = 0$, and it is odd for asymmetric motion, that is, $\Phi(\xi, \eta; \bar{\xi}) = -\Phi(-\xi, \eta; \bar{\xi})$. At the free surface, $\eta = 0$, condition (1.96) must be satisfied and yields the characteristic free-surface shapes, $w(\xi)$,

$$w(\xi) = \lambda \alpha \int_0^1 w(\bar{\xi}) \Phi(\xi, 0; \bar{\xi}) d\bar{\xi}, \quad 0 \leq \xi \leq 1 \quad (1.98)$$

The evaluation of the above integral requires knowledge of the kernel function $\Phi(\xi, 0; \bar{\xi}) \equiv G(\xi, \bar{\xi})$, which is determined by using conformal mapping through the successive transformations, $z \rightarrow \zeta \rightarrow W \rightarrow \varpi \rightarrow f$, shown in Figure 1.8,

$$f = \left[\frac{1-z}{2(1+z)} \right]^\gamma, \quad z = \xi + i\eta \quad (1.99)$$

where $\gamma = 2\pi/(\pi + 2\beta)$. Let f_1 and f_2 correspond to $\bar{\xi}$ and $-\bar{\xi}$, respectively. Introducing the complex potential,

$$F = G + i\psi = -\frac{1}{\pi} \log \left[\frac{f-f_1}{f-f_2} \right] + g(\bar{\xi}) \quad (1.100)$$

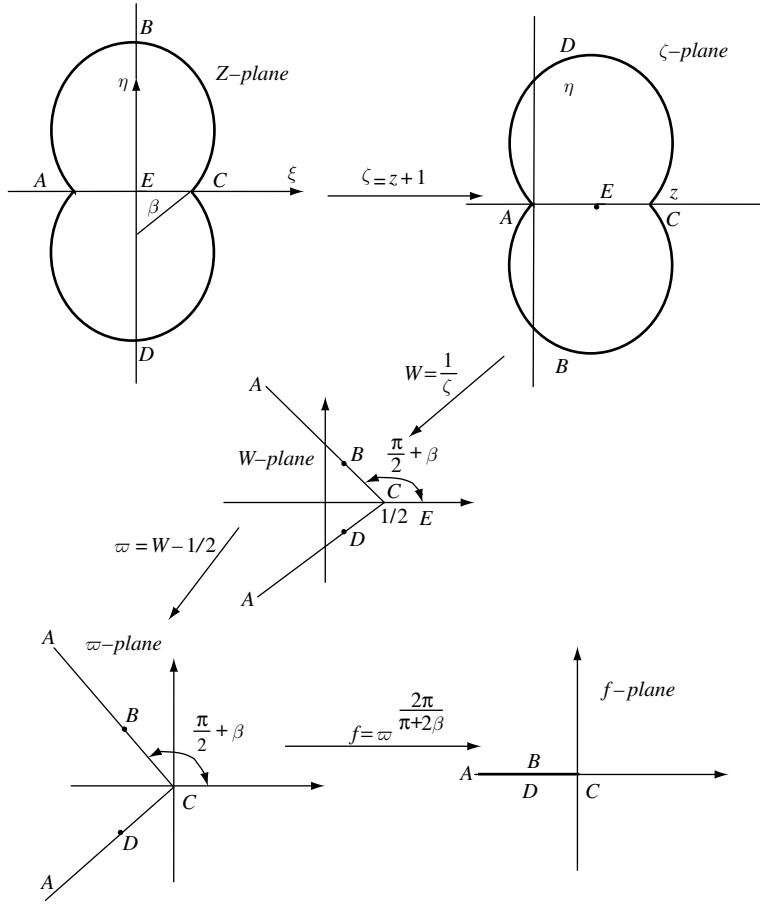


Figure 1.8 Successive transformation of conformal mapping sequence $z - \zeta - W - w - f$ planes.

where ψ is the stream function, of a sink at f_1 and a source at f_2 each of strength 2, g is an arbitrary function of $\bar{\xi}$. Substituting equation (1.99) into (1.100) gives the corresponding potential function

$$F(z) = -\frac{1}{\pi} \log \left\{ \frac{[(1-z)/(1+z)]^\gamma - [(1-\bar{\xi})/(1+\bar{\xi})]^\gamma}{[(1-z)/(1+z)]^\gamma - [(1-\bar{\xi})/(1+\bar{\xi})]^\gamma} \right\} + g(\bar{\xi}) \quad (1.101a)$$

The function g is selected such that $F(0) = 0$, and thus one can write

$$\begin{aligned} F(z) &= -\frac{1}{\pi} \log \left\{ \frac{[(1-z)/(1+z)]^\gamma - [(1-\bar{\xi})/(1+\bar{\xi})]^\gamma}{[(1-z)/(1+z)]^\gamma - [(1-\bar{\xi})/(1+\bar{\xi})]^\gamma} \right\} \\ &\quad \times \left\{ \frac{1 - [(1+\bar{\xi})/(1-\bar{\xi})]^\gamma}{1 - [(1-\bar{\xi})/(1+\bar{\xi})]^\gamma} \right\} \\ &= -\frac{1}{\pi} \log \left\{ \frac{(1-z)^\gamma (1+\bar{\xi})^\gamma - (1+z)^\gamma (1-\bar{\xi})^\gamma}{(1+z)^\gamma (1+\bar{\xi})^\gamma - (1-z)^\gamma (1-\bar{\xi})^\gamma} \right\} \end{aligned} \quad (1.101b)$$

The real part $G(\xi, \bar{\xi}) = \text{Re } F(z)$ is

$$G(\xi, \bar{\xi}) = -\frac{1}{\pi} \log \left\{ \frac{(1+\xi)^\gamma (1-\bar{\xi})^\gamma - (1-\xi)^\gamma (1+\bar{\xi})^\gamma}{(1+\xi)^\gamma (1+\bar{\xi})^\gamma - (1-\xi)^\gamma (1-\bar{\xi})^\gamma} \right\} \quad (1.102)$$

Note that the function G has singularity at $\xi = \bar{\xi}$ and $G(\xi, \bar{\xi}) = G(\bar{\xi}, \xi) = -G(-\xi, \bar{\xi})$. The kernel function given by equation (1.102) is then substituted in the integral equation (1.99) that gives the characteristic of free-surface shapes $w(\xi)$ and associated eigenvalues, $\lambda\alpha$. For ξ near $\bar{\xi}$, with $\xi \neq 1$, the kernel takes the approximate value

$$G(\xi, \bar{\xi}) \approx -\frac{1}{\pi} \log |\xi - \bar{\xi}| - \frac{1}{\pi} \log \left| \frac{2\gamma(1-\xi^2)^{\gamma-1}}{(1+\xi)^{2\gamma} - (1-\xi)^{2\gamma}} \right|. \quad (1.103)$$

At the wall, $\xi=1$, expression (1.103) takes the form

$$G(1, \bar{\chi}) \approx -\frac{\gamma}{\pi} \log |1 - \xi| + \gamma \log 2 \quad (1.104)$$

where $G(0, \bar{\xi}) = G(\xi, 0) = 0$, and $w(0)=0$. In order to generate a numerical algorithm to determine the normal modes, the integral on the right-hand side of equation (1.98) may be discretized in the form

$$\sum_{j=1}^{N-1} w(j\Delta) \int_{j\Delta-(\Delta/2)}^{j\Delta+(\Delta/2)} G(\xi, \bar{\xi}) d\bar{\xi} + w(1) \int_{1-(\Delta/2)}^1 G(\xi, \bar{\xi}) d\bar{\xi} \quad (1.105)$$

where $\Delta = 1/N$. Suppose $\xi = n\Delta$, then

$$\int_{j\Delta-(\Delta/2)}^{j\Delta+(\Delta/2)} G(\xi, \bar{\xi}) d\bar{\xi} \approx \Delta G(n\Delta, j\Delta) \quad n \neq j \neq N \quad (1.106a)$$

$$\int_{1-(\Delta/2)}^1 G(\xi, \bar{\xi}) d\bar{\xi} \approx \frac{\Delta}{2} G(n\Delta, 1) \quad n \neq N \quad (1.106b)$$

$$\begin{aligned} \int_{n\Delta-(\Delta/2)}^{n\Delta+(\Delta/2)} G(\chi, \bar{\xi}) d\bar{\xi} &\approx -\frac{1}{\pi} \log \left| \frac{2\gamma(1-(n\Delta)^2)^{\gamma-1}}{(1+n\Delta)^{2\gamma} - (1-n\Delta)^{2\gamma}} \right| \\ &\quad + \int_{-(\Delta/2)}^{(\Delta/2)} -(1/\pi) \log |t| dt \\ &= \Delta \left\{ -\frac{1}{\pi} \log \left| \frac{2\gamma(1-(n\Delta)^2)^{\gamma-1}}{(1+n\Delta)^{2\gamma} - (1-n\Delta)^{2\gamma}} \right| \right. \\ &\quad \left. + (1/\pi)[1 - \log(\Delta/2)] \right\}, \quad n \neq N \end{aligned}$$

$$\int_{1-(\Delta/2)}^1 G(1, \bar{\xi}) d\bar{\xi} \approx \frac{\Delta}{2} \{[(\gamma/\pi) \log 2] + (\gamma/\pi)[1 - \log(\Delta/2)]\} \quad (1.106d)$$

The integral equation (1.98) is approximated by the matrix equation

$$[A][T]\{w\} = \frac{N}{\lambda\alpha}\{w\} \quad (1.107)$$

where $\{w\}$ is the column vector with elements $w(n\Delta)$, ($n = 1, 2, \dots, N$), T is the integrating diagonal matrix

$$[T] = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 1/2 \end{bmatrix}$$

and $[A]$ is a symmetric square matrix with the general elements

$$\begin{aligned} A_{jn} &= G(j\Delta, n\Delta) & n \neq j \neq N \\ &= \chi(n\Delta) + (1/\pi)[1 - \log(\Delta/2)] & j = n \neq N \\ &= \frac{\gamma}{\pi} \log 2 + (1/\pi)[1 - \log(\Delta/2)] & j = n \neq N \end{aligned}$$

In the absence of a logarithmic singularity, the integration scheme is reduced to the trapezoidal rule. The orthogonality condition for distinct eigenfunctions w_k and the integral

$$w_l \int_0^1 w_k(\xi) w_l(\xi) d\xi = 0 \quad (1.108a)$$

is approximated by the expression

$$\{w_k\}^T [T] \{w_l\} = 0 \quad (1.108b)$$

where the superscript T denotes transpose.

Consider the special case of a half-full container, $e = 0$ and $\alpha = 1$, the kernel takes the form

$$G(\xi, \bar{\xi}) = -\frac{1}{\pi} \log \left| \frac{(\xi - \bar{\xi})(1 - \xi\bar{\xi})}{(\xi + \bar{\xi})(1 + \xi\bar{\xi})} \right| \quad (1.109)$$

A one-term Galerkin's solution for the lowest frequency is applied to equation (1.98)

$$\int_0^1 [w(\xi)]^2 d\xi = \lambda\alpha \int_0^1 \int_0^1 G(\xi, \bar{\xi}) w(\bar{\xi}) w(\xi) d\xi d\bar{\xi} \quad (1.110)$$

Table 1.3 *Liquid free-surface eigenvalues in circular canal for different fluid depth*

e	α	$\alpha\lambda_1$	$\alpha\lambda_2$	$\alpha\lambda_3$	λ_1	λ_2	λ_3
−1.0	0.0	0.0	0.0	0.0	1.0	6.0	15.0
−0.8	0.6	0.627	3.23	6.51	1.045	5.38	10.85
−0.6	0.8	0.879	3.98	7.3	1.099	4.97	9.13
−0.4	0.917	1.068	4.34	7.63	1.165	4.74	8.33
−0.2	0.980	1.224	4.56	7.82	1.249	4.65	7.99
0.0	1.0	1.360	4.70	7.96	1.360	4.70	7.96
0.2	0.98	1.482	4.81	8.06	1.513	4.91	8.23
0.4	0.917	1.596	4.89	8.15	1.742	5.34	8.89
0.6	0.8	1.706	4.97	8.22	2.13	6.22	10.23
0.8	0.6	1.822	5.05	8.30	3.04	8.42	13.84
1.0	0.0	2.018	5.20	8.44	∞	∞	∞

Source: Budiansky (1960).

For small liquid-free-surface oscillations, the mode shape may be expressed by the planar shape, $w = \xi$. Carrying out the integration (1.110), gives

$$\lambda_1 \approx \frac{1}{3} \left[\frac{2}{\pi} - \frac{\pi}{8} \right] = 1.367 \rightarrow \omega_1 = 1.169 \sqrt{g/R} \quad (1.111)$$

This result is identical to the one reported by Lamb and given by relation (1.91).

The first three modes and frequencies were evaluated numerically for different values of fluid depth ratio, $e = h/R$, using the 20×20 matrix formulation as described above. The numerical results are listed in Table 1.3 and plotted in Figure 1.9 by solid curves as a function of the fluid depth ratio e . It is seen that the fundamental mode has its lowest frequency in the nearly empty state, whereas the minimum frequency of a given higher mode appears to occur slightly below the half-full condition. As the fluid depth increases, the natural frequencies tend to shift to more closely packed spectra and the ratio $\lambda_{n+1} / \lambda_n$ decreases monotonically with increasing depth ratio. McCarty and Stephen (1960) obtained some numerical experimental results for the dependence of the liquid natural frequency parameter $\omega_n \sqrt{R/g}$ on the depth ratio $h/2R$. Figures 1.10 and 1.11 show this dependence for transverse and longitudinal sloshing modes, respectively.

Bipolar coordinate transformation McIver (1989) formulated the fluid field equations of a horizontal circular cylinder in terms of bipolar coordinates. Figure 1.7(a) shows a horizontal circular cylindrical tank partially filled with liquid to the depth h and the undisturbed free liquid surface coincides with the x -axis, instead of y , such that the positive y -axis, instead of z , is directed downward. With reference to the transformation to the bipolar coordinates η, ξ , outlined in the appendix of this chapter, the coordinates of the container center are $(x, y) = (0, a \cot \xi)$ where a is half the width of the free surface on the x -axis and

$$x + iy = a \tanh[(\eta + i\xi)/2] \quad (1.112)$$

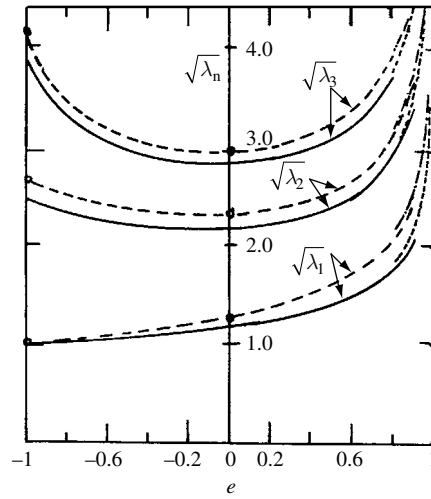
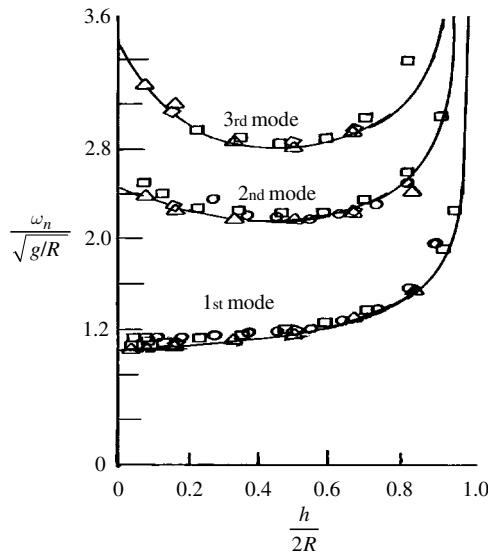


Figure 1.9 Dependence of the liquid natural frequencies on the liquid depth parameter e for circular canal (solid) and spherical tank (dashed). (Budiansky, 1960)



- | | |
|--------------------------------|--------------------------------|
| — theory (Budiansky, 1960) | ○ $R = 7$ cm, $L = 40.6$ cm |
| □ $R = 10.9$ cm, $L = 38.1$ cm | ◇ $R = 15.2$ cm, $L = 30.5$ cm |
| △ $R = 15.2$ cm, $L = 61.0$ cm | |

Figure 1.10 Dependence of liquid natural frequency on depth ratio for transverse modes in circular canal. (McCarty and Stephens, 1960)

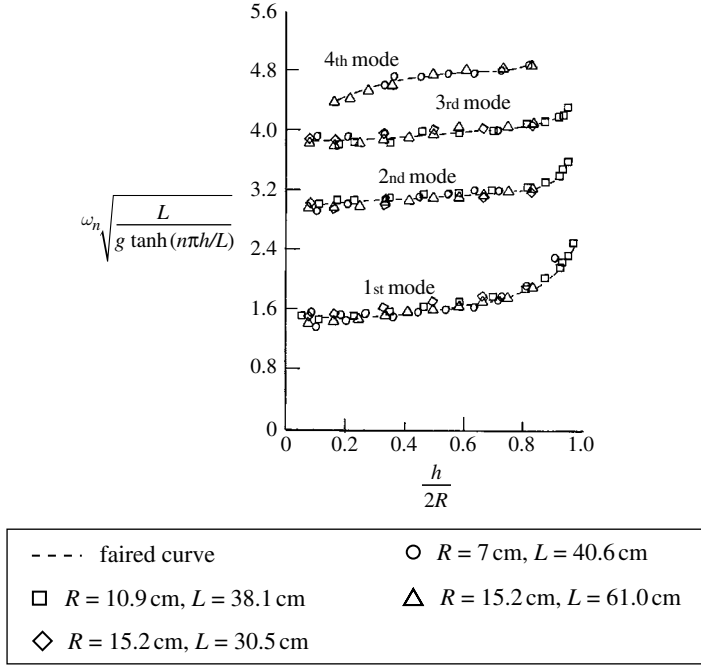


Figure 1.11 Dependence of liquid natural frequency on depth ratio for longitudinal modes in circular canal. (McCarty and Stephens, 1960)

Note that the cylinder surface coincides with the coordinate, $\xi = \xi_0$, where $0 < \xi_0 \leq \pi$, and ξ_0 is given by the relationship

$$\cos \xi_0 = 1 - \frac{h}{R} \quad (1.113)$$

The free surface coincides with $\xi = 0$. Laplace's equation for the velocity potential function for the two-dimensional case in terms of bipolar coordinates, η and ξ , is

$$\frac{\partial^2 \Phi}{\partial \eta^2} + \frac{\partial^2 \Phi}{\partial \xi^2} = 0, \quad -\infty < \eta < \infty, \quad 0 < \xi < \xi_0 \quad (1.114)$$

It is seen that equation (1.114) is separable and can be solved by setting $\Phi(\eta, \xi) = H(\eta)Z(\xi)$, which gives $(H''/Z = -Z''/H = \tau)$, where τ is a constant. The solution must satisfy the following boundary conditions:

The velocity normal to the wetted wall must vanish, that is,

$$\left. \frac{\partial \Phi}{\partial \xi} \right|_{\xi=\xi_0} = 0 \quad (1.115)$$

The combined free-surface condition is

$$\frac{\partial \Phi}{\partial \xi} + \frac{\lambda}{1 + \cosh \eta} \Phi = \big|_{\xi=0} 0. \quad (1.116)$$

where $\lambda = \omega^2 a/g$. Two types of solutions are considered: anti-symmetric and symmetric.

(1) *Anti-symmetric sloshing modes*

For anti-symmetric sloshing modes, the general solution of equation (1.114) satisfying the wall boundary condition (1.115) may be written in the integral form

$$\Phi(\eta, \xi) = \int_0^\infty A(\tau) \cosh[\tau(\eta - \eta_0)] \sin(\tau\xi) d\tau \quad (1.117)$$

Substituting equation (1.117) into the free-surface boundary condition (1.116) and carrying out Fourier sine transform with respect to ξ , gives

$$A(\tau')(\tau') \sinh(\tau'\eta_0) = \lambda \int_0^\infty A(\tau) \cosh(\tau'\eta_0) I_A(\tau, \tau') d\tau, \quad 0 < \tau' < \infty \quad (1.118)$$

where

$$I_A(\tau, \tau') = \frac{2}{\pi} \int_0^\infty \frac{\sin(\tau\xi) \sin(\tau'\xi)}{1 + \cosh \xi} d\xi = \frac{\tau - \tau'}{\sinh[\pi(\tau - \tau')]} - \frac{\tau + \tau'}{\sinh[\pi(\tau + \tau')]} \quad (1.119)$$

The integral formula $\int_0^\infty \frac{\cos(\tau\xi)}{\cosh \xi} d\xi = \frac{\tau\pi}{2 \sinh(\tau\pi/2)}$ is taken from Gradshteyn and Ryzhik (1980, formula 3.982(1)). Introducing the new variable

$$B(\tau) = A(\tau) \sqrt{\tau \sinh \tau\xi_0 \cosh \tau\xi_0} \quad (1.120)$$

equation (1.120) takes the form

$$B(\tau') = \lambda \int_0^\infty B(\tau) K_A(\tau, \tau') d\tau, \quad 0 < \tau' < \infty \quad (1.121)$$

where

$$K_A(\tau, \tau') = \frac{I_A(\tau, \tau')}{\sqrt{\tau\tau' \tanh \tau\beta_0 \tanh \tau'\beta_0}} \quad (1.122)$$

The eigenvalue problem has been reduced to the integral equation (1.121) as in the previous analysis by Budiansky (1960). The symmetry of the kernel given by expression (1.122) guarantees the real values of the natural frequencies.

(2) *Symmetric sloshing modes*

For symmetric sloshing modes, the solution of equation (1.114) satisfying condition (1.115) takes the form

$$\Phi(\eta, \xi) = M + \int_0^\infty A(\tau) \cosh[\tau(\eta - \eta_0)] \cos(\tau\xi) d\tau \quad (1.123)$$

where M is a constant, which may be obtained by setting the mean level of the free surface to zero. The combined free-surface condition (1.116), after carrying out the Fourier cosine transform and introducing $B(\tau)$, gives,

Table 1.4 Natural frequencies for different values of liquid depth ratio h/R

h/R	Antisymmetric	Symmetric	h/R	Antisymmetric	Symmetric
0.2	1.04385	2.92908	1.2	1.50751	3.21640
	5.35498	8.03025		4.85091	6.46747
	10.76724	13.48837		8.07834	9.68639
	16.1798	18.8477		11.2932	12.8989
0.4	1.09698	2.89054	1.4	1.73463	3.53751
	4.93704	6.99058		5.27678	6.99993
	9.00749	11.00134		8.72206	10.43884
	12.9835	14.9595		12.1571	13.8722
0.6	1.16268	2.88924	1.6	2.12372	4.14328
	4.69867	6.46064		6.13932	8.10314
	8.19875	9.92610		10.08074	12.04189
	11.6490	13.3691		14.0138	15.9749
0.8	1.24606	2.93246	1.8	3.02140	5.62694
	4.60670	6.23613		8.31388	10.90612
	7.85373	9.46499		13.55955	16.15857
	11.07407	12.6813		18.7997	21.4033
1.0	1.35573	3.03310	2.0	2.00612	3.4533
	4.65106	6.23920		5.12530	6.62861
	7.81986	9.39668		8.25995	9.7839
	19.9718	12.5457		11.3982	12.9330

Source: McIver (1989).

$$B(\tau') = \lambda \int_0^{\infty} B(\tau) K_s(\tau, \tau') d\tau, \quad 0 < \tau' < \infty \quad (1.124)$$

where

$$K_s(\tau, \tau') = \frac{I_s(\tau, \tau')}{\sqrt{\tau \tau' \tanh \tau \beta_0 \tanh \tau' \beta_0}} \quad (1.125)$$

and

$$I_s(\tau, \tau') = \frac{2}{\pi} \int_0^{\infty} \frac{\cos(\tau \xi) \cos(\tau' \xi)}{1 + \cosh \xi} d\xi = \frac{\tau - \tau'}{\sinh[\pi(\tau - \tau')]} + \frac{\tau + \tau'}{\sinh[\pi(\tau + \tau')]} \quad (1.126)$$

McIver (1989) estimated the natural frequencies $\omega_{mn}^2 R/g$ corresponding to the lowest four symmetric and four anti-symmetric modes and the results are listed in Table 1.4 The natural frequencies are given for discrete values of liquid depth ratio h/R . McIver employed the Gaussian–Laguerre quadrature formula (see “Toroidal coordinate transformation” later) for discretizing the integral equations.

Spherical container

Conformal mapping With reference to Figure 1.7(a), the cylindrical coordinate frame (r, θ, z) is introduced, where $x = r \cos \theta$, $y = r \sin \theta$, and the plane $x-y$ coincides with the undisturbed surface whose radius is αR . If the linear coordinates r and z are normalized with respect to the radius, one can establish the new nondimensional coordinate frame (ρ, η, θ) , where $\rho = r/\alpha R$, and $\eta = z/\alpha R$. The Laplace equation takes the form

$$\frac{\partial^2 \Phi}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial \Phi}{\partial \rho} + \frac{\partial^2 \Phi}{\partial \eta^2} + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \theta^2} = 0 \quad (1.127)$$

The solution of this equation subject to the free-surface boundary condition (1.96) may be written in the form

$$\Phi(\rho, \eta, \theta) = \sum_{k=1}^{\infty} \Psi(\rho, \eta) \cos k\theta \quad (1.128)$$

The analysis is restricted to the modes associated with $k = 1$, which are induced by lateral accelerations in the x -direction. Budiansky developed an integral equation similar to equation (1.98) by considering the volume of the fluid enclosed by the wetted surface and its reflection about the undisturbed free surface, $\eta = 0$. With reference to Figure 1.7(d), which represents the plane of the undisturbed free surface, a distribution of three-dimensional sinks along the annulus $(\bar{\rho}, \bar{\rho} + d\bar{\rho})$ of strength $2f(\bar{\rho}) \cos \gamma$ per unit area is introduced. The resulting velocity potential is $f(\bar{\rho})\Omega(\rho, \eta, \theta; \bar{\rho})\bar{\rho} d\bar{\rho}$. Just below the plane $\eta = 0$ the velocity potential is $f(\bar{\rho}) \cos \theta$, and the vertical velocity just below that plane is zero everywhere except in the annulus. At $\theta = 0$ the free-surface condition (1.96) gives

$$f(\rho) = \lambda \alpha \int_0^1 \Omega(\rho, 0, 0; \bar{\rho}) \bar{\rho} f(\bar{\rho}) d\bar{\rho} \quad (1.129)$$

where the vertical velocity just below the plane $\eta = 0$ is zero everywhere except in the annulus in which its value is $f(\bar{\rho}) \cos \theta$. Setting $g(\rho) = \sqrt{\rho} f(\rho)$, one can write

$$g(\rho) = \lambda \alpha \int_0^1 H(\rho, \bar{\rho}) g(\bar{\rho}) d\bar{\rho} \quad (1.130)$$

where $\alpha = a/R$ is bounded between the values 0 and 1, a is half the width of the free surface on the x -axis, and

$$H(\rho, \bar{\rho}) = \sqrt{\rho \bar{\rho}} \Omega(\rho, 0, 0; \bar{\rho}) \quad (1.131)$$

For a nearly full tank, the symmetrized region is the entire space and the potential of a sink in the plane $\eta = 0$ is unmodified by the presence of a rigid boundary at $\eta = 0$. Since the potential of a simple sink (source) of strength m is $\phi = m/(4\pi r)$, the potential of a sink of strength 2 at $(\bar{\rho}, 0, \gamma)$ is

$$\phi = \frac{1}{2\pi} \frac{1}{\sqrt{(\rho \cos \theta - \bar{\rho} \cos \gamma)^2 + (\rho \sin \theta - \bar{\rho} \sin \gamma)^2 + \eta^2}} \quad (1.132)$$

Accordingly one may write

$$\Omega(\rho, 0, 0; \bar{\rho}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\cos \gamma \, d\gamma}{\sqrt{\rho^2 + \bar{\rho}^2 - 2\rho\bar{\rho} \cos \gamma}} \quad (1.133)$$

The kernel $H(\rho, \bar{\rho})$ is given by the expression

$$H(\rho, \bar{\rho}) = \frac{\sqrt{\rho\bar{\rho}}}{\pi} \int_0^\pi \frac{\cos \gamma \, d\gamma}{\sqrt{\rho^2 + \bar{\rho}^2 - 2\rho\bar{\rho} \cos \gamma}} \quad (1.134a)$$

Let $\rho < \bar{\rho}$, $\mu = \rho/\bar{\rho}$, $q = 2\sqrt{m}/(m+1)$, the above kernel takes the form

$$H(\mu) = \frac{q}{\pi} \int_0^{\pi/2} \frac{[2 \sin^2(\gamma/2) - 1] \, d(\gamma/2)}{\sqrt{1 - q^2 \sin^2(\gamma/2)}} \quad (1.134b)$$

Evaluating this integral in terms of complete elliptic integrals, gives

$$H = \frac{1}{\pi} \left\{ \left[\frac{2}{q} - q \right] K(q) - \frac{2}{q} E(q) \right\} \quad (1.135)$$

where

$$K(q) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - q^2 \sin^2 \theta}}, \quad E(q) = \int_0^{\pi/2} \sqrt{1 - q^2 \sin^2 \theta} \, d\theta \quad (1.136)$$

are the complete elliptic integrals of the first and second kind, respectively. The kernel (1.135) has a logarithmic singularity at $\rho = \bar{\rho}$.

For a half-full tank, the kernel function is evaluated on the basis of the known Neumann function or Dirichlet function of the second kind for the sphere. With reference to Figure 1.7(e) the function of the position Q is given by the expression

$$N(Q) = \frac{1}{2\pi} \left\{ \frac{1}{\overline{PQ}} + \frac{1}{\overline{PQ} \, \overline{QP'}} + \ln \frac{2}{1 - \overline{OQ} \, \overline{OP} \cos \vartheta + \overline{PQ} \, \overline{QP'}} \right\} \quad (1.137a)$$

This function is harmonic and has a sink of strength 2 at P in the unit sphere. Furthermore, the normal derivative of $N(Q)$ has the constant value of $-(1/2\pi)$ at the surface of the sphere. With reference to Figure 1.7(e), let Q be the point $(\rho, 0, 0)$ in the equatorial plane of the sphere, and let P be the point $(\bar{\rho}, 0, \gamma)$. In terms of these points the position function (1.137a) takes the form

$$N(\rho, 0, 0) = \frac{1}{2\pi} \left\{ \frac{1}{\sqrt{\rho^2 + \bar{\rho}^2 - 2\rho\bar{\rho}\cos\gamma}} + \frac{1}{\sqrt{(\rho\bar{\rho})^2 + 1 - 2\rho\bar{\rho}\cos\gamma}} \right. \\ \left. + \ln \frac{2}{1 - \rho\bar{\rho}\cos\gamma + \sqrt{(\rho\bar{\rho})^2 + 1 - 2\rho\bar{\rho}\cos\gamma}} \right\} \quad (1.137b)$$

and the kernel $H(\rho, \bar{\rho})$ takes the form

$$H(\rho, \bar{\rho}) = \frac{\sqrt{\rho\bar{\rho}}}{\pi} \left\{ \int_0^\pi \frac{\cos\gamma d\gamma}{\sqrt{\rho^2 + \bar{\rho}^2 - 2\rho\bar{\rho}\cos\gamma}} + \int_0^\pi \frac{\cos\gamma d\gamma}{\sqrt{(\rho\bar{\rho})^2 + 1 - 2\rho\bar{\rho}\cos\gamma}} \right. \\ \left. + \int_0^\pi \cos\gamma \left[\ln \frac{2}{1 - \rho\bar{\rho}\cos\gamma + \sqrt{(\rho\bar{\rho})^2 + 1 - 2\rho\bar{\rho}\cos\gamma}} \right] d\gamma \right\} \quad (1.137c)$$

In view of the cosine variation of local sink strength at the radius $\bar{\rho}$, the normal flow at the spherical surface vanishes as expected, although the Neumann function (1.137a) for a single sink produces a constant flux at the surface. The evaluation of the first two integrals in equation (1.137c) follows immediately from the results of a nearly full container. The last integral may be evaluated by setting $s = \rho\bar{\rho}$, and letting

$$I(s) = \int_0^\pi \cos\gamma \left[\ln \frac{2}{1 - s\cos\gamma + \sqrt{s^2 + 1 - 2s\cos\gamma}} \right] d\gamma \\ = - \int_0^\pi \cos\gamma \ln \left(1 - s\cos\gamma + \sqrt{s^2 + 1 - 2s\cos\gamma} \right) d\gamma \quad (1.138)$$

Subtracting the zero expression $\int_0^\pi \cos\gamma \ln s d\gamma = 0$ from (1.138) and differentiating both sides with respect to s , gives

$$I'(s) = \frac{1}{s} \int_0^\pi \frac{\cos\gamma d\gamma}{\sqrt{s^2 + 1 - 2s\cos\gamma}} \quad (1.139a)$$

Equation (1.139a) can be written in terms of complete elliptic integrals (1.136) in the form

$$I'(s) = \frac{2}{s^2} [K(s) - E(s)] \quad (1.139b)$$

Integrating equation (1.139b) with respect to s gives

$$I(s) = \int_0^s \frac{2}{s^2} [K(s) - E(s)] ds = \frac{2}{s} [E(s) - (1 - s^2)K(s)] \quad (1.140)$$

Table 1.5 *Liquid free-surface eigenvalues in spherical tank for different fluid depth*

e	α	$\alpha\lambda_1$	$\alpha\lambda_2$	$\alpha\lambda_3$	λ_1	λ_2	λ_3
-1	0.0	0.0	0.0	0.0	1.0	7.0	17.0
0	1.0	1.565	5.34	8.66	1.565	5.34	8.56
1	0.0	2.78	5.99	9.25	∞	∞	∞

Source: Budiansky (1960).

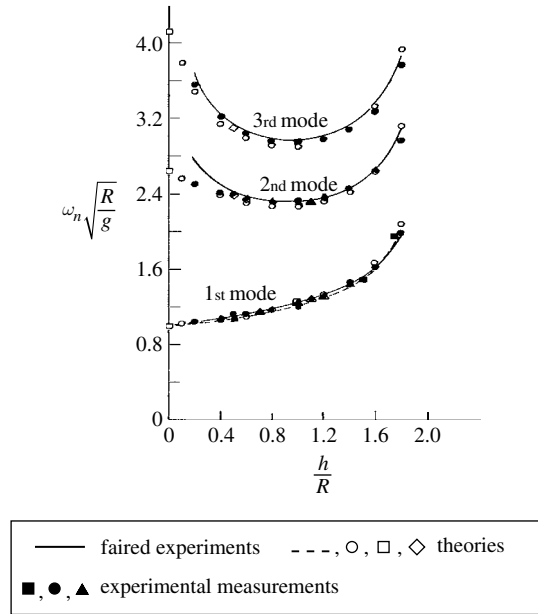


Figure 1.12 Dependence of liquid natural frequencies on depth ratio in a spherical tank as obtained experimentally and analytically. (Abramson, 1966a)

Substituting this result in terms of ρ and $\bar{\rho}$, equation (1.137c) takes the form

$$H(\rho, \bar{\rho}) = \frac{1}{\pi} \left\{ \left(2\sqrt{\bar{\rho}/\rho} \right) [K(\rho/\bar{\rho}) - E(\rho/\bar{\rho})] + 2(\rho\bar{\rho})^{3/2} K(\rho\bar{\rho}) \right\} \quad (1.141)$$

with $\rho < \bar{\rho}$, and $H(\rho, \bar{\rho}) = H(\bar{\rho}, \rho)$.

The solution of the integral equation (1.110) with kernel equations (1.135) and (1.141) was obtained numerically as outlined for the case of the circular canal. The values of λ_n for nearly full ($e = 1$) and half-full ($e = 0$) for the first three modes are listed in Table 1.5. Figure 1.9 shows the asymptotic behavior of $\sqrt{\lambda_n}$ near $e = 1$ by the dot-dash curves. And the values of $\sqrt{\lambda_n}$ for $e = 0$ and 1 are shown as circles.

Abramson (1966a) documented a set of results reported by Mikishev and Dorozhkin (1961), Leonard and Walton (1961), Stofan and Armstead (1962), Abramson, *et al.* (1963), and Chu (1964a). A comparison of theoretical and measured values of the first three natural frequencies is shown in Figure 1.12. Based on the measured natural frequency of the first mode the

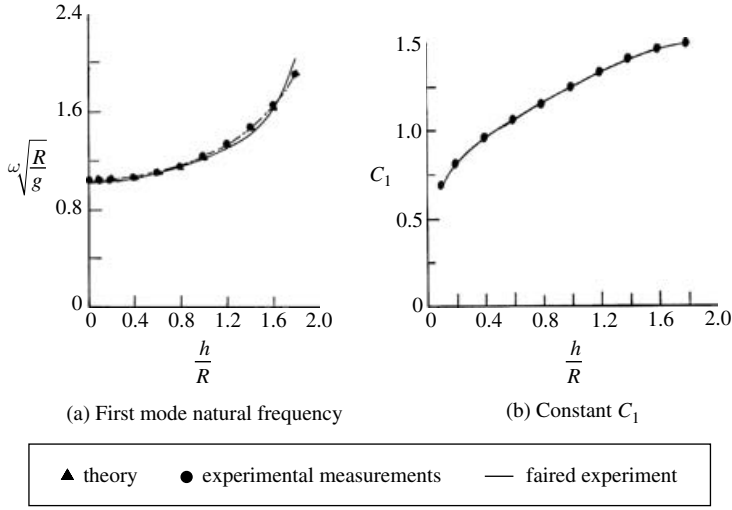


Figure 1.13 Dependence of the liquid first mode natural frequency and constant C_1 on depth ratio. (Abramson, 1966a)

following empirical equations were obtained by Mikishev and Dorozhkin (1961) for certain ranges of the liquid depth parameter h/R

$$\sqrt{\lambda_1} = \frac{\sqrt{(1.84)^2 - (1.46 - (h/R)^2) - 0.56}}{\sqrt{2(h/R) - (h/R)^2}}, \quad 0.1 < h/R < 1.0 \quad (1.142a)$$

$$\sqrt{\lambda_1} = \frac{\sqrt{(1.84)^2 - (1.97 - (h/R)^2) - 0.34}}{\sqrt{2(h/R) - (h/R)^2}}, \quad 1 < h/R < 2.0 \quad (1.142b)$$

$$\sqrt{\lambda_1} = \frac{C_1}{\sqrt{2(h/R) - (h/R)^2}}, \quad 0.1 < h/R < 2.0 \quad (1.142c)$$

Figure 1.13 shows the dependence of the first mode natural frequency, $\sqrt{\lambda_1} = \omega\sqrt{R/g}$, and the constant C_1 in relation (1.142c), on the depth ratio h/R , as measured by McCarty and Stephens (1960), Mikishev and Dorozhkin (1961), and Abramson and Ransleben (1961e). The following empirical formula was obtained by Rattayya (1965) for the first mode

$$\lambda_1 = \frac{1}{3} \frac{240 - 220(h/R) + 72(h/R)^2 - 9(h/R)^3}{80 - 100(h/R) + 44(h/R)^2 - 9(h/R)^3 + (h/R)^4} \sin \left[\frac{\pi}{4} \left(2 - \frac{h}{R} \right) \left(1 + \frac{h}{3R} \right) \right] \quad (1.143)$$

Table 1.6 shows a comparison of the first natural frequency of a liquid free surface in a spherical tank as reported by Abramson (1966a). Riley and Trembath (1961) developed a computer program to compute the sloshing in axially symmetric tanks of a general shape. Their program applies a Rayleigh–Ritz procedure using three deep-tank eigenfunctions and five odd powers of the radius vector to minimize the quantity (Hieatt and Riley, 1959 and Troesch, 1960),

Table 1.6 Comparison of the first natural frequency $\sqrt{\lambda_1}$ of a liquid in a spherical tank as reported by different sources

H/R	Mikishev and Dorozhkin (1961)	Budiansky (1960)	Abramson, <i>et al.</i> (1963)	Chu (1964a)	Rattayya (1965)
0.05	1.02	1.0	1.01		1.01
0.5	1.09	1.06	1.08	1.1	1.1
1.0	1.22	1.25	1.22		1.22
1.8	1.92	2.04	2.03		1.83

Source: Abramson, *et al.* (1963).

$$\int_{\text{Volume}} \int \int (\nabla \Phi)^2 dv / \int_S \int \Phi^2 ds$$

Their algorithm computes the lowest three eigenvalues and the corresponding mode shapes of the free surface.

Toroidal coordinate transformation With reference to Figure 1.7(a), McIver (1989) introduced cylindrical coordinates, (r, θ, z) , where the z -axis is pointed vertically downwards and the origin is at the center of the free-surface circle of radius a . The toroidal coordinates, η, ξ, ϕ , outlined in the appendix to this chapter, are related to the Cartesian coordinates through the following transformations

$$x = a \frac{\sinh \eta \cos \varphi}{\cosh \eta - \cos \xi}, \quad y = a \frac{\sinh \eta \sin \varphi}{\cosh \eta - \cos \xi}, \quad z = a \frac{\sin \xi}{\cosh \eta - \cos \xi} \quad (1.144)$$

$$r + iz = a \tanh[(\eta + i\xi)/2], \quad 0 \leq \eta \leq \infty, \quad -\pi < \xi \leq \pi \quad (1.145)$$

If x is replaced by r , then the geometry represents a vertical cross-section containing the axis of the spherical tank. The circle of intersection of the container with the free surface is at $\eta = \infty$ and the z -axis coincides with $\eta = 0$. It is obvious that the free surface coincides with the coordinate surface $\xi = 0$ and the container wall with the coordinate surface $\xi = \xi_0$, $0 < \xi \leq \pi$. The fluid depth ratio h/R is related to ξ_0 by equation (1.113) but in toroidal coordinates. The boundary conditions are the same as those given by equations (1.115) and (1.116) for the horizontal cylinder. Note that Laplace's equation in toroidal coordinates (see the appendix to this chapter) is not fully separable. The general solution of Laplace's equation with azimuthal wave number $m = 0, 1, 2, \dots$ is

$$\begin{aligned} \Phi(\eta, \xi, \varphi) &= \cos m\varphi \sqrt{\cosh \eta + \cos \xi} \\ &\times \int_0^\infty [\bar{A}(\tau) \cosh \tau \xi + \bar{B}(\tau) \sinh \tau \xi] P_{-\frac{1}{2}+ir}^m(\cosh \eta) d\tau \end{aligned} \quad (1.146)$$

where $P_\nu^\mu(\cosh \eta)$ is the associated Legendre function of the first kind with degree μ , order ν , and argument $\cosh \eta$. For the limiting case of a fluid-filled half-space bounded by a solid plane

containing a circular aperture, $\xi_0 = \pi$, further conditions are necessary to ensure solutions having no flow to infinity. Henrici, *et al.* (1970) indicated that there is no such flow provided

$$\int_S \int \Phi \, ds = \int_0^{2\pi} \int_0^\infty \frac{\Phi a^2 \sinh \eta}{(1 + \cosh \eta)^2} d\eta d\varphi = 0, \quad \beta = 0 \quad (1.147)$$

where S is the free-surface area. For $m \neq 0$, condition (1.147) is satisfied trivially because of the form of variation in φ . For $m = 0$, the behavior of the gradient of the velocity potential for large $R = \sqrt{r^2 + z^2}$, that is, $(\eta, \xi) = (0, \pi)$, may be obtained from equation (1.146) using the approximate form of relation (1.145), that is,

$$\eta + i(\xi - \pi) = \frac{2a}{r + iz} \quad (1.148)$$

McIver (1989) found that the condition (Henrici, *et al.*, 1970)

$$|\nabla \Phi| = O(R^{-2}), \quad R \rightarrow \infty \quad (1.149)$$

is violated unless the function $\bar{A}(\tau)$ satisfies a condition determined by substituting equation (1.146) into equation (1.147). This process gives

$$\int_0^\infty \frac{\bar{A}(\tau) \, d\tau}{\sinh \tau \pi} = 0 \quad (1.150)$$

where the integral

$$\int_0^\infty \frac{P_{-\frac{1}{2}+ir}(\cosh \eta) \sinh \eta}{(1 + \cosh \eta)^{3/2}} d\eta = \frac{2^{3/2} \pi}{\sinh \tau \pi} \quad (1.151)$$

is obtained from equation 7.135(2) of Gradshteyn and Ryzhik, 1980. Condition (1.150) is satisfied by taking

$$\bar{A}(\tau) = A(\tau) + \sqrt{2}M \operatorname{sech} \tau \pi \quad (1.152)$$

where $M = -2^{5/2} \int_0^\infty \frac{A(\tau) \, d\tau}{\sinh \tau \pi}$ is a constant. Substituting equation (1.152) into equation (1.146) gives

$$\begin{aligned} \Phi(\eta, \xi, \varphi) = & M \delta_{m0} + \cos m\varphi \sqrt{\cosh \eta + \cos \xi} \\ & \times \int_0^\infty [A(\tau) \cosh \tau \xi + B(\tau) \sinh \tau \xi] P_{-\frac{1}{2}+ir}^m(\cosh \eta) \, d\tau \end{aligned} \quad (1.153)$$

where δ_{mn} is the Kronecker delta. The introduction of M might be interpreted as shifting the mean free surface by a prescribed amount to ensure that it coincides with $\xi = 0$ and thus condition (1.147) is satisfied. Substituting equation (1.153) into the wall boundary condition (1.115), gives

$$\begin{aligned}
& \int_0^\infty \tau [A(\tau) \sinh \tau \xi_0 + B(\tau) \cosh \tau \xi_0] P_{-\frac{1}{2}+ir}^m(\cosh \eta) d\tau \\
&= \frac{1}{2} \frac{\sin \xi_0}{\cosh \eta + \cos \xi_0} \int_0^\infty [A(\tau) \cosh \tau \xi_0 + B(\tau) \sinh \tau \xi_0] P_{-\frac{1}{2}+ir}^m(\cosh \eta) d\tau \quad (1.154)
\end{aligned}$$

Substituting equation (1.153) into the free-surface boundary condition (1.116) gives

$$\begin{aligned}
& \int_0^\infty \tau B(\tau) P_{-\frac{1}{2}+ir}^m(\cosh \eta) d\tau + \frac{\lambda}{(1 + \cosh \eta)^{3/2}} \\
& \times \left[-2^{5/2} \delta_{mn} \int_0^\infty \frac{\tau A(\tau)}{\sinh \tau \pi} d\tau + \sqrt{1 + \cosh \eta} \int_0^\infty A(\tau) P_{-\frac{1}{2}+ir}^m(\cosh \eta) d\tau \right] = 0 \quad (1.155)
\end{aligned}$$

McIver (1989) applied the Mehr–Fock transform of order m (see Sneddon, 1972) to equations (1.154) and (1.155). The transform of equation (1.154) is

$$\begin{aligned}
A(\tau') \sinh \tau' \xi_0 + B(\tau') \cosh \tau' \xi_0 &= \frac{1}{2} \tanh \tau' \pi \sin \xi_0 \\
&\times \int_0^\infty [A(\tau) \cosh \tau \xi_0 + B(\tau) \sinh \tau \xi_0] g_m(\tau) I_m(\xi_0; \tau, \tau') d\tau \quad (1.156)
\end{aligned}$$

The transform of equation (1.155) is, after using relation (1.151),

$$B(\tau') = -\lambda \tanh \tau' \pi \int_0^\infty A(\tau) g_m(\tau) \left[I_m(0; \tau, \tau') - \delta_{mn} \frac{16\tau\tau'}{\sinh \tau \pi \sinh \tau' \pi} \right] d\tau \quad (1.157)$$

where (see Gradshteyn and Ryzhik, 1980)

$$g_m(\tau) = \left(\tau^2 + \frac{1}{4} \right) \left(\tau^2 + \frac{9}{4} \right) \dots \left(\tau^2 + \frac{1}{4} (2m-1)^2 \right). \quad (1.158a)$$

$$I_m(\xi_0; \tau, \tau') = \int_0^\infty \frac{P_{-\frac{1}{2}+ir}^{-m}(\cosh \eta) P_{-\frac{1}{2}+ir}^{-m}(\cosh \eta) \sinh \eta}{\cosh \eta + \cos \xi_0} d\tau \quad (1.158b)$$

$$P_{-\frac{1}{2}+ir}^m(\cosh \eta) = (-1)^m g_m(\tau) P_{-\frac{1}{2}+ir}^{-m}(\cosh \eta) \quad (1.158c)$$

Introducing the new functions

$$C(\tau) = [A(\tau) \cosh \tau \xi_0 + B(\tau) \sinh \tau \xi_0] \sqrt{\frac{g_m(\tau) \tanh \tau \xi_0}{\tanh \tau \pi}} \quad (1.159a)$$

$$D(\tau) = A(\tau) \sqrt{\frac{g_m(\tau) \tanh \tau \xi_0}{\tanh \tau \pi}} \quad (1.159b)$$

Equations (1.156) and (1.157) take the form

$$C(\tau') \coth^2 \tau' \xi_0 - D(\tau') \coth \tau' \xi_0 \operatorname{cosech} \tau' \xi_0 = \frac{1}{2} \sin \xi_0 \int_0^\infty C(\tau) H_m(\xi_0; \tau, \tau') d\tau \quad (1.160)$$

$$C(\tau') \coth \tau' \xi_0 \operatorname{cosech} \tau' \xi_0 - D(\tau') \coth^2 \tau' \xi_0 = -\lambda \int_0^\infty D(\tau) L_m(\xi_0; \tau, \tau') d\tau \quad (1.161)$$

where

$$H_m(\xi_0; \tau, \tau') = G_m(\tau) G_m(\tau') I_m(\xi_0; \tau, \tau') \quad (1.162a)$$

$$L_m(\xi_0; \tau, \tau') = G_m(\tau) G_m(\tau') \left[I_m(0; \tau, \tau') - \delta_{m0} \frac{16\tau\tau'}{\sinh \tau\pi \sinh \tau'\pi} \right] \quad (1.162b)$$

$$G_m(\tau) = \sqrt{\frac{g_m(\tau) \tanh \tau\pi}{\tanh \tau\xi_0}} \quad (1.162c)$$

The natural frequencies are determined by estimating the values of $\lambda = \omega^2 a/g$ for which the coupled integral equations (1.160) and (1.161) possess nontrivial solution for the functions $C(\tau)$ and $D(\tau)$. McIver (1989) adopted an algorithm for solving the integral equations by using an appropriate quadrature rule and obtained a matrix eigenvalues problem. The integrals in equations (1.160) and (1.161) are approximated using the Gauss–Laguerre quadrature formula in the form

$$\int_0^\infty f(\tau) d\tau \approx \sum_{i=1}^N w_i f(\tau_i) \quad (1.163)$$

where w_i , $i = 1, 2, \dots, N$, are adjusted weights containing the factor $e^{2\pi\tau}$ to account for the exponential decay of the integrands with large τ . This approximation yields the matrix equations

$$\mathbf{SC} - \mathbf{TD} = \delta \mathbf{HC} \quad (1.164)$$

$$\mathbf{TC} - \mathbf{SD} = -\lambda \mathbf{LD} \quad (1.165)$$

where $\delta = \frac{1}{2} \sin \xi_0$. The elements of the diagonal matrices $\mathbf{S}$ and $\mathbf{T}$ are

$$S_{ii} = \coth^2 \tau_i \xi_0, \quad T_{ii} = \coth \tau_i \xi_0 \operatorname{cosech} \tau_i \xi_0 \quad (1.166a, b)$$

The elements of the matrices $\mathbf{H}$ and $\mathbf{L}$ are

$$H_{ij} = w_j H_m(\xi_0; \tau_i, \tau_j), \quad L_{ij} = w_j L_m(\xi_0; \tau_i, \tau_j) \quad (1.167a, b)$$

The elements of the column vectors $\mathbf{C}$ and $\mathbf{D}$ are

$$C_i = C(\tau_i), \quad D_i = D(\tau_i) \quad (1.168a, b)$$

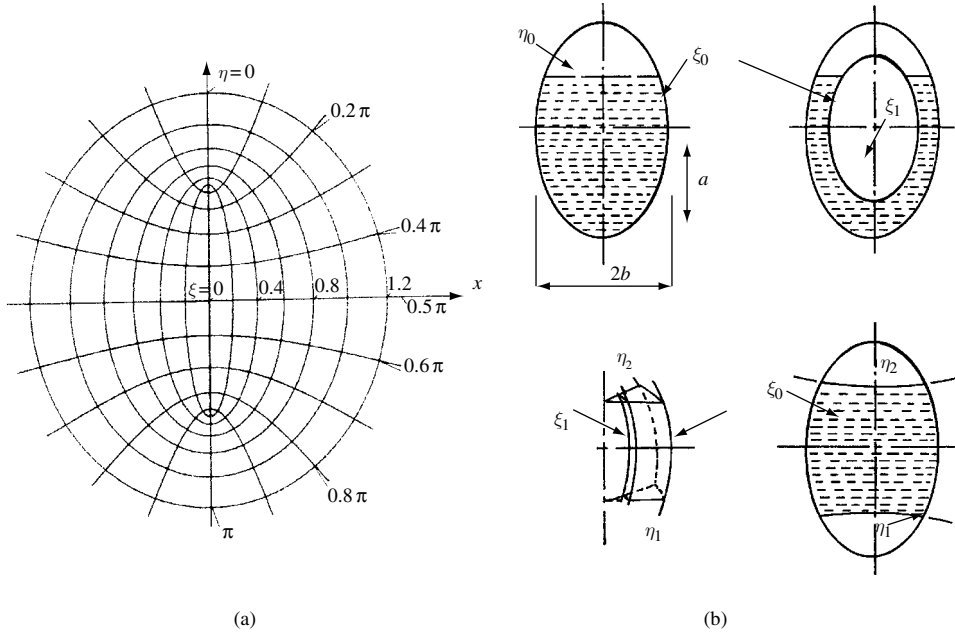


Figure 1.14 (a) Geometry of prolate container showing the coordinate system (b) four different geometries of prolate spheroidal containers.

Eliminating the vector $\mathbf{C}$ from equations (1.164) and (1.165) gives the matrix eigenvalues equation

$$[\mathbf{T}[\delta\mathbf{H} - \mathbf{S}]^{-1}\mathbf{T} + \mathbf{S}]\mathbf{D} = \lambda\mathbf{LD} \quad (1.169)$$

The elements of matrices $\mathbf{H}$ and $\mathbf{L}$ depend on the integral (1.158b), which can be estimated numerically. Table 1.7 lists the numerical values of the natural frequency parameter $\omega_{ij}^2 R/g$ for the first four azimuthal wave numbers $m = 0, 1, 2, 3$ and the first lowest modes. The natural frequencies are estimated for discrete values of fluid depth ratio h/R .

1.6.3 Prolate spheroidal tank

Prolate containers are usually subdivided by annular and radial walls and different configurations as shown in Figure 1.14. They also have the advantage of a remarkable reduction in the dynamic contribution of sloshing loads, higher values of liquid natural frequencies, and smaller liquid mass participating in the liquid motion. Bauer and Eidel (1989b) examined the free oscillations of the free surface in an annular prolate spheroidal sector. A limiting case is the slim ellipse, which resembles a special case of a cylinder.

With reference to the appendix to this chapter and Figure 1.14(a), which shows the prolate spheroidal coordinates ξ , η , and ϕ , the surfaces $\xi = \xi_0$ and $\xi = \xi_1$ represent the annular walls such that any particular value of ξ gives a measure of the semi-axes ratio of the ellipse, b/a , while $\phi = 0$ and $\phi = 2\pi\alpha$ are sector walls. Note that as $\xi \rightarrow 1$ the prolate spheroid approaches a regular spherical container. The bottom may be represented by the surface $\eta = \eta_1$ while the

Table 1.7 Natural frequency parameter $\omega_{ij}^2 R/g$ for spherical container for the first four azimuthal wave numbers $m=0, 1, 2, 3$ and the first four lowest modes

h/R	$m=0$	$m=1$	$m=2$	$m=3$
0.2	3.82612	1.07232	2.10792	3.12949
	9.25613	6.20081	8.39523	10.48832
	14.75561	11.88212	14.2944	16.58021
	20.118	17.3589	19.8090	22.1585
0.4	3.70804	1.15826	2.23491	3.28209
	7.91895	5.67422	7.42178	9.05998
	11.94118	9.85513	11.6525	13.36738
	15.9077	13.8685	15.6994	17.4637
0.6	3.65014	1.26251	2.38767	3.46642
	7.26596	5.36832	6.88669	8.31214
	10.74498	8.94181	10.5082	12.0064
	14.1964	12.4233	14.0217	15.5636
0.8	3.65836	1.39239	2.57671	3.6961
	6.98858	5.24058	6.65230	7.98544
	10.23113	8.55088	10.1555	11.4186
	13.4553	11.7995	13.2950	14.7386
1.0	3.74617	1.56016	2.81969	3.99416
	6.97636	5.27555	6.65941	7.97281
	10.14748	8.50444	9.94129	11.31996
	13.3042	11.6835	13.1499	14.5669
1.2	3.9381	1.7881	3.1492	4.4031
	7.2188	5.4930	6.9145	8.2674
	10.452	8.7793	10.251	11.665
	13.673	12.021	13.521	14.971
1.4	4.3010	2.1232	3.6336	5.0123
	7.8006	5.973	7.509	8.971
	11.2559	9.476	11.058	12.579
	14.698	12.938	14.547	16.104
1.6	5.0075	2.6864	4.4512	6.0547
	9.01565	6.9571	8.7390	10.433
	12.9748	10.956	12.782	14.536
	16.9191	14.916	16.769	18.562
1.8	6.76418	3.9593	6.3155	8.4631
	12.1139	9.4535	11.858	14.137
	17.396	14.755	17.210	19.565
	22.657	20.022	22.509	24.912
2.0	4.1213	2.7547	4.1213	5.4000
	7.3421	5.8922	7.3421	8.7183
	10.5171	9.0328	10.517	11.941
	13.6773	12.174	13.677	15.129

Source: McIver (1989).

undisturbed free surface is at $\eta = \eta_0$. Note that the hyperbolas are selected to provide a description for possible bottom curved shapes.

For incompressible and irrotational flow, the continuity equation is (see the appendix to this chapter),

$$\begin{aligned} \nabla^2 \Phi = & \frac{1}{c^2 (\sinh^2 \xi + \sin^2 \eta) \sinh \xi} \frac{\partial}{\partial \xi} \left(\sinh \xi \frac{\partial \Phi}{\partial \xi} \right) \\ & + \frac{1}{c^2 (\sinh^2 \xi + \sin^2 \eta) \sin \eta} \frac{\partial}{\partial \eta} \left(\sin \eta \frac{\partial \Phi}{\partial \eta} \right) + \frac{1}{c^2 \sinh^2 \xi \sin^2 \eta} \frac{\partial^2 \Phi}{\partial \phi^2} = 0 \end{aligned} \quad (1.170a)$$

Introducing the new variables $\bar{\xi} = \cosh \xi$, $\bar{\eta} = \cos \eta$, $a/2 = c$, $1 \leq \bar{\xi}$, $-1 \leq \bar{\eta} \leq 1$ and their relations to the x, y, z coordinates:

$$x = \frac{a}{2} \sqrt{(\bar{\xi}^2 - 1)(1 - \bar{\eta}^2)} \cos \phi, \quad y = \frac{a}{2} \sqrt{(\bar{\xi}^2 - 1)(1 - \bar{\eta}^2)} \sin \phi, \quad \text{and} \quad z = \frac{a}{2} \bar{\xi} \bar{\eta}$$

Laplace's equation can be written in terms of the new variables

$$\nabla^2 \Phi = \frac{4}{a^2 (\bar{\xi}^2 - \bar{\eta}^2)} \left[\frac{\partial}{\partial \bar{\xi}} (\bar{\xi}^2 - 1) \frac{\partial \Phi}{\partial \bar{\xi}} + \frac{\partial}{\partial \bar{\eta}} (1 - \bar{\eta}^2) \frac{\partial \Phi}{\partial \bar{\eta}} + \frac{\bar{\xi}^2 - \bar{\eta}^2}{(\bar{\xi}^2 - 1)(1 - \bar{\eta}^2)} \frac{\partial^2 \Phi}{\partial \phi^2} \right] \quad (1.170b)$$

The surface, $\bar{\xi} = \text{const}$, is a prolate spheroid with interfocal distance a . The major axis is $a\bar{\xi}/2$, and the minor axis is $(a\sqrt{\bar{\xi}^2 - 1})/2$. The surfaces, $\bar{\eta} = \text{const}$, are two sheets of a hyperboloid of revolution with foci at $z = \pm a/2$ asymptotic to the cone which has its generating line at an angle $\theta = \cos^{-1} \bar{\eta}$. The surface $\phi = \text{const}$ is a plane through the vertical axis at an angle ϕ to the x, z plane.

Equation (1.170a) must be solved subject to the following boundary conditions

$$\left. \frac{\partial \Phi}{\partial \xi} \right|_{\xi=\xi_0, \xi_1} = 0 \quad \text{at the annular container walls} \quad (1.171a)$$

$$\left. \frac{\partial \Phi}{\partial \phi} \right|_{\phi=0, 2\pi} = 0 \quad \text{at the container sector wall} \quad (1.171b)$$

$$\left. \frac{\partial \Phi}{\partial \eta} \right|_{\eta=\eta_1} = 0 \quad \text{at the tank bottom} \quad (1.171c)$$

The combined free-surface condition, which is obtained from the unsteady Bernoulli equation and the kinematic condition, is

$$\frac{\partial^2 \Phi}{\partial t^2} - \frac{g \cosh \xi \sin \eta}{c (\sinh^2 \xi + \sin^2 \eta)} \frac{\partial \Phi}{\partial \eta} = 0 \quad \text{at } \eta = \eta_0 \quad (1.171d)$$

The presence of the coordinate ξ in the free-surface condition (1.171d) creates some difficulty. However, for a small value of ξ , one can set $\cosh \xi \approx 1$ and $\sinh \xi \approx 0$. In this case, condition (1.171d) takes the approximate form

$$\frac{\partial^2 \Phi}{\partial t^2} - \frac{g}{c \sin \eta} \frac{\partial \Phi}{\partial \eta} = 0 \quad (1.171e)$$

Now, the solution of Laplace's equation (1.170b) may be obtained using the separation of variables in the form

$$\Phi(\bar{\xi}, \bar{\eta}, \phi) = F(\phi)Z(\bar{\xi})H(\bar{\eta}) \quad (1.172)$$

where

$$\frac{d^2 \Phi}{d\phi^2} = -m^2 \Phi \quad (1.173a)$$

$$\frac{d}{d\bar{\eta}} \left[(1 - \bar{\eta}^2) \frac{dH}{d\bar{\eta}} \right] + n(n+1)H - \frac{m^2}{(1 - \bar{\eta}^2)} H = 0, \quad -1 \leq \bar{\eta} \leq 1 \quad (1.173b)$$

$$\frac{d}{d\bar{\xi}} \left[(1 - \bar{\xi}^2) \frac{dZ}{d\bar{\xi}} \right] + n(n+1)Z - \frac{m^2}{(1 - \bar{\xi}^2)} Z = 0, \quad \bar{\xi} \geq 1 \quad (1.173c)$$

The solutions of equation (1.173a), for periodic boundary conditions on ϕ , are in the form of $\cos m\phi$ and $\sin m\phi$ (if the tank is not sectorized), where m is zero or a positive integer. Equations (1.173b) and (1.173c) are solved by using the spherical harmonics P_n^m and their second solutions Q_n^m , defined as follows

$$P_n^m(x) = \frac{(1 - x^2)^{m/2}}{2^n n!} \frac{d^{m+n}}{dx^{m+n}} (x^2 - 1)^n, \quad m, n = 0, 1, 2, \dots, \quad m \leq n \quad (1.174a)$$

$$Q_n^m(x) = (-1)^m (x^2 - 1)^{m/2} \frac{d^m}{dx^m} Q_n(x) \quad (1.174b)$$

$$Q_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left[(x^2 - 1)^n \ln \left(\frac{x+1}{x-1} \right) \right] - \frac{1}{2} P_n(x) \ln \left(\frac{x+1}{x-1} \right) \quad (1.174c)$$

For the function H to be finite in equation (1.173b), n must be zero or a positive integer and H must be proportional to the Legendre function of the first kind, $P_n^m(\bar{\eta})$. The variable $\bar{\xi}$ goes from $+1$, along the line between the two foci, to infinity. There is no solution which is finite over the entire range for most values of n and m . For the motion of fluid inside this type of container the solution for the potential function must satisfy the boundary conditions (1.171).

For a simple prolate spheroidal tank with no truncation or walls, the velocity potential function in terms of the coordinates ξ, η, ϕ takes the form, after dropping condition (1.171b),

$$\Phi(\xi, \eta, \phi, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} A_{mn} \exp(i\omega_{mn}t) P_{\lambda_{mn}}^m(\cosh \xi) P_{\lambda_{mn}}^m(-\cos \eta) \cos(m\phi) \quad (1.175)$$

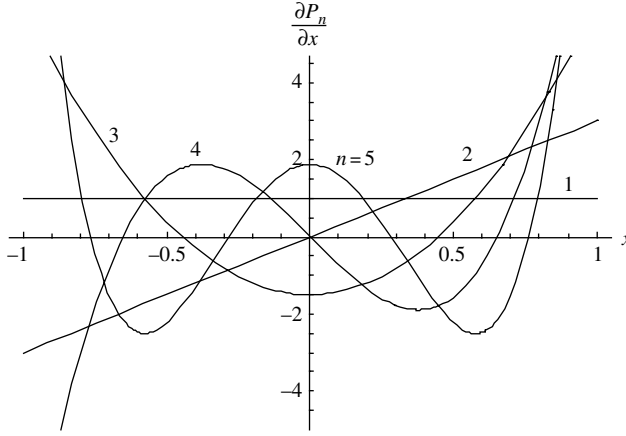


Figure 1.15 Plot of $\partial P_n(x)/\partial x$ showing the zero roots.

In order to determine the natural frequency we introduce equation (1.175) into equation (1.171d). The corresponding approximate natural frequency is

$$\omega_{mn}^2 = -\frac{g}{c} \frac{P_{\lambda_{mn}}^{\prime m}(-\cos \eta_0)}{P_{\lambda_{mn}}^m(-\cos \eta_0)} \quad (1.176)$$

where λ_{mn} are the roots of $P_{\lambda}^{\prime m}(\cosh \xi_0) = 0$. The zeros of this function are obtained graphically using MATHEMATICA by plotting the following expression

$$\frac{\partial P_n(x)}{\partial x} = \frac{-nP_{n-1}(x) + nxP_n(x)}{-1+x^2} = 0 \quad (1.177)$$

Figure 1.15 shows a plot for the derivative of this function for $n = 1, 2, \dots, 5$.

For a sectoried tank, one may assume the following solution (see Bauer and Eidel, 1989b)

$$\Phi(\xi, \eta, \phi, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} A_{mn} \exp(i\omega_{mn}t) L_{mn}(\xi) \bar{L}_{mn}(\eta) \cos(m\phi/2\alpha) \quad (1.178)$$

where ω_{mn} is the natural frequency of the sloshing mode mn ,

$$L_{mn}(\xi) = P_{\lambda_{mn}}^{m/2\alpha}(\cosh \xi) Q_{\lambda_{mn}}^{\prime m/2\alpha}(\cosh \xi_0) - P_{\lambda_{mn}}^{\prime m/2\alpha}(\cosh \xi_0) Q_{\lambda_{mn}}^{m/2\alpha}(\cosh \xi) \quad (1.179a)$$

$$\bar{L}_{mn}(\eta) = P_{\lambda_{mn}}^{m/2\alpha}(\cosh \eta) Q_{\lambda_{mn}}^{\prime m/2\alpha}(\cosh \eta_1) - P_{\lambda_{mn}}^{\prime m/2\alpha}(\cosh \eta_1) Q_{\lambda_{mn}}^{m/2\alpha}(\cosh \eta) \quad (1.179b)$$

Functions $P_{\lambda_{mn}}^{m/2\alpha}$ and $Q_{\lambda_{mn}}^{m/2\alpha}$ are the associated Legendre functions of the first and second kind, and λ_{mn} are the roots of $L_{mn}(\xi_1) = 0$, which are obtained numerically. The constants A_{jk} are obtained from the initial conditions. In order to determine the natural frequencies of the free surface, equation (1.178) is introduced into the free-surface condition (1.171d)

$$\begin{aligned}
& - \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \omega_{mn}^2 A_{mn} \exp(i\omega_{mn}t) L_{mn}(\xi) \bar{L}_{mn}(\eta) \cos(m\phi/2\alpha) \\
& = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{g \cosh \xi \sin \eta_0}{c(\sinh^2 \xi + \sin^2 \eta_0)} A_{mn} \exp(i\omega_{mn}t) L_{mn}(\xi) \\
& \quad \times \left(\frac{\partial \bar{L}_{mn}(\eta)}{\partial \eta} \Big|_{\eta=\eta_0} \right) \cos(m\phi/2\alpha)
\end{aligned} \tag{1.180}$$

The right-hand side is expanded into a series of functions L_{mn} using the orthogonality property

$$\int_{\xi_0}^{\xi_1} L_{mn}(\xi) L_{mk}(\xi) \sinh \xi \, d\xi = \begin{cases} 0 & \text{if } n \neq k \\ S_{mn} & \text{if } n = k \end{cases} \tag{1.181}$$

where

$$S_{mn} = \frac{\sinh \xi_1 L_{mn}(\xi_1) (\partial^2 L_{mn}(\xi_1) / \partial \lambda_{mn} \partial \xi) - \sinh \xi_0 L_{mn}(\xi_0) (\partial^2 L_{mn}(\xi_0) / \partial \lambda_{mn} \partial \xi)}{2\lambda_{mn} + 1}.$$

The result of the expansion yields an infinite hierarchy set of linear homogenous equations, and the natural frequencies are evaluated by solving the determinant:

$$\det \left| \gamma_{n\ell} \left(m, \frac{g}{c}, \xi_0, \xi_1, \eta_0, \eta_1 \right) + \omega_{mn}^2 \delta_{n\ell} \right| = 0 \tag{1.182}$$

where $m = 1, 2, \dots$, $\ell = 1, 2, \dots$, $j = 0, 1, 2, \dots$, $\delta_{n\ell}$ is the Kronecker symbol, and

$$\gamma_{n\ell} \left(m, \frac{g}{c}, \xi_0, \xi_1, \eta_0, \eta_1 \right) = \frac{g \sin \eta_0 \bar{L}'_{m\ell}(\eta_0)}{c L_{jk}(\eta_0)} \frac{\int_{\xi_0}^{\xi_1} \frac{\cosh \xi \sinh \xi}{(\sinh^2 \xi + \sin^2 \eta_0)} L_{mn}(\xi) L_{m\ell}(\xi) \, d\xi}{\int_{\xi_0}^{\xi_1} L_{mn}^2(\xi) \sinh \xi \, d\xi} \tag{1.183}$$

The dependency on the coordinate ξ may be neglected for $0 < \xi < \xi_0 \ll 1$. In this case one can obtain the approximate expression for the natural frequency

$$\omega_{mn}^2 = - \frac{g}{c} \frac{\bar{L}'_{mn}(\eta_0)}{\sin \eta_0 L_{mn}(\eta_0)} \tag{1.184}$$

The dependence of the natural frequency ratio, $\omega_{jk}/\sqrt{g/c}$, on the coordinate η_0/π , which represents a fluid depth parameter, and for the sloshing modes $jk = (0, 1), (0, 2), (1, 1)$ and $(1, 2)$ is shown in Figures 1.16(a,b) for $b/a = 0.2$ and 0.5 , respectively. With reference to Figure 1.16(a) the dashed curves represent the natural frequencies as estimated using the simplified expression given by equation (1.184). The solid curves represent the enhanced results for the natural frequencies by expanding all functions that depend on ξ into series of associated Legendre functions in the free-surface condition (1.171d). This method is lengthy and yields an infinite hierarchy determinant, which is truncated to order three. Another set of results

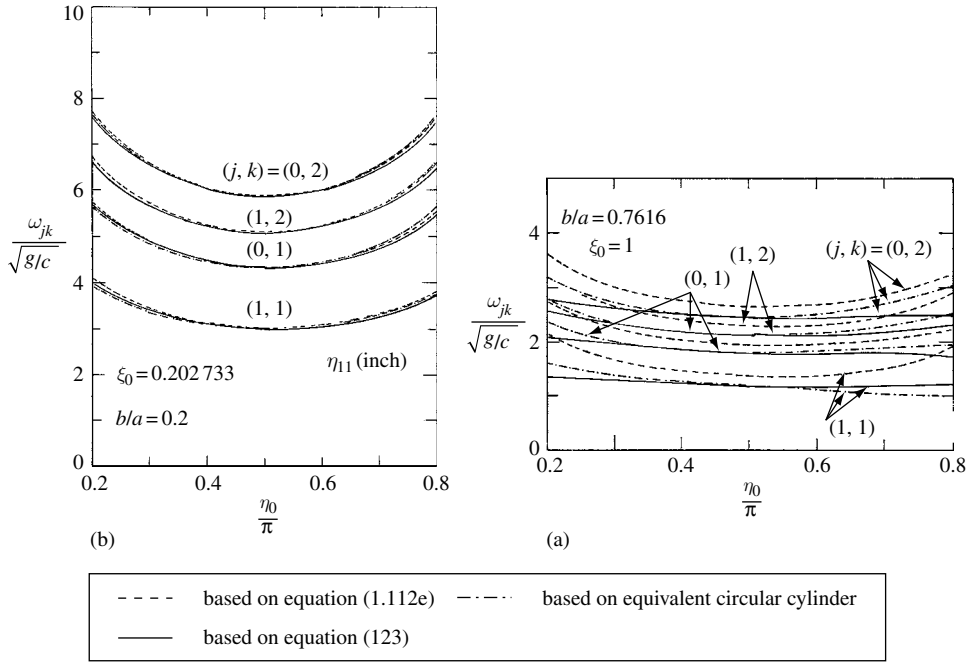


Figure 1.16 Dependence of liquid natural frequencies on height coordinate in prolate spheroidal tank. (Bauer and Eidel, 1989b)

shown by dash-dot curves in Figures 1.16 was obtained by employing the natural frequency expression of an equivalent circular cylindrical container whose radius is

$$R_e = c \cosh \xi_0 \sin \eta_0$$

and liquid depth

$$h_e = h[1 + (4a - h)/(2a - h)]/6$$

The equivalent natural frequency is given in the form

$$\omega_{jk}^2 = \frac{g \varepsilon_{jk}}{R_e} \tanh(\varepsilon_{jk} h_e / R_e) \quad (1.185)$$

The equivalence is taken on the basis of equal free-surface area and equal fluid volume. The values of ε_{jk} are the zeros of the derivative of Bessel functions of the first kind and j th-order, that is, $J'_j(\varepsilon_{jk}) = 0$. It is seen that all methods agree very well in the vicinity of $\eta_0 = \pi/2$ where the actual free surface is close to the coordinate plane η_0 at which the free-surface condition has been satisfied. For large and small values of η_0 , the deviation is noticed but not significant. Another feature is that as the liquid depth decreases, that is, by increasing η_0 , the frequency decreases to depth level $\eta_0 = \pi/2$, above which it increases again. Note that the natural frequency for a relatively larger fluid depth, for example, $\eta_0 = 0.2\pi$, is larger than for a smaller fluid depth, as may be compared with $\eta_0 = 0.8\pi$. The mode $j = 1$ is associated with a nodal line at $\pi/2$ and $3\pi/2$. A similar trend is also observed in Figure 1.16(b) for $b/a = 0.7616$, but there is a

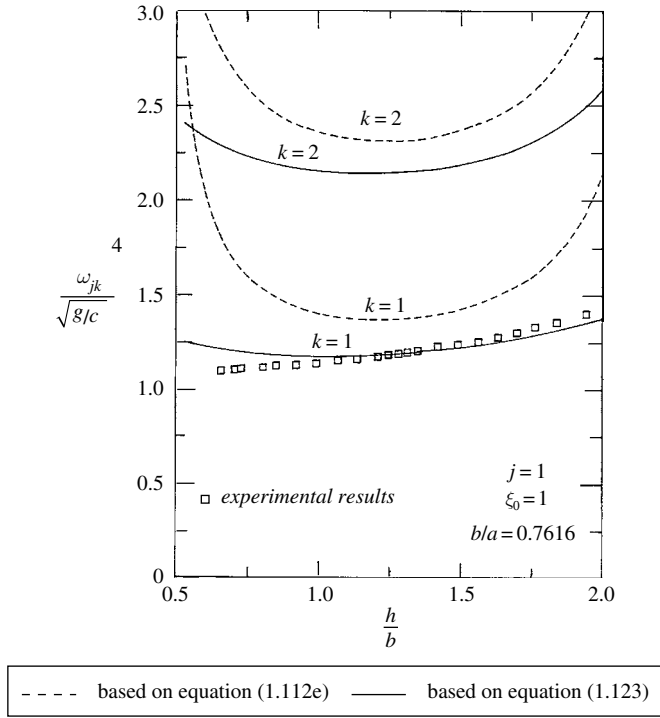


Figure 1.17 Dependence of liquid natural frequencies on depth ratio in prolate spheroidal tank. (Bauer and Eidel, 1989b)

drop in the natural frequencies. Figure 1.17 shows another set of results including those measured experimentally as a function of the fluid depth ratio h/b for $b/a = 0.7616$, that is, $\xi_0 = 1.0$. The measured results agree very well with the equivalent cylinder expression given by relation (1.185).

1.6.4 Oblate spheroidal tank

Geometrically, oblate spheroidal tanks are flattened spheres and, depending on the direction of excitation, they can assume different orientations, referred to as longitudinal, transverse, and horizontal, as shown in Figure 1.18. Leonard and Walton (1961) experimentally measured the natural frequencies and mode shapes in spheroidal tanks of different orientations. The dependence of the measured natural frequencies of the first three sloshing modes on the fluid depth ratio, $h/2b$ or $h/2a$, is shown in Figure 1.19 for the three tank orientations.

For the horizontal orientation case, the measured results were compared with the natural frequencies of a liquid in an upright circular cylinder of radius, r , equal to the radius of the liquid surface and a liquid depth $h_c = h(3b - h)/[3(2b - h)]$, which yields the same liquid volume. For this case, the natural frequency is given by an expression similar to relation (1.43a), that is,

$$\omega_n = \sqrt{\frac{g\xi_n}{r} \tanh(\xi_n h_c / r)} \quad (1.186)$$

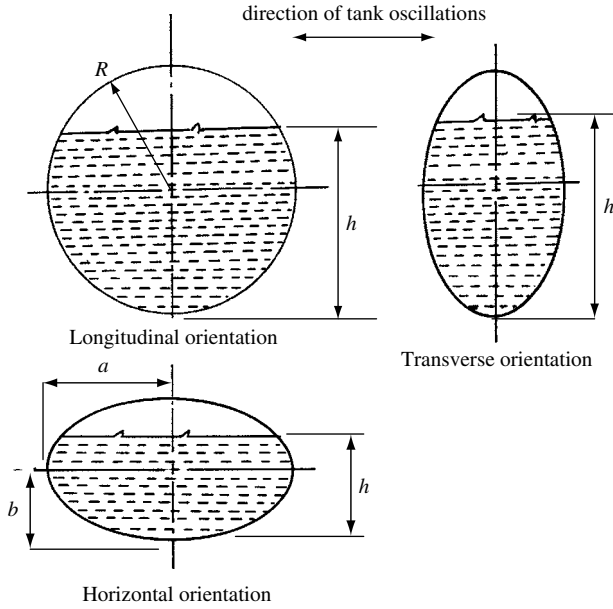


Figure 1.18 Orientation and dimensions of spheroidal tanks.

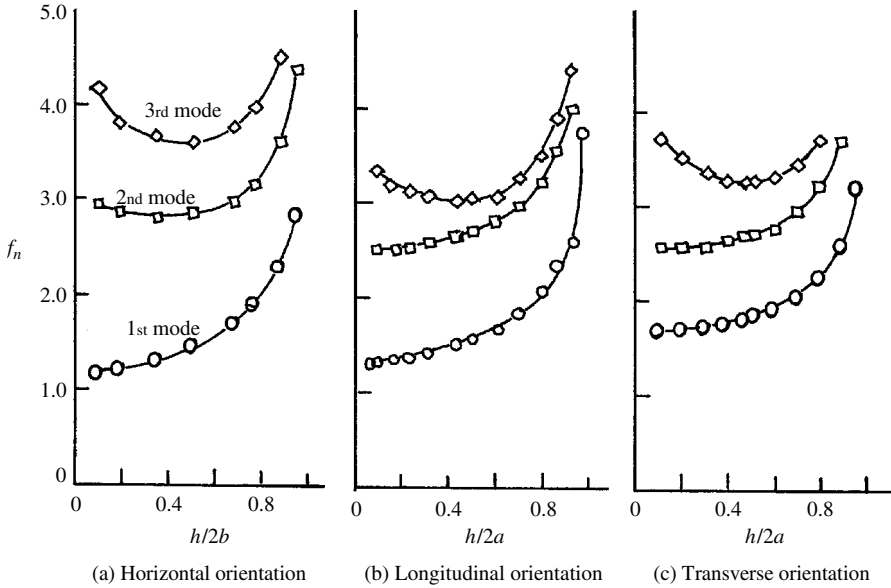


Figure 1.19 Dependence of liquid natural frequencies on depth ratio for three orientations of a spheroidal tank. (Leonard and Walton, 1961)

where ξ_n is the n th zero of $dJ_1(\xi_n \tilde{r}/r)/d\tilde{r}|_{\tilde{r}=r} = 0$. Leonard and Walton (1961) plotted the ratio $\sigma_n = \omega_{n(\text{measured})}/\omega_n$ versus the fluid depth ratio $h/2b$ and the results are shown in Figure 1.20. Any deviation from unity reflects the accuracy of the equivalent analytical expression given by relation (1.186). It is seen that the deviation is significant for the first mode particularly as the fluid depth

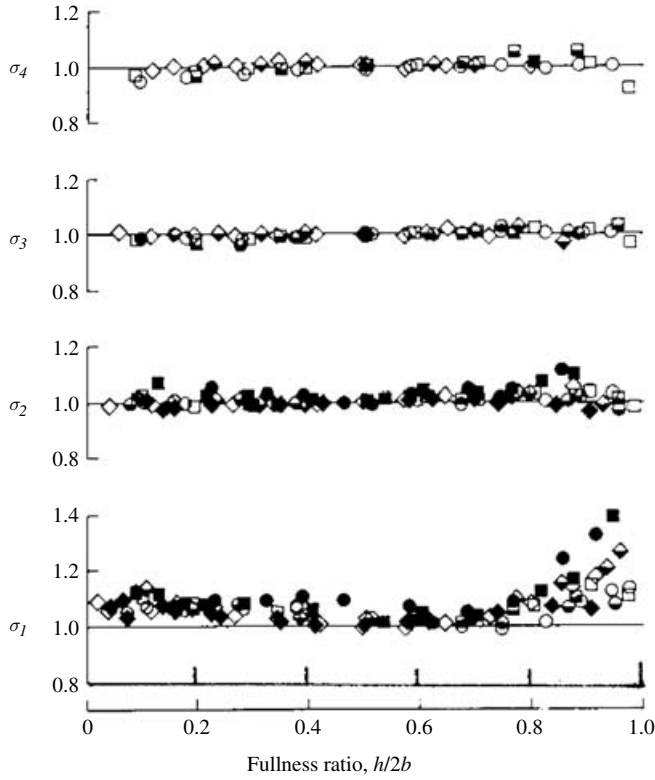


Figure 1.20 Dependence of the frequency parameter $\sigma_n = \omega_n / \sqrt{\frac{r}{g} \xi_n \tanh(\xi_n h_c / r)}$ on the filling ratio in spheroids. (Leonard and Walton, 1961)

ratio exceeds 0.8. Rattayya (1965) calculated the natural frequencies and mode shapes in horizontally oriented oblate spheroidal tanks using a variational approach. Figure 1.21 shows the dependence of the frequency parameter $\omega \sqrt{a/g}$ on the fluid depth ratio for the first three modes and for different values of the ratio of the ellipse axes b/a as estimated analytically by Rattayya.

Mikishev and Dorozhkin (1961) developed the following two empirical formulas for the liquid-free-surface natural frequencies in longitudinal and transverse orientations, respectively

$$\omega_{n,l} = \sqrt{\frac{g}{r} k_{l,n} \tanh(h_c k_{l,n} / r)} \quad (1.187)$$

$$\omega_{n,t} = \sqrt{\frac{g}{r} k_{t,n} \tanh(h_c k_{t,n} / r)} \quad (1.188)$$

where r is the radius of the free surface, $h_c = (h/3)(3a - h)/(2a - h)$ and the parameters $k_{l,n}$ and $k_{t,n}$ are proportional to the positive parametric zeros of the first derivative of the Mathieu function (see, for example, Chu, 1960a, and Abramowitz and Segun, 1968). Approximate values for the first mode parameters $k_{l,1}$ and $k_{t,1}$ are plotted in Figure 1.22. Figures 1.23 and 1.24 show the dependence of the natural frequency parameter on the fluid depth ratio for different values of the ellipse axes ratio b/a , for longitudinal and transverse orientations, respectively.

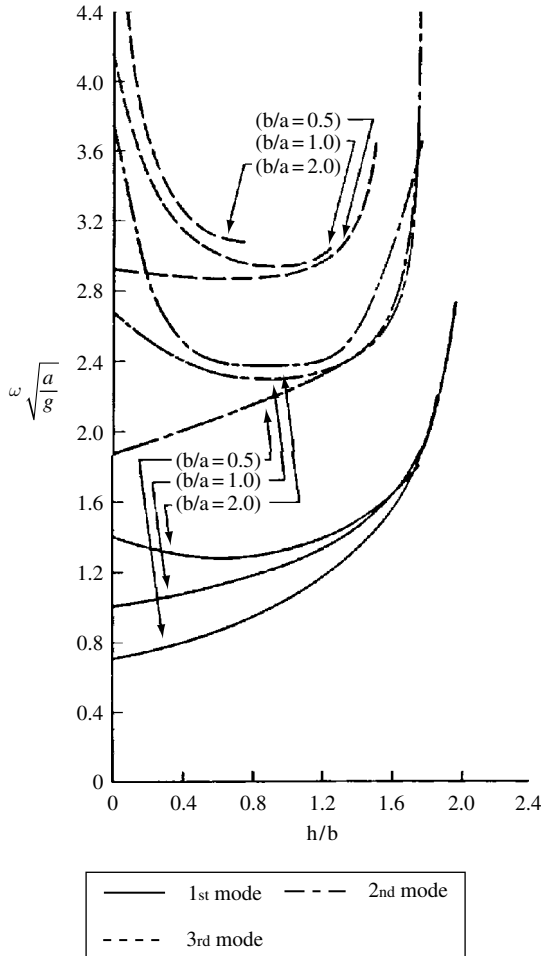


Figure 1.21 Dependence of sloshing frequency on the liquid depth in oblate spheroidal tanks. (Rattayya, 1965)

1.6.5 Conical container

Closed form solutions for the case of a conical container have been obtained by Green (1957, 1959), Green and Landau (1957), Levin (1957, 1963), and Bauer (1982i). It was noted that when the tank is very deep, the bottom shape has no influence on the liquid-free-surface motion and the bottom can be treated as being flat, based on equal liquid volumes (Abramson and Ransleben, 1961a). On the other hand, for extremely shallow liquid depths, the bottom shape governs the fluid motion and the problem is reduced to being two-dimensional. The two limiting cases were analyzed by Lawrence, *et al.* (1958) using a variational approach to obtain an approximate solution for tanks of intermediate depth.

First mode analysis

Simple formulations to estimate the first mode natural frequency in a conical container were developed by Troesch (1960) and Levin (1963). For the case of a conical container whose vertex

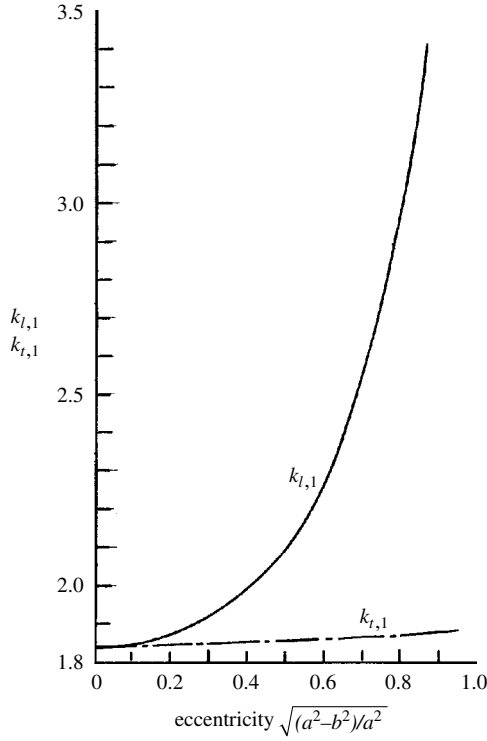


Figure 1.22 Dependence of the parameters $k_{l,1}$ and $k_{t,1}$ on the eccentricity. (Leonard and Walton, 1961)

is below the liquid free surface, see Figure 1.25, a possible solution of the velocity potential function in terms of cylindrical coordinates may be written in the form

$$\Phi = \exp(i\omega t)F(r, z) \cos m\theta \quad (1.189)$$

In this case, Laplace's equation takes the form

$$r^2 \frac{\partial^2 F}{\partial r^2} + r \frac{\partial F}{\partial r} + r^2 \frac{\partial^2 F}{\partial z^2} - m^2 F = 0, \quad z < 0 \text{ and } r > 0 \quad (1.190)$$

This equation should be solved subject to the boundary conditions

$$\frac{\partial F}{\partial n} = 0 \quad \text{on the container wall} \quad (1.191)$$

$$\frac{\partial F}{\partial z} - \frac{\omega^2}{g} F = 0 \quad \text{on the free surface} \quad (1.192)$$

For the case of free-surface motion with one nodal diameter, that is, $m = 1$, Troesch (1960) used a general series expansion and chose the following form for F that satisfies equation (1.190)

$$F(r, z) = r + Arz \quad (1.193)$$

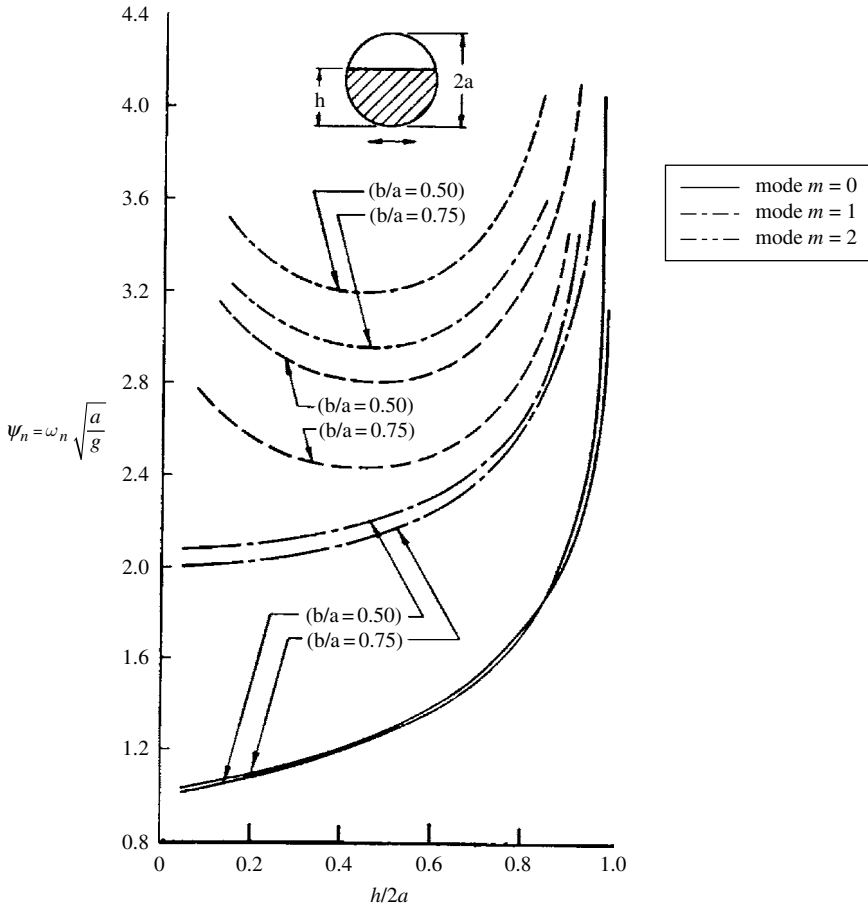


Figure 1.23 Dependence of frequency parameter with depth ratio in spheroids with longitudinal orientation.

Substituting equation (1.193) into (1.192) gives $A = \omega^2/g$. The shape of the container wall belonging to the function F should be compatible with the ordinary differential equation

$$\frac{\partial F}{\partial z} dr - \frac{\partial F}{\partial r} dz = 0 \quad (1.194)$$

Substituting equation (1.193) into equation (1.194), and integrating yields the shape of the container

$$r = \pm \left(z + \frac{g}{\omega^2} \right) \quad (1.195)$$

where the constant of integration is set to zero. Equation (1.195) represents the shape of a 45° cone with a depth of $h = g/\omega^2$, and radius $R = g/\omega^2$ at the free surface, ($z = 0$). In this case, the frequency of the free surface is

$$\omega = \sqrt{g/h} \quad (1.196)$$

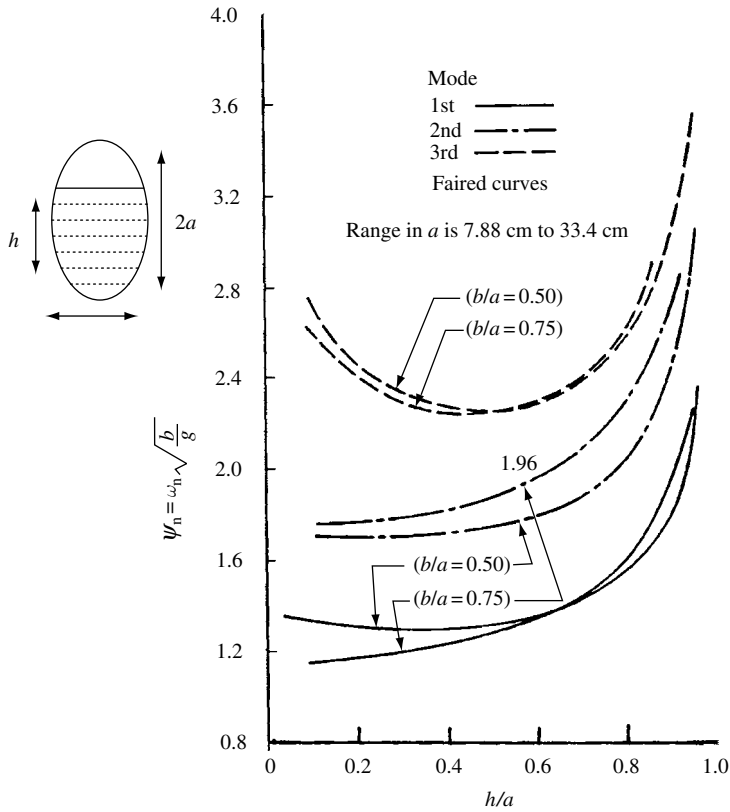


Figure 1.24 Dependence of liquid frequency parameter on the depth ratio in spheroids with transverse orientation. (Leonard and Walton, 1961)

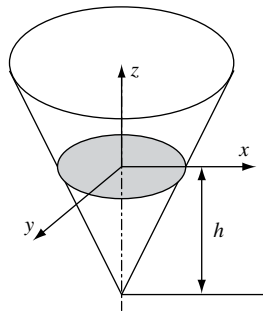


Figure 1.25 Coordinate system of a conical tank.

Levin (1963) obtained the same result by using a direct approach based on the geometry of the container and the appropriate coordinate system. Lawrence, *et al.* (1958) developed a variational approach based on Hamilton's variational formulation to obtain approximate estimation of sloshing frequencies and mode shapes of axially symmetrical containers. This approach is described in the next section.

Variational approach

It is possible to follow the variational method outlined in Section 1.3. The fluid wave height η can be expressed in terms of the velocity potential function using the linearized Bernoulli equation $\eta = \frac{1}{g} \frac{\partial \Phi}{\partial t}$. Upon substituting this expression in equation (1.27), and performing the variation gives

$$0 = \delta \int_{t_1}^{t_2} dt \left\{ - \int_v dv \nabla^2 \Phi \delta \Phi + \int_{S_w} dS_w \frac{\partial \Phi}{\partial n} \delta \Phi + \int_{S_f} dS_f \left(\frac{\partial \Phi}{\partial n} + \frac{1}{g} \frac{\partial^2 \Phi}{\partial t^2} \right) \delta \Phi \right\} - \frac{1}{g} \int_{S_f} dS_f \frac{\partial \Phi}{\partial t} \delta \Phi \Big|_{t_1}^{t_2} \quad (1.197)$$

Note that the fluid density ρ has been absorbed in the definition of the Lagrangian $L = T - V$. Furthermore, the last term in the above equation vanishes at the end points of the time interval. For the variation δI to vanish for all admissible variations of the potential function Φ the following equations can be extracted from equation (1.197)

$$\nabla^2 \Phi = 0 \quad (1.198a)$$

$$\frac{\partial \Phi}{\partial n} = 0 \quad \text{on the wetted wall} \quad (1.198b)$$

$$\frac{\partial \Phi}{\partial n} + \frac{1}{g} \frac{\partial^2 \Phi}{\partial t^2} = 0 \quad \text{on the free surface} \quad (1.198c)$$

These equations constitute the fluid field equations. The sloshing modes and natural frequencies may be determined as solutions of the extremum problem for the integral I

$$I = \frac{1}{2} \int_{t_1}^{t_2} dt \left\{ \int_v |\nabla \Phi|^2 dv - \frac{1}{g} \int_S \left(\frac{\partial \Phi}{\partial t} \right)^2 ds \right\} \quad (1.199)$$

Now, let the velocity potential function be harmonic in time and given by the expression

$$\Phi(r, z, \theta, t) = F(r, z, \theta) \sin \omega t \quad (1.200)$$

Substituting equation (1.200) into equation (1.199) gives

$$I = \frac{1}{2} \int_v |\nabla F|^2 dv - \frac{\omega^2}{2g} \int_S F^2 dS \quad (1.201)$$

A constant factor $\pi\rho/\omega$ in equation (1.201) has been set to unity as it is absorbed in the Lagrangian. For axially symmetric containers, one may use the cylindrical coordinates and express the potential function in the form

$$F = G(r, z) \cos m\theta \quad (1.202)$$

Substituting (1.202) into equation (1.201) gives

$$I = \int_0^{R_0} r dr \int_{-h}^0 \left(\left(\frac{\partial G}{\partial r} \right)^2 + \left(\frac{\partial G}{\partial z} \right)^2 + \frac{m^2}{r^2} G^2 \right) dz - \frac{\omega^2}{g} \int_0^{R_0} r G^2 dr \quad (1.203)$$

where the constant factor $\pi/2$ has been omitted.

In order to estimate the mode shapes and natural frequencies one may substitute equation (1.200) into equations (1.198) or take the variation of equation (1.201). In both cases the following equations are obtained

$$\nabla^2 F = 0 \quad (1.204a)$$

$$\frac{\partial F}{\partial n} = 0 \quad \text{on the wetted wall} \quad (1.204b)$$

$$\frac{\partial F}{\partial n} - \frac{\omega^2}{g} F = 0 \quad \text{on the free surface} \quad (1.204c)$$

Multiplying each of the above equations by the function F and integrating over the corresponding domains by parts, gives

$$\int_{S_w} F \frac{\partial F}{\partial n} dS_w + \int_{S_f} F \frac{\partial F}{\partial n} dS_f - \int_v |\nabla F|^2 dv = 0 \quad (1.205a)$$

$$\int_{S_w} F \frac{\partial F}{\partial n} dS_w = 0 \quad (1.205b)$$

$$\int_{S_f} F \frac{\partial F}{\partial n} dS_f - \frac{\omega^2}{g} \int_{S_f} F^2 dS_f = 0 \quad (1.205c)$$

Introducing equation (1.205b) into (1.205a) and solving the resulting equation together with equation (1.205c) gives the following expression for the natural frequency

$$\omega^2 = g \int_v |\nabla F|^2 dv / \int_{S_f} F^2 dS_f \quad (1.206)$$

It is not difficult to show that this value is a minimum when the integral expression (1.201c) is a superlative. The natural frequency given by expression (1.206) can be estimated once the potential function F is defined. Lawrence, *et al.* (1958) determined the natural frequencies for two cases of tank filling, the shallow and deep tank fillings.

For the shallow tank, the depth of the fluid is sufficiently smaller than the distance between mode crests, and equation (1.206) can be written in the form

$$\omega^2 = g \frac{\int_{S_f} dS_f \int_{-h}^0 dz \left((\partial F / \partial r)^2 + (\partial F / r \partial \theta)^2 + (\partial F / \partial z)^2 \right)}{\int_{S_f} F^2 dS_f} \quad (1.207)$$

One may assume that the potential function F is independent of z and, thus, it is convenient to denote this special function by $E(r, \theta)$ and the corresponding sloshing frequency by ω_e . In this case, expression (1.207) takes the form

$$\omega_e^2 = gh \frac{\int_{S_f} \left((\partial E / \partial r)^2 + (\partial E / r \partial \theta)^2 \right) dS_f}{\int_{S_f} E^2 dS_f} \quad (1.208)$$

For axially symmetric containers, the potential is a function of r only, $G(r, \theta) = \bar{G}(r)$, and one may express equation (1.203) in the form

$$I = \int_0^{R_0} \left(\left[\left(\frac{\partial \bar{G}}{\partial r} \right)^2 + \frac{m^2}{r^2} \bar{G}^2 \right] - \frac{\omega^2}{g} \bar{G}^2 \right) r dr \quad (1.209)$$

To solve this equation, the function $\bar{G}(r)$ may be represented by the polynomial, (Lamb, 1945)

$$\bar{G}(r) = \sum_{n=0}^N A_n (r/R_0)^{m+2n} \quad (1.210)$$

It is important to note that the presence of m in the exponent is consistent with the fact that when $m \neq 0$ the singularity of equation (1.203) at $r=0$ requires that $\bar{G}(r)$ contain no constant terms. The depth, h , will be represented by a polynomial in the form

$$h = h_0 \sum_{i=0}^K B_i (r/R_0)^i \quad (1.211)$$

The polynomial coefficients A_n are determined such that the integral (1.209) has a stationary value. This condition requires

$$\frac{\partial I}{\partial A_n} = 0, \quad n = 0, 1, 2, \dots, N \quad (1.212)$$

This condition yields a set of $N+1$ simultaneous homogeneous algebraic equations of the general form

$$\sum_{\ell=0}^N \left\{ \left[\sum_{i=0}^K \frac{m^2 + (n+\ell)m + 2n\ell}{m+n+\ell+i/2} k_i \right] - \frac{\beta^2/2}{m+n+\ell+1} \right\} A_\ell = 0 \quad (1.213)$$

where $\beta^2 = \frac{\omega^2 R_0^2}{g h_0}$. These equations are consistent if the following determinant vanishes

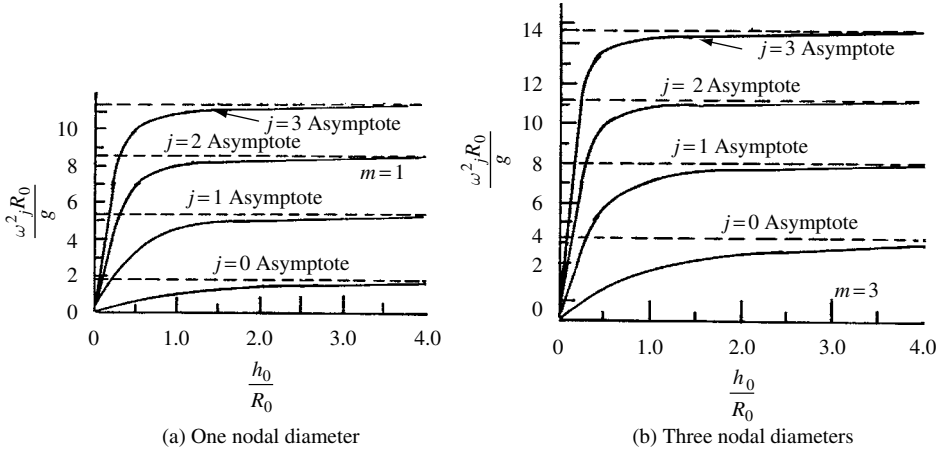


Figure 1.26 Dependence of liquid natural frequencies in a conical tank. (Lawrence, *et al.*, 1958)

$$|D_{n\ell}| = 0 \quad (1.214)$$

where

$$D_{n\ell} = \left[\sum_{i=0}^K \frac{m^2 + (n + \ell)m + 2n\ell}{m + n + \ell + i/2} k_i \right] - \frac{\beta^2/2}{m + n + \ell + 1}$$

The solution of equation (1.214) provides $N + 1$ frequencies. The corresponding mode shapes are given by equations (1.210) and (1.213).

For the case of a conical bottom, the height is described by the expression

$$h = h_0 \left(1 - \frac{r}{R_0} \right) \quad (1.215)$$

For $m = 1$ and 3 (i.e., sloshing modes with one and three nodal diameters), Lawrence, *et al.* (1958) numerically determined the dependence of the first four ($j = 0, 1, 2, 3$) natural frequencies $\omega_j^2 R_0 / g$ on the depth ratio h_0 / R_0 and the results are plotted in Figure 1.26(a, b). Mikishev and Doroshkin (1961) and Dokuchaev (1964) extended the work of Lawrence, *et al.* and determined analytically and experimentally the natural frequencies in upright and inverted cones. For cones with small semi-vertex angles Mikishev and Doroshkin (1961) expressed the natural frequencies by the empirical relationship

$$\frac{\omega_j^2 R_0}{g} = 1.84 C_3^2 \quad (1.216)$$

where the constant C_3 depends on the semi-vertex angle and is plotted in Figure 1.27 for the upward and downward cones. This result is valid for $h_0 / R_0 > 2.75$ and liquid oscillation amplitudes less than $0.01 R_0$.

Transformation to conical spherical coordinates

The analysis of free oscillations of a liquid free surface in conical tanks was considered by Bauer (1999) in terms of spherical coordinates, (r, ϑ, φ) , where r emanates from the vertex of

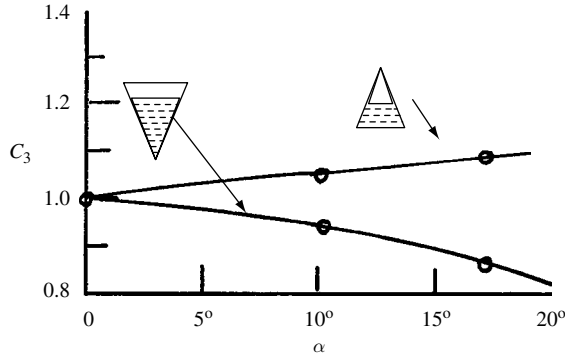


Figure 1.27 Dependence of the constant C_3 on the cone semi-angle. (Mikishev and Dorozhkin, 1961)

the cone (center of the sphere), ϑ is the angular coordinate with the vertical, and φ the angular coordinate around the vertical. For an incompressible and inviscid flow, the liquid flow is governed by the Laplace equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \Phi}{\partial \vartheta} \right) + \frac{1}{r^2 \sin^2 \vartheta} \frac{\partial^2 \Phi}{\partial \varphi^2} = 0 \quad (1.217)$$

The solution of equation (1.217) for conical tanks, whose wall converges downward (∇), may be written in the form

$$\Phi(r, \vartheta, \varphi, t) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{mn} \left(\frac{r}{a} \right)^{\lambda_{mn}} P_{\lambda_{mn}}^m(\cos \vartheta) \cos m\varphi e^{i\omega t} \quad (1.218)$$

where $P_{\lambda_{mn}}^m(\cos \vartheta)$ are associated Legendre functions of the first kind of degree λ and of angular mode m , and a is the radius to the fluid surface. The velocity potential function Φ must satisfy the combined free-surface condition involving the effect of surface tension, at $r = a$

$$\frac{\partial^2 \Phi}{\partial t^2} + g \frac{\partial \Phi}{\partial r} - \frac{\sigma}{\rho a^2} \left[2 \frac{\partial \Phi}{\partial r} + \frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial^2 \Phi}{\partial r \partial \vartheta} \right) + \frac{1}{\sin^2 \vartheta} \frac{\partial^3 \Phi}{\partial \varphi^2 \partial r} \right] = 0 \quad (1.219)$$

For free-liquid oscillations, the velocity normal to wall, $\vartheta = \alpha$, vanishes, that is,

$$\frac{1}{r} \frac{\partial \Phi}{\partial \vartheta} = 0 \quad (1.220)$$

This condition yields

$$\frac{\partial}{\partial \vartheta} \left[P_{\lambda_{mn}}^m(\cos \vartheta) \right] \Big|_{\vartheta=\alpha} = 0 \quad (1.221)$$

The roots λ_{mn} of equation (1.221) depend on the cone vertex angle β and are listed by Bauer (1999) for several cone angles. Samples of these roots are given for $m = (0, 1)$ and $n = (1, 2)$ in Table 1.8. Applying the surface condition (1.219), gives the natural frequencies of the free surface

$$\omega_{nm}^2 = \frac{g \lambda_{mn}}{a} + \frac{\sigma}{\rho a^3} \lambda_{mn} (\lambda_{mn} - 1) (\lambda_{mn} + 2) \quad (1.222)$$

Table 1.8 Zeros of the derivative of the associated Legendre function, $dP_{\lambda_{nm}}^m(\cos \vartheta)/d\vartheta = 0$

α	m	$n = 1$	$n = 2$
$\alpha = 10^\circ$	0	21.4598	39.6995
	1	10.0835	30.0567
$\alpha = 15^\circ$	0	14.1446	26.3023
	2	6.5842	19.8793
$\alpha = 20^\circ$	0	10.4885	19.6044
	1	4.8432	14.7931
$\alpha = 25^\circ$	0	8.2960	15.5864
	1	3.8056	11.7434
$\alpha = 30^\circ$	0	6.8354	12.9083
	2	3.1196	9.7121
$\alpha = 35^\circ$	0	5.7930	10.9958
	1	2.6347	8.2626

Source: Bauer, 1999

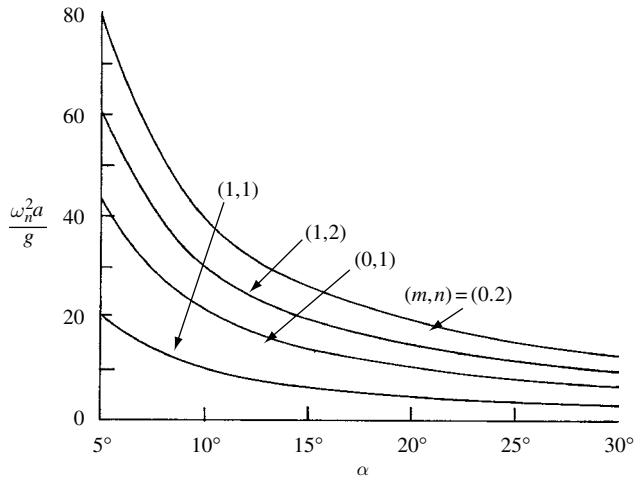


Figure 1.28 Dependence of liquid natural frequencies on vertex angle in a conical tank (∇). (Bauer, 1982i, 1999)

The dependence of the natural frequency ratio $\omega_{nm}^2/(g/a)$ on the cone semi-vertex angle α is shown in Figure 1.28 for $(m, n) = (0, 1), (0, 2), (1, 1)$, and $(1, 2)$. It is seen that as the vertex angle increases the natural frequency decreases.

For inverted conical containers (Δ), the following boundary conditions must be satisfied

$$\frac{\partial \Phi}{\partial r} = 0 \quad \text{at the tank bottom } (r = a) \quad (1.223)$$

At $\vartheta = \alpha$

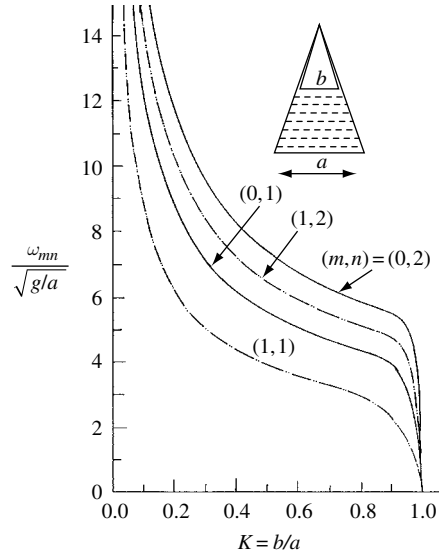


Figure 1.29 Dependence of liquid natural frequencies on the depth ratio. (Bauer, 1982i, 1999)

$$\frac{1}{r} \frac{\partial \Phi}{\partial \vartheta} = 0 \quad (1.224)$$

The combined free-surface condition, ignoring surface tension effect is

$$\frac{\partial^2 \Phi}{\partial t^2} - g \frac{\partial \Phi}{\partial r} = 0 \quad \text{at } r = b \quad (1.225)$$

The solution of Laplace's equation is

$$\begin{aligned} \Phi(r, \vartheta, \varphi, t) = & \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \left[A_{mn} \left(\frac{r}{a} \right)^{\lambda_{mn}} + \frac{\lambda_{mn}}{(\lambda_{mn} + 1)} \left(\frac{r}{a} \right)^{-(\lambda_{mn} + 1)} \right] \\ & \times P_{\lambda_{mn}}^m(\cos \vartheta) \cos m\varphi e^{i\omega t} \end{aligned} \quad (1.226)$$

where λ_{mn} are obtained from $(\partial/\partial\vartheta) [P_{\lambda_{mn}}^m(\cos \vartheta)] \big|_{\vartheta=\alpha} = 0$. The natural frequencies are given by the expression

$$\omega_{mn}^2 = \frac{g\lambda_{mn}}{ak} \frac{(1 - k^{2\lambda_{mn}+1})}{(\lambda_{mn}/(1 + \lambda_{mn})) + k^{2\lambda_{mn}+1}}, \quad k = b/a \quad (1.227)$$

Figure 1.29 shows the dependence of the natural frequency ratio, $\omega_{mn}/\sqrt{g/a}$, on the radii ratio, $k = b/a$, which measures the fluid depth, for the modes $(m, n) = (0, 1)$, $(0, 2)$, $(1, 1)$, and $(1, 2)$, and for semi-cone angle $\alpha = 15^\circ$. As k increases the fluid natural frequencies drop from infinity to zero as $k \rightarrow 1$.

1.6.6 Toroidal containers

Toroidal tanks exist in the form of a ring whose cross-section is circular hollow. Figure 1.30 shows a schematic diagram of a toroidal tank in three possible orientations with respect to a

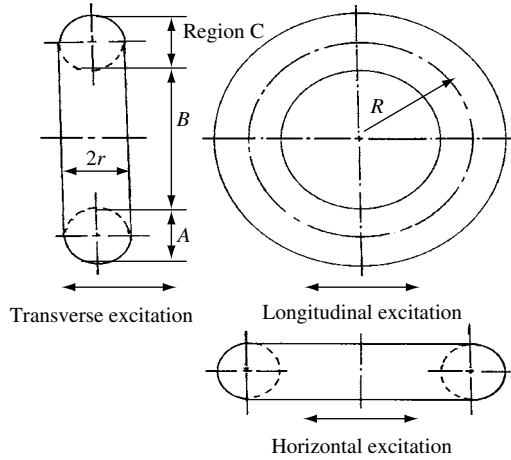


Figure 1.30 Toroidal tank.

given excitation, longitudinal, transverse, and horizontal. This type of container is found to be very convenient for carrying liquid propellant in large space vehicles such as the Space Transportation System (STS). In particular, a reusable liquid hydrogen/liquid oxygen-fueled orbit transfer vehicle (OTV) for boosting large satellites to geosynchronous orbit must have a large volume for the low-density liquid hydrogen fuel. In view of the lack of analytical closed form solutions, McCarty, *et al.* (1960) and Sumner (1963) measured the natural frequencies and mode shapes of liquid for different tank orientations. Meserole and Fortini (1987) and Shi (1987) conducted experimental and analytical studies to derive equivalent models for the slosh modes.

In adopting a nondimensional frequency parameter, certain regular tank shapes that approximate every boundary region of the toroidal tank may be considered. The following are different cases considered by McCarty, *et al.* (1960).

Horizontal modes

A frequency parameter may be taken as the ratio of the measured natural frequency to the natural frequency of a liquid contained in an upright annular circular cylinder having the same inner and outer radii of the toroid, (r_i and r_o), and a liquid depth h_c necessary to produce the same liquid volume in the toroid. This frequency is given by equation (1.60) and should read in the present case as

$$\omega_{ne} = \sqrt{\frac{g}{r_o} \xi_{1n} \tanh(\xi_{1n} h_c / r_o)} \quad (1.228)$$

and the frequency parameter is

$$\lambda_n = \omega_n / \omega_{ne} \quad (1.229)$$

where ω_n is the measured frequency. Figure 1.31 shows the dependence of the frequency parameter λ_n on the fluid depth ratio $h/2r$ for the first four sloshing modes and different values of minor and major radii. It is seen that at a given fluid fullness ratio, the frequency parameter for a given mode is the same for all toroids examined. This implies that the frequency

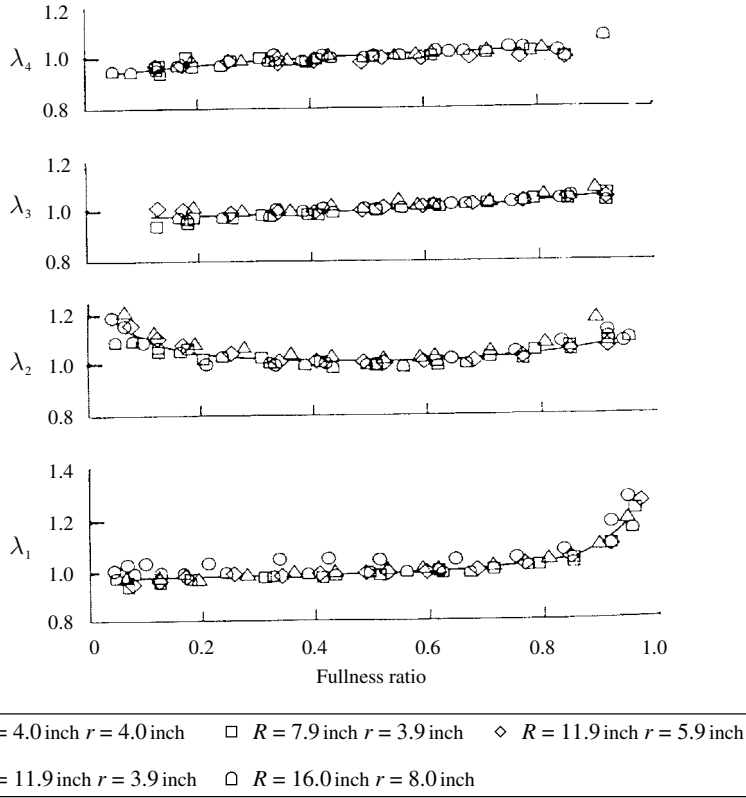


Figure 1.31 Dependence of liquid natural frequency parameter $\lambda_n = \omega_n/\omega_{ne}$ on fullness ratio for horizontal toroidal tanks. (McCarty, *et al.*, 1960)

parameter λ_n is independent of the tank dimensions. However, at the near-full and near-empty conditions, the value of λ_n deviates from 1.

Vertical modes

The liquid boundaries will depend on the fluid depth, and McCarty, *et al.* divided the tank into three regions labeled “A” bounded by the region ($0 < h < 2r$), “B” bounded by ($2r < h < 2R$), and “C” given by ($2R < h < 2(R + r)$), as show in Figure 1.30.

Under transverse disturbance, the liquid in region A or C has boundaries that are somewhat similar to spheres or horizontal cylinders undergoing transverse oscillations. In this case, the following frequency parameter can be used

$$\sigma_n = \omega_n \sqrt{r/g} \quad (1.230)$$

In region B, the liquid boundary is similar to that of an upright circular cylinder, and the corresponding frequency is similar to relation (1.43a) and should read as

$$\omega_{ne} = \sqrt{\frac{g\xi_{1n}}{r} \tanh(\xi_{1n}h/r)}, \text{ or } \sqrt{\xi_{1n}g/r} \text{ as } \tanh(\xi_{1n}h/r) \rightarrow 1 \quad (1.231)$$

where the symbols follow equation (1.43a) but with radius r .

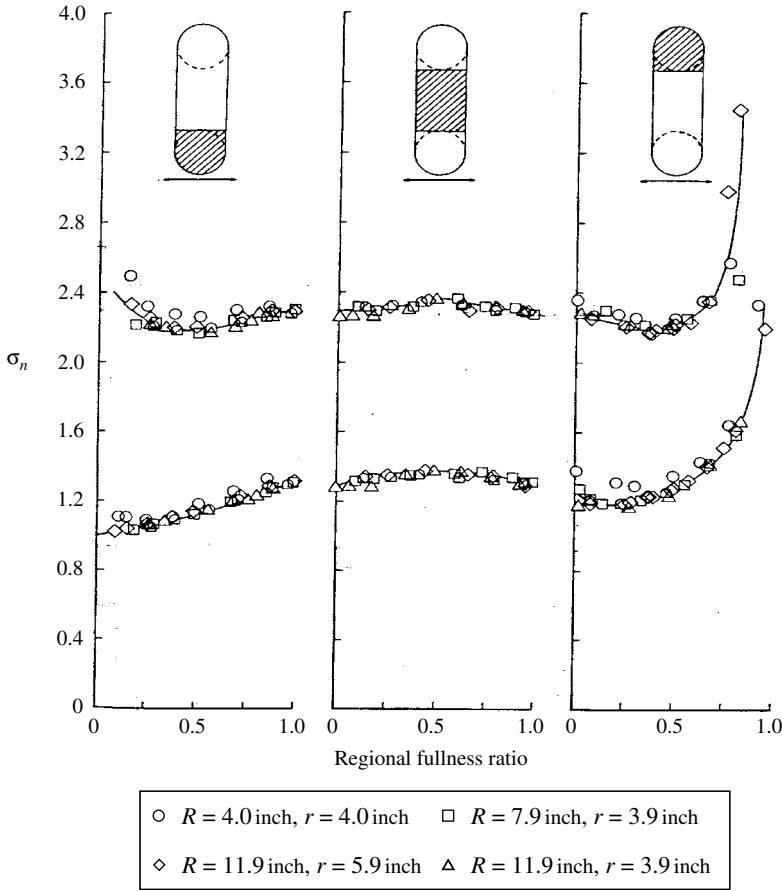


Figure 1.32 Dependence of liquid natural frequency parameter $\sigma_n = \omega_n \sqrt{r/g}$ on fullness ratio for transverse modes of vertical toroidal tanks. (McCarty, *et al.*, 1960)

Figure 1.32 shows the dependence of the frequency parameter, σ_n , on the regional fluid fullness ratio [region “A” $h/2r$, $(h - 2r)/2(R - r)$, region “C” $(h - 2R)/2r$] for the first two modes.

Under longitudinal disturbance, the frequency parameter in region A or C is taken as the ratio of the measured natural frequency to the frequency of a simple pendulum whose length equals the distance from the center of the toroid to the center of mass of a mass distributed uniformly along the arc of the peripheral circle subtended by the liquid surface as shown in Figure 1.33. The corresponding frequency of the pendulum is

$$\omega_p = \sqrt{\frac{g\phi}{(R + r) \sin \phi}} \quad (1.232)$$

where ϕ is the angle between the vertical and the radius to the point of intersection of the fluid surface with the peripheral circle. The resulting frequency parameter of the first mode is

$$\psi_1 = \omega_1 / \omega_p \quad (1.233)$$

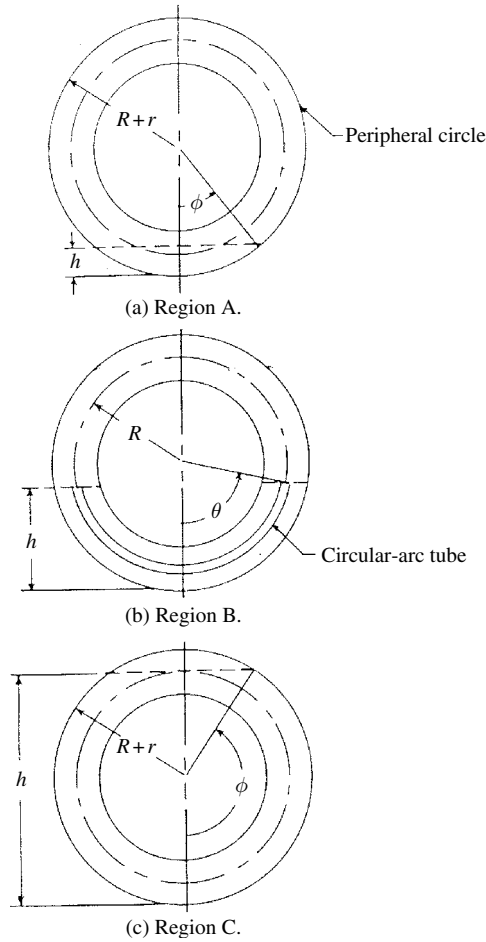


Figure 1.33 Regions of liquid in vertical toroidal tanks for longitudinal sloshing modes.

For higher modes in these two toroidal regions, the frequency parameter can be established through similarity of liquid boundaries in a sphere of radius $R + r$. The resulting parameter takes the form

$$\psi_n = \frac{\omega_n}{\sqrt{g/(R + r)}} \quad (1.234)$$

where ω_1 and ω_n are measured natural frequencies of the first and n th modes, respectively. Figure 1.34(a) shows the dependence of the frequency parameter ψ_n on the fluid regional fullness ratio for the first four modes for region A. The values of this parameter for each of the three regions are plotted separately as a function of the region fullness ratio. It is seen that for a given fullness ratio, only the first mode frequency parameter is independent of the tank size. However, for higher modes, the frequency parameter is dependent on the radii ratio R/r . For a given fullness ratio, as R/r increases the frequency parameter increases as well. Figure 1.31(b) shows the frequency depth relationship for region B for the first four modes. This region is also characterized by the dependence of the frequency parameter on the radii ratio R/r for a given

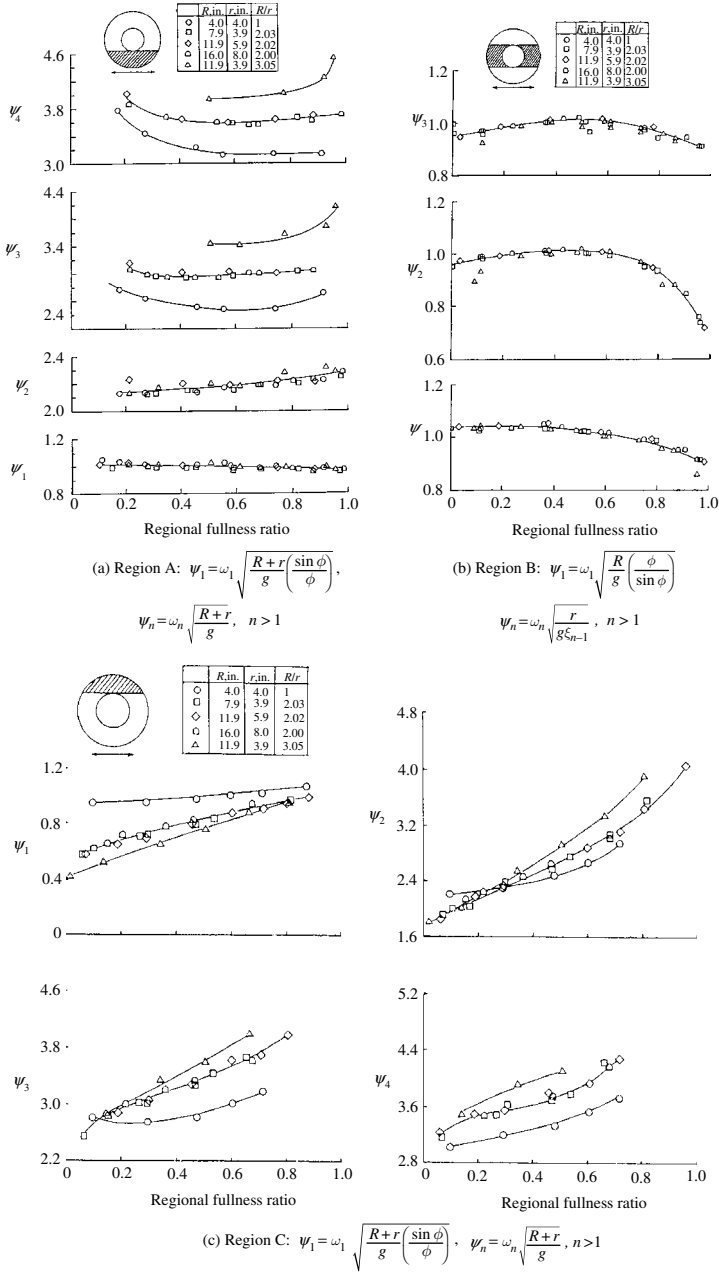


Figure 1.34 Dependence of liquid frequency parameter on the fullness ratio for longitudinal modes of vertical toroidal tank regions. (McCarty, *et al.*, 1960)

fullness ratio. For regions A and C, the frequency parameter is not effective in combining the effects of various tank size and geometry.

For region B, the frequency can be nondimensionalized by using the frequency of the circular arc tube of small cross-section shown in Figure 1.33, that is,

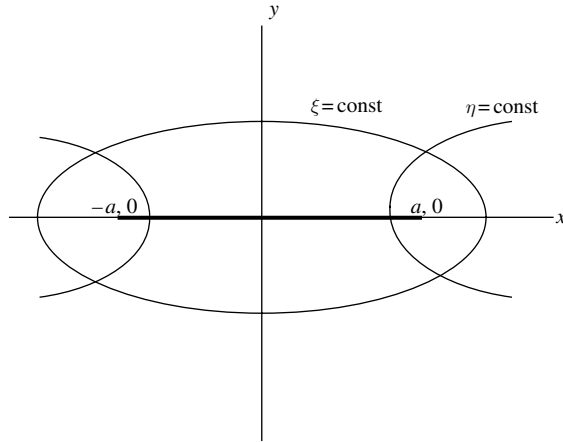


Figure 1.35 Elliptic cylindrical coordinates.

$$\omega_{1e} = \sqrt{\frac{g}{r}} \sqrt{\frac{\sin \theta}{\theta}} \quad (1.235)$$

where θ is the angle between the vertical and the radius R that terminates in the liquid surface. The corresponding frequency parameter is

$$\psi_1 = \omega_1 / \omega_{1e} \quad (1.236)$$

For higher modes, the reference frequency of the liquid in an upright circular cylinder is taken. The order of the analogous cylindrical frequency is decreased by 1 because the manometer-type mode not present in the upright cylinder is considered to be the fundamental mode for this region. It is assumed that the equivalent fluid depth is sufficiently large to ensure that $\tanh(\xi_{1n}h/r) \rightarrow 1$. The corresponding frequency parameter is

$$\psi_n = \omega_n \sqrt{\frac{r}{g\xi_{n-1}}}, \quad n > 1$$

Figure 1.34(b) presents the dependence of the frequency ratio, ψ_n , on the fluid regional fullness ratio for the first three modes for the mid-region of vertical toroids. For a given fullness ratio, the frequency parameter is independent of the tank dimensions.

1.6.7 Upright elliptic containers

Elliptic containers are different to circular ones, because the ellipticity of the container cross-sections affects the natural frequencies of the free liquid surface (Chu, 1960a and Bauer and Knölker, 1981). Bauer (1999) presented a brief analysis of free oscillations of liquid free in elliptic containers. With reference to Figure 1.35, the relationships between elliptic and Cartesian coordinates are (Fogiel, 1994)

$$x = a \cosh \xi \cos \eta, \quad y = a \sinh \xi \sin \eta, \quad z = z \quad (1.237)$$

where $\xi \geq 0$, $0 \leq \eta < 2\pi$, $-\infty < z < \infty$, and at the side wall, $\xi = \xi_0$, the equation of the ellipse takes the form

$$\frac{x^2}{a^2 \cosh^2 \xi_0} + \frac{y^2}{a^2 \sinh^2 \xi_0} = 1 \quad (1.238)$$

The Laplace equation is

$$\nabla^2 \Phi(\xi, \eta, z, t) = \frac{\partial^2 \Phi}{\partial \xi^2} + \frac{\partial^2 \Phi}{\partial \eta^2} + a^2 (\cosh \xi - \cos \eta) \frac{\partial^2 \Phi}{\partial z^2} = 0 \quad (1.239)$$

The velocity potential function may be written in the form

$$\Phi(\xi, \eta, z, t) = \phi(\xi, \eta) \cosh(k(z+h)) e^{i\omega_n t} \quad (1.240)$$

Equation (1.240) satisfies the boundary condition at the tank bottom, $\partial \Phi / \partial z|_{z=-h} = 0$, where k is a factor to be determined from other boundary conditions. Equation (1.239) takes the form

$$\frac{\partial^2 \phi}{\partial \xi^2} + \frac{\partial^2 \phi}{\partial \eta^2} + a^2 k^2 (\cosh \xi - \cos \eta) \phi = 0 \quad (1.241)$$

The solution of (1.241) may be written in the form

$$\phi(\xi, \eta) = Z(\xi) H(\eta) \quad (1.242)$$

Introducing (1.242) into equation (1.241) gives the following ordinary differential equations

$$-\frac{d^2 G(\xi)}{d\xi^2} + (\nu - k^2 a^2 \cosh \xi) G(\xi) = 0 \quad (1.243a)$$

$$\frac{d^2 H(\eta)}{d\eta^2} + (\nu - k^2 a^2 \cos \eta) H(\eta) = 0 \quad (1.243b)$$

Equation (1.243b) is Mathieu's equation, while equation (1.243a) is a modified Mathieu's equation since it has imaginary argument. The solutions of equation (1.243b) are periodic in η since η varies from 0 to 2π . The solutions of equation (1.243a) are not periodic in ξ . Therefore we need the periodic eigenfunctions for η together with two solutions for ξ , one is proportional to the sine and cosine of Mathieu's functions $\text{Se}_m(\eta, ka)$, and $\text{Ce}_m(\eta, ka)$, respectively, and the other solution is independent of $\text{Se}_m(\eta, ka)$, and $\text{Ce}_m(\eta, ka)$. The second solution is given by the radial function of the first kind (Bessel functions) $\text{Je}_m(\xi, ka)$ and the second kind (Neumann functions) $\text{Ne}_{2m}(\xi, ka)$, (Morse and Feshbach, 1953). Two general solutions of equation (1.239) may be written in the form

$$\Phi(\eta, \xi, z, t) = \sum_{m=1}^{\infty} [A_m \text{Se}_m(\eta, ka) + B_m \text{Ce}_m(\eta, ka)] \left\{ \begin{matrix} \text{Je}_m(\xi, ka) \\ \text{Ne}_m(\xi, ka) \end{matrix} \right\} \cosh(k(z+h)) e^{i\omega_n t} \quad (1.244)$$

These solutions must satisfy the boundary condition at the tank wall

$$\left. \frac{\partial \Phi}{\partial \xi} \right|_{\xi=\xi_0} = 0 \quad (1.245)$$

The natural frequency is obtained from the combined free-surface condition

$$\frac{\partial^2 \Phi}{\partial t^2} + g \frac{\partial \Phi}{\partial z} = \Big|_{z=0} 0 \quad (1.246)$$

Bauer (1999) obtained the following expression for the free-surface natural frequency for oscillations in the x -direction

$$\omega_{mn}^2 = \frac{2g}{eR_1} \sqrt{ak_{mn}} \tanh\left(\frac{2\sqrt{ak_{mn}}}{e} \frac{h}{R_1}\right) \quad (1.247a)$$

For oscillations in the y -direction, the natural frequency is given by the expression

$$\omega_{mny}^2 = \frac{2g}{eR_1} \sqrt{a\bar{k}_{mn}} \tanh\left(\frac{2\sqrt{a\bar{k}_{mn}}}{e} \frac{h}{R_1}\right) \quad (1.247b)$$

where $e = \sqrt{R_1^2 - R_2^2}/R_1$ is the ellipse eccentricity, R_1 and R_2 are the major and minor axes of the ellipse, q_{mn} and $\bar{q}_{mn}$ are the roots of the modified Mathieu functions, $\text{Ne}'_m(\xi_0, q_{mn}) = 0$ and $\text{Ne}'_m(\xi_0, a\bar{k}_{mn}) = 0$, respectively. These functions are defined in Gradshteyn and Ryzhik (1980, Section 8.61). The natural frequencies in the x -direction are smaller than those in the y -direction depending on the eccentricity e .

1.6.8 Paraboloid container

Paraboloid containers usually form the lower portion of cylindrical containers. Bauer (1984d, 1999) used coordinate surfaces as boundaries to approximate the liquid-free-surface motion. The approximation is accurate if the shape of the free surface is close to the actual free surface in the container. With reference to the paraboloid coordinates, ξ, η, φ , shown in Figure 1.36, the free surface at $\xi = \xi_0$ exhibits the value $\xi_0 \gg 1$, when the container wall at $\eta = \eta_0 \ll 1$. The coordinates, ξ, η, φ , are related to the Cartesian coordinates by the following relations

$$x = \xi\eta \cos \varphi, \quad y = \xi\eta \sin \varphi, \quad z = \frac{1}{2}(\xi^2 - \eta^2) \quad (1.248a)$$

The radius in the xy plane is

$$r = \sqrt{x^2 + y^2} = \xi\eta \quad (1.248b)$$

The spatial radius is

$$R = \sqrt{x^2 + y^2 + z^2} = \frac{1}{2}(\xi^2 + \eta^2) \quad (1.248c)$$

The container wall, $\eta = \eta_0$, satisfies the equation

$$z = -\frac{\eta_0^2}{2} + \frac{1}{2\eta_0^2}(x^2 + y^2) \quad (1.248d)$$

The container bottom, $\xi = \xi_0$, and the free surface, $\xi = \xi_1$, are described by the following relation